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CALCULUS MADE E



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THE MACMILLAN CO. OF CANADA,
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AND THE

INTEGRAL CALCULUS.

BY

F. R. S.

SECOND EDITION, ENLARG

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What one fool can do, another can

(*Ancient Sim*

further explanations would be useful.

The author acknowledges with gratitude many va
and letters received from teachers, students, and—c

October, 1914.

PROLOGUE . . . . . .
I. To deliver you from the Prelimina
II. On Different Degrees of Smallnes
III. On Relative Growings . . . . . .
IV. Simplest Cases . . . . . .
V. Next Stage. What to do with Cons
VI. Sums, Differences, Products and Q
VII. Successive Differentiation . . . . . .
VIII. When Time Varies . . . . . .
IX. Introducing a Useful Dodge . . . . . .
X. Geometrical Meaning of Different
XI. Maxima and Minima . . . . . .
XII. Curvature of Curves . . . . . .
XIII. Other Useful Dodges . . . . . .
XIV. On true Compound Interest and the
ganic Growth . . . . . .

Table of Standard Forms . . . . . .

Answers to Exercises . . . . . .

Some calculus-tricks are quite easy. Some are er
The fools who write the textbooks of advanced math
are mostly clever fools—seldom take the trouble to
the easy calculations are. On the contrary, they
impress you with their tremendous cleverness by go
most difficult way.

Being myself a remarkably stupid fellow, I ha
myself the difficulties, and now beg to present to
parts that are not hard. Master these thoroughly
follow. What one fool can do, another can.

THE preliminary terror, which chokes off most fif
even attempting to learn how to calculate, can be
all by simply stating what is the meaning—in comm
the two principal symbols that are used in calculati

These dreadful symbols are:

(1) d which merely means "a little bit of."

Thus dx means a little bit of x; or du means a
dinary mathematicians think it more polite to say
instead of "a little bit of." Just as you please. But
these little bits (or elements) may be considered to be

(2) $\int$ which is merely a long S, and may be calle
sum of."

Thus $\int dx$ means the sum of all the little bits o
the sum of all the little bits of t. Ordinary mathe
symbol "the integral of." Now any fool can see tha
as made up of a lot of little bits, each of which i
add them all up together you get the sum of all the

That's all.

small quantities of various degrees of smallness.

We shall have also to learn under what circumst
sider small quantities to be so minute that we ma
consideration. Everything depends upon relative mi

Before we fix any rules let us think of some fan
are 60 minutes in the hour, 24 hours in the day, 7
There are therefore 1440 minutes in the day and 10
week.

Obviously 1 minute is a very small quantity of ti
a whole week. Indeed, our forefathers considered
pared with an hour, and called it "one minùte,"
fraction—namely one sixtieth—of an hour. When
quire still smaller subdivisions of time, they divide
60 still smaller parts, which, in Queen Elizabeth's
"second minùtes" (*i.e.* small quantities of the secon
ness). Nowadays we call these small quantities of t
smallness "seconds." But few people know *why* the

Now if one minute is so small as compared with

portion which for any purpose we call relatively sm
state other fractions of a higher degree of smallnes
purpose of time, $\frac{1}{60}$ be called a *small* fraction, ther
small fraction of a *small* fraction) may be regarded
of the second order of smallness.*

Or, if for any purpose we were to take 1 per c
small fraction, then 1 per cent. of 1 per cent. (*i.e.*
small fraction of the second order of smallness; and
a small fraction of the third order of smallness, be
1 per cent. of 1 per cent.

Lastly, suppose that for some very precise purpos
$\frac{1}{1{,}000{,}000}$ as "small." Thus, if a first-rate chronom
or gain more than half a minute in a year, it mu
an accuracy of 1 part in $1,051,200$. Now if, for s

*The mathematicians talk about the second order of "m
ness) when they really mean second order of *smallness*. Th
beginners.

But, it must be remembered, that small quantit
our expressions as factors multiplied by some other f
important if the other factor is itself large. Even a
important if only it is multiplied by a few hundred.

Now in the calculus we write dx for a little bit
such as dx, and du, and dy, are called "differentia
of x, or of u, or of y, as the case may be. [You *rea*
or *dee-you*, or *dee-wy*.] If dx be a small bit of x, and
itself, it does not follow that such quantities as $x \cdot dx$
are negligible. But $dx \times dx$ would be negligible, bei
of the second order.

A very simple example will serve as illustration.

Let us think of x as a quantity that can grow by
as to become $x + dx$, where dx is the small incremen
The square of this is $x^2 + 2x \cdot dx + (dx)^2$. The s
negligible because it is a first-order quantity; while t
the second order of smallness, being a bit of, a bit

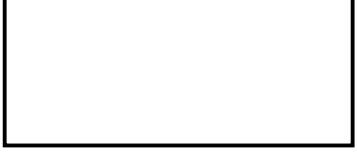


FIG. 1.

Geometrically this may be depicted as follows (Fig. 1) the side of which we will take to represen the square to grow by having a bit dx added to The enlarged square is made up of the original s rectangles at the top and on the right, each of whi (or together $2x \cdot dx$), and the little square at the top which is $(dx)^2$. In Fig. 2 we have taken dx as q of x—about $\frac{1}{5}$. But suppose we had taken it on thickness of an inked line drawn with a fine pen. Th square will have an area of only $\frac{1}{10{,}000}$ of x^2, and be p Clearly $(dx)^2$ is negligible if only we consider the i itself small enough.

Let us consider a simile.

Suppose a millionaire were to say to his secretar
give you a small fraction of any money that comes
that the secretary were to say to his boy: I will give y
of what I get. Suppose the fraction in each case to
if Mr. Millionaire received during the next week £
would receive £10 and the boy 2 shillings. Ten p
small quantity compared with £1000; but two shilli
quantity indeed, of a very secondary order. But v
disproportion if the fraction, instead of being $\frac{1}{100}$, h
$\frac{1}{1000}$ part? Then, while Mr. Millionaire got his £1
would get only £1, and the boy less than one farthi

The witty Dean Swift* once wrote:

> "So, Nat'ralists observe, a Flea
> "Hath smaller Fleas that on him prey.
> "And these have smaller Fleas to bite
> "And so proceed *ad infinitum*."

* *On Poetry: a Rhapsody* (p. 20), printed 1733—usually

ing, and with rates of growth. We classify all quantit
constants and *variables*. Those which we regard as
call *constants*, we generally denote algebraically by l
ginning of the alphabet, such as a, b, or c; while those
as capable of growing, or (as mathematicians say) o
note by letters from the end of the alphabet, such
or sometimes t.

Moreover, we are usually dealing with more th
once, and thinking of the way in which one variab
other: for instance, we think of the way in which
by a projectile depends on the time of attaining t
are asked to consider a rectangle of given area, and
increase in the length of it will compel a correspond
breadth of it. Or we think of the way in which a
slope of a ladder will cause the height that it reache

Suppose we have got two such variables that
other. An alteration in one will bring about an alte
because of this dependence. Let us call one of the v

angled triangle (Fig. 4), of which the slope of the

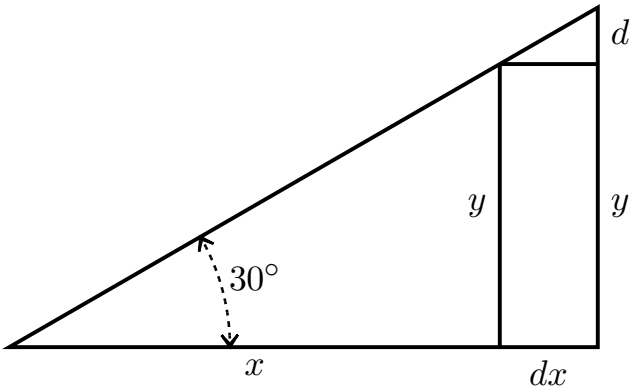


FIG. 4.

at 30°. If we suppose this triangle to expand and
the same as at first, then, when the base grows so as
the height becomes $y + dy$. Here, increasing x res
of y. The little triangle, the height of which is dy, an
is dx, is similar to the original triangle; and it is obv
of the ratio $\frac{dy}{dx}$ is the same as that of the ratio $\frac{y}{x}$.
it will be seen that here

$$\frac{dy}{dx} = \frac{1}{1.73}.$$

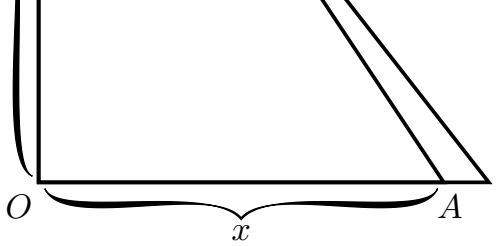


FIG. 5.

height it reaches up the wall. Now y clearly depends see that, if we pull the bottom end A a bit further top end B will come down a little lower. Let us sta language. If we increase x to $x + dx$, then y will bec when x receives a positive increment, the increment is negative.

Yes, but how much? Suppose the ladder was so bottom end A was 19 inches from the wall the top 15 feet from the ground. Now, if you were to pull t 1 inch more, how much would the top end come do inches: $x = 19$ inches, $y = 180$ inches. Now the inc we call dx, is 1 inch: or $x + dx = 20$ inches.

Now y is 180, so that dy is $180 - 179.89 = 0.11$ inch

So we see that making dx an increase of 1 in making dy a decrease of 0.11 inch.

And the ratio of dy to dx may be stated thus:

$$\frac{dy}{dx} = -\frac{0.11}{1}.$$

It is also easy to see that (except in one particul be of a different size from dx.

Now right through the differential calculus we ar hunting for a curious thing, a mere ratio, namely, th dy bears to dx when both of them are indefinitely s

It should be noted here that we can only find t y and x are related to each other in some way, so tha y does vary also. For instance, in the first exampl base x of the triangle be made longer, the height becomes greater also, and in the second example, i the foot of the ladder from the wall be made to inc

was built, any change in x would naturally cause n
in y; in this case $\frac{dy}{dx}$ has no meaning whatever, an
to find an expression for it. Whenever we use d
dz, etc., the existence of some kind of relation betw
implied, and this relation is called a "function" in x
expressions given above, for instance, namely $\frac{y}{x} = \text{ta}$
l^2, are functions of x and y. Such expressions conta
is, contain without distinctly showing it) the means
x in terms of y or y in terms of x, and for this rea
implicit functions in x and y; they can be respec
forms

$$y = x \tan 30^\circ \quad \text{or} \quad x = \frac{y}{\tan 30^\circ}$$

and

$$y = \sqrt{l^2 - x^2} \quad \text{or} \quad x = \sqrt{l^2 - y^2}$$

These last expressions state explicitly (that is, d
of x in terms of y, or of y in terms of x, and they
called *explicit functions* of x or y. For example x^2

are the independent variables, and u is the depende

Sometimes the exact relation between several qu
ther is not known or it is not convenient to state it
or convenient to state, that there is some sort of rela
variables, so that one cannot alter either x or y o
affecting the other quantities; the existence of a f
is then indicated by the notation $F(x, y, z)$ (impli
$x = F(y, z)$, $y = F(x, z)$ or $z = F(x, y)$ (explicit fun
the letter f or ϕ is used instead of F, so that $y = f$
$y = \phi(x)$ all mean the same thing, namely, that the
on the value of x in some way which is not stated.

We call the ratio $\dfrac{dy}{dx}$ "the *differential coefficien*
to x." It is a solemn scientific name for this very
we are not going to be frightened by solemn name
themselves are so easy. Instead of being frightened
nounce a brief curse on the stupidity of giving long
and, having relieved our minds, will go on to the

of the differential coefficient is that to which it ap
limiting case when each of them is considered as infi

Let us now learn how to go in quest of $\frac{dy}{dx}$.

$\frac{dy}{dx}$ is read "*dee-wy by dee-eks,*" or "*dee-wy o*

So also $\frac{du}{dt}$ is read "*dee-you by dee-tee.*"

Second differential coefficients will be met with like this:

$\frac{d^2y}{dx^2}$; which is read "*dee-two-wy over dee-ek*

and it means that the operation of differentiating y has been (or has to be) performed twice over.

Another way of indicating that a function has be by putting an accent to the symbol of the function. which means that y is some unspecified function of x write $F'(x)$ instead of $\frac{d\big(F(x)\big)}{dx}$. Similarly, $F''(x)$ original function $F(x)$ has been differentiated twice to x.

algebraical expression.

Case 1.

Let us begin with the simple expression $y = x$ that the fundamental notion about the calculus is t Mathematicians call it *varying*. Now as y and x another, it is clear that if x grows, x^2 will also grow then y will also grow. What we have got to find ou between the growing of y and the growing of x. In ot is to find out the ratio between dy and dx, or, in brie of $\frac{dy}{dx}$.

Let x, then, grow a little bit bigger and become y will grow a bit bigger and will become $y+dy$. Then be true that the enlarged y will be equal to the squar Writing this down, we have:

$$y + dy = (x + dx)^2.$$

$$y + dy = x^2 + 2x \cdot dx.$$

Now $y = x^2$; so let us subtract this from the equ
left

$$dy = 2x \cdot dx.$$

Dividing across by dx, we find

$$\frac{dy}{dx} = 2x.$$

Now *this** is what we set out to find. The ratio
to the growing of x is, in the case before us, found t

**N.B.*—This ratio $\frac{dy}{dx}$ is the result of differentiating y w
ferentiating means finding the differential coefficient. Suppo
function of x, as, for example, $u = 7x^2 + 3$. Then if we were to
with respect to x, we should have to find $\frac{du}{dx}$, or, what is the sa
On the other hand, we may have a case in which time was the
(see p. 14), such as this: $y = b + \frac{1}{2}at^2$. Then, if we were told t
means we must find its differential coefficient with respect t
business would be to try to find $\frac{dy}{dt}$, that is, to find $\frac{d(b+\frac{1}{2}a}{dt}$

dx

But, you will say, we neglected a whole unit.

Well, try again, making dx a still smaller bit.

Try $dx = \frac{1}{10}$. Then $x + dx = 100.1$, and

$$(x + dx)^2 = 100.1 \times 100.1 = 10,020.$$

Now the last figure 1 is only one-millionth part
is utterly negligible; so we may take $10,020$ withou
at the end. And this makes $dy = 20$; and $\frac{dy}{dx} = \frac{2}{0}$
still the same as $2x$.

Case 2.

Try differentiating $y = x^3$ in the same way.

We let y grow to $y + dy$, while x grows to $x + d$

Then we have

$$y + dy = (x + dx)^3.$$

Doing the cubing we obtain

$$y + dy = x^3 + 3x^2 \cdot dx + 3x(dx)^2 + (d$$

dx

Case 3.

Try differentiating $y = x^4$. Starting as before by l
grow a bit, we have:

$$y + dy = (x + dx)^4.$$

Working out the raising to the fourth power, we

$$y + dy = x^4 + 4x^3\,dx + 6x^2(dx)^2 + 4x(dx)^3$$

Then striking out the terms containing all the h
as being negligible by comparison, we have

$$y + dy = x^4 + 4x^3\,dx.$$

Subtracting the original $y = x^4$, we have left

$$dy = 4x^3\,dx,$$

and

$$\frac{dy}{dx} = 4x^3.$$

Just look at these results: the operation of diffe
to have had the effect of diminishing the power of x
in the last case reducing x^4 to x^3), and at the same t
a number (the same number in fact which original
power). Now, when you have once seen this, you mig
how the others will run. You would expect that diffe
give $5x^4$, or differentiating x^6 would give $6x^5$. If yo
of these, and see whether the conjecture comes righ

Try $y = x^5$.

Then
$$\begin{aligned} y + dy &= (x + dx)^5 \\ &= x^5 + 5x^4\,dx + 10x^3(dx)^2 + 10x \\ &\qquad + 5x(dx)^4 + (dx)^5 \end{aligned}$$

Neglecting all the terms containing small quant
orders, we have left

$$y + dy = x^5 + 5x^4\,dx,$$

Let $y = x^n$,

then, we should expect to find that

$$\frac{dy}{dx} = nx^{(n-1)}.$$

For example, let $n = 8$, then $y = x^8$; and differ
give $\frac{dy}{dx} = 8x^7$.

And, indeed, the rule that differentiating x^n gives
is true for all cases where n is a whole number and po
$(x + dx)^n$ by the binomial theorem will at once sh
question whether it is true for cases where n has neg
values requires further consideration.

Case of a negative power.

Let $y = x^{-2}$. Then proceed as before:

$$\begin{aligned} y + dy &= (x + dx)^{-2} \\ &= x^{-2}\left(1 + \frac{dx}{x}\right)^{-2}. \end{aligned}$$

Subtracting the original $y = x^{-2}$, we find

$$dy = -2x^{-3}dx,$$
$$\frac{dy}{dx} = -2x^{-3}.$$

And this is still in accordance with the rule inferred

Case of a fractional power.

Let $y = x^{\frac{1}{2}}$. Then, as before,

$$y + dy = (x + dx)^{\frac{1}{2}} = x^{\frac{1}{2}}\left(1 + \frac{dx}{x}\right)^{\frac{1}{2}}$$
$$= \sqrt{x} + \frac{1}{2}\frac{dx}{\sqrt{x}} - \frac{1}{8}\frac{(dx)^2}{x\sqrt{x}} + \text{terms wi}$$
powers

Subtracting the original $y = x^{\frac{1}{2}}$, and neglecting have left:

$$dy = \frac{1}{2}\frac{dx}{\sqrt{x}} = \frac{1}{2}x^{-\frac{1}{2}} \cdot dx,$$

(3) $y = x^{2a}$

(4) $u = t^{2.4}$

(5) $z = \sqrt[3]{u}$

(6) $y = \sqrt[3]{x^-$

(7) $u = \sqrt[5]{\dfrac{1}{x^8}}$

(8) $y = 2x^a$

(9) $y = \sqrt[q]{x^3}$

(10) $y = \sqrt[n]{\frac{}{x}$

You have now learned how to differentiate power

is!

being made to grow y also changed its value and
think of x as a quantity that we can vary; and, rega
of x as a sort of *cause*, we consider the resulting v
effect. In other words, we regard the value of y as
of x. Both x and y are variables, but x is the one tha
and y is the "dependent variable." In all the precedi
been trying to find out rules for the proportion wh
variation in y bears to the variation independently

Our next step is to find out what effect on the p
ating is caused by the presence of *constants*, that is
don't change when x or y change their values.

Added Constants.

Let us begin with some simple case of an added

Let $$y = x^3 + 5.$$

Just as before, let us suppose x to grow to $x+dx$ and

$$dy = 3x^2\,dx.$$

$$\frac{dy}{dx} = 3x^2.$$

So the 5 has quite disappeared. It added noth
of x, and does not enter into the differential coefficie
or 700, or any other number, instead of 5, it would
So if we take the letter a, or b, or c to represent a
simply disappear when we differentiate.

If the additional constant had been of negat
-5 or $-b$, it would equally have disappeared.

Multiplied Constants.

Take as a simple experiment this case:

Let $y = 7x^2$.

Then on proceeding as before we get:

$$\begin{aligned} y + dy &= 7(x + dx)^2 \\ &= 7\{x^2 + 2x \cdot dx + (dx)^2\} \\ &= 7x^2 + 14x \cdot dx + 7(dx)^2. \end{aligned}$$

of $\frac{dy}{dx}$.

These values we tabulate as follows:

x	0	1	2	3	4	5	-
y	0	7	28	63	112	175	
$\frac{dy}{dx}$	0	14	28	42	56	70	–

Now plot these values to some convenient scale, two curves, Figs. 6 and 6a.

Carefully compare the two figures, and verify the height of the ordinate of the derived curve, Fig. to the *slope* of the original curve,* Fig. 6, at the c of x. To the left of the origin, where the original curv (that is, downward from left to right) the correspo the derived curve are negative.

Now if we look back at p. 18, we shall see tha ating x^2 gives us $2x$. So that the differential coeffi

*See p. 76 about *slopes* of curves.

−3 −2 −1 0 1 2 3 4 5 −50

FIG. 6.—Graph of $y = 7x^2$. FIG. 6*a*.—G

7 times as big as that of x^2. If we had taken $8x^2$, th
cient would have come out eight times as great as th
$y = ax^2$, we shall get

$$\frac{dy}{dx} = a \times 2x.$$

If we had begun with $y = ax^n$, we should have ha
So that any mere multiplication by a constant re
multiplication when the thing is differentiated. And,
multiplication is equally true about *division*: for
above, we had taken as the constant $\frac{1}{7}$ instead of 7, v
the same $\frac{1}{7}$ come out in the result after differentiati

Some Further Examples.

The following further examples, fully worked out
master completely the process of differentiation as a

(2) Differentiate $y = a\sqrt{x} - \frac{1}{2}\sqrt{a}$.

The term $\frac{1}{2}\sqrt{a}$ vanishes, being an added consta

the index form, is written $ax^{\frac{1}{2}}$, we have

$$\frac{dy}{dx} = a \times \frac{1}{2} \times x^{\frac{1}{2}-1} = \frac{a}{2} \times x^{-\frac{1}{2}},$$

or

$$\frac{dy}{dx} = \frac{a}{2\sqrt{x}}.$$

(3) If $ay + bx = by - ax + (x + y)\sqrt{a^2 - b^2}$,

find the differential coefficient of y with respect to x

As a rule an expression of this kind will need a lit

than we have acquired so far; it is, however, always

whether the expression can be put in a simpler form

First we must try to bring it into the form $y =$

involving x only.

The expression may be written

$$(a - b)y + (a + b)x = (x + y)\sqrt{a^2 -$$

hence $y = \sqrt{\frac{}{a-b}}x$ and $\frac{}{dx} = \sqrt{\frac{}{a-}}$

(4) The volume of a cylinder of radius r and h
the formula $V = \pi r^2 h$. Find the rate of variation
radius when $r = 5.5$ in. and $h = 20$ in. If $r = h$, f
of the cylinder so that a change of 1 in. in radius
400 cub. in. in the volume.

The rate of variation of V with regard to r is

$$\frac{dV}{dr} = 2\pi r h.$$

If $r = 5.5$ in. and $h = 20$ in. this becomes
that a change of radius of 1 inch will cause a ch
690.8 cub. inch. This can be easily verified, for
$r = 5$ and $r = 6$ are 1570 cub. in. and 2260.8 cub. i
$2260.8 - 1570 = 690.8$.

Also, if

$$r = h, \quad \frac{dV}{dr} = 2\pi r^2 = 400 \quad \text{and} \quad r = h = \sqrt{\frac{4}{}}$$

was 1000° C.

The sensitiveness is the rate of variation of the temperature, that is $\frac{d\theta}{dt}$. The formula may be writt

$$\theta = \frac{\theta_1}{t_1^4} t^4 = \frac{25t^4}{1000^4},$$

and we have

$$\frac{d\theta}{dt} = \frac{100t^3}{1000^4} = \frac{t^3}{10,000,000,000}.$$

When $t = 800$, 1000 and 1200, we get $\frac{d\theta}{dt} = 0.05$ respectively.

The sensitiveness is approximately doubled from becomes three-quarters as great again up to 1200°.

Exercises II. (See p. 252 for Answers.)

Differentiate the following:

(1) $y = ax^3 + 6$.

(2) $y = 13x^{\frac{3}{2}}$

(8) It has been found that if c be the candle pow
cent electric lamp, and V be the voltage, $c = aV^b$,
constants.

Find the rate of change of the candle power wit
calculate the change of candle power per volt at 80
in the case of a lamp for which $a = 0.5 \times 10^{-10}$ and

(9) The frequency n of vibration of a string of dia
and specific gravity σ, stretched with a force T, is g

$$n = \frac{1}{DL}\sqrt{\frac{gT}{\pi\sigma}}.$$

Find the rate of change of the frequency when
varied singly.

(10) The greatest external pressure P which a tub
out collapsing is given by

$$P = \left(\frac{2E}{1-\sigma^2}\right)\frac{t^3}{D^3},$$

(c) the lateral area of a cone of slant dimensi

(d) the volume of a cone of radius r and heig

(e) the area of a sphere of radius r;

(f) the volume of a sphere of radius r.

(12) The length L of an iron rod at the temperatu $L = l_t\big[1+0.000012(T-t)\big]$, where l_t is the length at find the rate of variation of the diameter D of an ir being shrunk on a wheel, when the temperature T

We have learned how to differentiate simple algebra
as $x^2 + c$ or ax^4, and we have now to consider how
of two or more functions.

For instance, let

$$y = (x^2 + c) + (ax^4 + b);$$

what will its $\frac{dy}{dx}$ be? How are we to go to work on t

The answer to this question is quite simple: just
one after the other, thus:

$$\frac{dy}{dx} = 2x + 4ax^3. \quad (Ans.)$$

If you have any doubt whether this is right, try a
working it by first principles. And this is the way.

Let $y = u+v$, where u is any function of x, and v
of x. Then, letting x increase to $x + dx$, y will incre
u will increase to $u + du$; and v to $v + dv$.

And we shall have:

$$y + dy = u + du + v + dv.$$

preceding paragraph, and put in the values of the

shall have, using the notation shown (p. 16),

$$\begin{aligned}\frac{dy}{dx} &= \frac{d(x^2+c)}{dx} + \frac{d(ax^4+b)}{dx} \\ &= 2x \qquad\quad + 4ax^3,\end{aligned}$$

exactly as before.

If there were three functions of x, which we may

that

$$y = u + v + w;$$

then $$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} + \frac{dw}{dx}.$$

As for *subtraction*, it follows at once; for if the fu

had a negative sign, its differential coefficient woul

so that by differentiating

$$y = u - v,$$

we should get $$\frac{dy}{dx} = \frac{du}{dx} - \frac{dv}{dx}.$$

Now there are two ways in which we may go to

First way. Do the multiplying first, and, having
differentiate.

Accordingly, we multiply together $x^2 + c$ and ax

This gives $ax^6 + acx^4 + bx^2 + bc$.

Now differentiate, and we get:

$$\frac{dy}{dx} = 6ax^5 + 4acx^3 + 2bx.$$

Second way. Go back to first principles, and con

$$y = u \times v;$$

where u is one function of x, and v is any other func
x grows to be $x + dx$; and y to $y + dy$; and u beco
becomes $v + dv$, we shall have:

$$\begin{aligned} y + dy &= (u + du) \times (v + dv) \\ &= u \cdot v + u \cdot dv + v \cdot du + du \cdot \end{aligned}$$

and, dividing through by dx, we get the result:

$$\frac{dy}{dx} = u\,\frac{dv}{dx} + v\,\frac{du}{dx}.$$

This shows that our instructions will be as follow

the product of two functions, multiply each function

coefficient of the other, and add together the two pr

You should note that this process amounts to the

as constant while you differentiate v; then treat v as

differentiate u; and the whole differential coefficient

of these two treatments.

Now, having found this rule, apply it to the conc

was considered above.

We want to differentiate the product

$$(x^2 + c) \times (ax^4 + b).$$

Call $(x^2 + c) = u$; and $(ax^4 + b) = v$.

Think of this example, $y = \dfrac{bx^5 + c}{x^2 + a}$. In such a
try to work out the division beforehand, because x^2
into $bx^5 + c$, neither have they any common factor.
for it but to go back to first principles, and find a r

So we will put $$y = \frac{u}{v};$$

where u and v are two different functions of the inde
Then, when x becomes $x + dx$, y will become $y + dy$;
$u + du$; and v will become $v + dv$. So then

$$y + dy = \frac{u + du}{v + dv}.$$

$$\frac{v}{} \qquad \frac{v}{}$$

$$-\frac{u \cdot dv}{v} - \frac{u \cdot dv \cdot dv}{v^2}$$

$$-\frac{du \cdot dv}{v} + \frac{u \cdot}{}$$

As both these remainders are small quantities o
they may be neglected, and the division may stop her
remainders would be of still smaller magnitudes.

So we have got:

$$y + dy = \frac{u}{v} + \frac{du}{v} - \frac{u \cdot dv}{v^2};$$

which may be written

$$= \frac{u}{v} + \frac{v \cdot du - u \cdot dv}{v^2}.$$

the differential coefficient of the divisor function; a
divide by the square of the divisor function.

Going back to our example $y = \dfrac{bx^5 + c}{x^2 + a}$,

write $bx^5 + c = u;$

and $x^2 + a = v.$

Then

$$\frac{dy}{dx} = \frac{(x^2 + a)\,\dfrac{d(bx^5 + c)}{dx} - (bx^5 + c)\,\dfrac{d(}{}}{(x^2 + a)^2}$$

$$= \frac{(x^2 + a)(5bx^4) - (bx^5 + c)(2x)}{(x^2 + a)^2},$$

$$\frac{dy}{dx} = \frac{3bx^6 + 5abx^4 - 2cx}{(x^2 + a)^2}. \quad (Answer.)$$

The working out of quotients is often tedious, b
difficult about it.

Some further examples fully worked out are give

(2) Differentiate $y = 2a\sqrt{b}x^3 - \dfrac{3b\sqrt[3]{a}}{x} - 2\sqrt{ab}$.

Putting x in the index form, we get

$$y = 2a\sqrt{b}x^{\frac{3}{2}} - 3b\sqrt[3]{a}x^{-1} - 2\sqrt{ab}.$$

Now

$$\frac{dy}{dx} = 2a\sqrt{b} \times \tfrac{3}{2} \times x^{\frac{3}{2}-1} - 3b\sqrt[3]{a} \times (-1) \times$$

or,
$$\frac{dy}{dx} = 3a\sqrt{bx} + \frac{3b\sqrt[3]{a}}{x^2}.$$

(3) Differentiate $z = 1.8\sqrt[3]{\dfrac{1}{\theta^2}} - \dfrac{4.4}{\sqrt[5]{\theta}} - 27°$.

This may be written: $z = 1.8\,\theta^{-\frac{2}{3}} - 4.4\,\theta^{-\frac{1}{5}} - 27$

The $27°$ vanishes, and we have

$$\frac{dz}{d\theta} = 1.8 \times -\tfrac{2}{3} \times \theta^{-\frac{2}{3}-1} - 4.4 \times \left(-\tfrac{1}{5}\right)\theta$$

or,
$$\frac{dz}{d\theta} = -1.2\,\theta^{-\frac{5}{3}} + 0.88\,\theta^{-\frac{6}{5}};$$

or,
$$\frac{dz}{d\theta} = \frac{0.88}{\sqrt[5]{\theta^6}} - \frac{1.2}{\sqrt[3]{\theta^5}}.$$

(5) Differentiate $y = (2x-3)(x+1)^2$.

$$\frac{dy}{dx} = (2x-3)\,\frac{d\big[(x+1)(x+1)\big]}{dx} + (x+1)^2$$

$$= (2x-3)\left[(x+1)\,\frac{d(x+1)}{dx} + (x+1)\right.$$

$$\left. + (x+1)^2\right.$$

$$= 2(x+1)\big[(2x-3)+(x+1)\big] = 2(x+$$

or, more simply, multiply out and then differentiate

(6) Differentiate $y = 0.5x^3(x-3)$.

$$\frac{dy}{dx} = 0.5\left[x^3\frac{d(x-3)}{dx} + (x-3)\frac{d(x^3)}{dx}\right]$$

$$= 0.5\left[x^3 + (x-3)\times 3x^2\right] = 2x^3 -$$

Same remarks as for preceding example.

(7) Differentiate $w = \left(\theta + \frac{1}{\theta}\right)\left(\sqrt{\theta} + \frac{1}{\sqrt{\theta}}\right)$.

This, again, could be obtained more simply by n
factors first, and differentiating afterwards. This is n
possible; see, for instance, p. 170, example 8, in
differentiating a product *must* be used.

(8) Differentiate $y = \dfrac{a}{1 + a\sqrt{x} + a^2 x}$.

$$\frac{dy}{dx} = \frac{(1 + ax^{\frac{1}{2}} + a^2 x) \times 0 - a\dfrac{d(1 + ax^{\frac{1}{2}}}{dx}}{(1 + a\sqrt{x} + a^2 x)^2}$$

$$= -\frac{a(\frac{1}{2}ax^{-\frac{1}{2}} + a^2)}{(1 + ax^{\frac{1}{2}} + a^2 x)^2}.$$

(9) Differentiate $y = \dfrac{x^2}{x^2 + 1}$.

$$\frac{dy}{dx} = \frac{(x^2 + 1)\, 2x - x^2 \times 2x}{(x^2 + 1)^2} = \frac{2x}{(x^2 +}$$

(10) Differentiate $y = \dfrac{a + \sqrt{x}}{a - \sqrt{x}}$.

Now $$\theta = \frac{1 - at^{\frac{3}{2}}}{1 + at^{\frac{3}{2}}}.$$

$$\frac{d\theta}{dt} = \frac{(1 + at^{\frac{3}{2}})(-\frac{2}{3}at^{-\frac{1}{3}}) - (1 - at^{\frac{2}{3}}) \times}{(1 + at^{\frac{3}{2}})^2}$$

$$= \frac{5a^2\sqrt[6]{t^7} - \dfrac{4a}{\sqrt[3]{t}} - 9a\sqrt[2]{t}}{6(1 + a\sqrt[2]{t^3})^2}.$$

(12) A reservoir of square cross-section has sides
of 45° with the vertical. The side of the bottom is
expression for the quantity pouring in or out when
varies by 1 foot; hence find, in gallons, the quantity
when the depth is reduced from 14 to 10 feet in 24

The volume of a frustum of pyramid of height
and a, is $V = \frac{H}{3}(A + a + \sqrt{Aa})$. It is easily seen th
45°, if the depth be h, the length of the side of th
the water is $200 + 2h$ feet, so that the volume of wa

$$\frac{h}{3}[200^2 + (200 + 2h)^2 + 200(200 + 2h)] = 40,000h$$

the temperature at 100° C.

Expand the numerator by the binomial theorem

$$P = \frac{1}{140^5}(40^5 + 5 \times 40^4 t + 10 \times 40^3 t^2 + 10 \times 40^2 t^3$$

hence $$\frac{dP}{dt} = \frac{1}{537,824 \times 10^5}$$
$$(5 \times 40^4 + 20 \times 40^3 t + 30 \times 40^2 t^2 + 20$$

when $t = 100$ this becomes 0.036 atmosphere per change of temperature.

Exercises III. (See the Answers on p. 253.)

(1) Differentiate

(a) $u = 1 + x + \frac{x^2}{1 \times 2} + \frac{x^3}{1 \times 2 \times 3} + \cdots$.

(b) $y = ax^2 + bx + c$. (c) $y = (x + a$

(d) $y = (x + a)^3$.

(5) If $x = (y+3) \times (y+5)$, find $\frac{dy}{dx}$.

(6) Differentiate $y = 1.3709x \times (112.6 + 45.202x$

Find the differential coefficients of

(7) $y = \dfrac{2x+3}{3x+2}$.

(8) $y = \dfrac{1+}{1}$

(9) $y = \dfrac{ax+b}{cx+d}$.

(10) $y = \dfrac{x^n}{x^{-n}}$

(11) The temperature t of the filament of an in
lamp is connected to the current passing through t
lation

$$C = a + bt + ct^2.$$

Find an expression giving the variation of the cur
to a variation of temperature.

(12) The following formulae have been proposed
lation between the electric resistance R of a wire a

(13) The electromotive-force E of a certain type c
been found to vary with the temperature t accordin

$$E = 1.4340\big[1 - 0.000814(t - 15) + 0.000007(t$$

Find the change of electromotive-force per d
and 25°.

(14) The electromotive-force necessary to maintai
length l with a current of intensity i has been found
be

$$E = a + bl + \frac{c + kl}{i},$$

where a, b, c, k are constants.

Find an expression for the variation of the
(a) with regard to the length of the arc; (b) with reg
of the current.

differentiating a function (see p. 13). Begin with a

Let $y = x^5$.

First differentiation,	$5x^4$.
Second differentiation,	$5 \times 4x^3$
Third differentiation,	$5 \times 4 \times 3x^2$
Fourth differentiation,	$5 \times 4 \times 3 \times 2x$
Fifth differentiation,	$5 \times 4 \times 3 \times 2 \times 1$
Sixth differentiation,	

There is a certain notation, with which we are (see p. 14), used by some writers, that is very co to employ the general symbol $f(x)$ for any functi symbol $f(\ )$ is read as "function of," without sayi function is meant. So the statement $y = f(x)$ merel a function of x, it may be x^2 or ax^n, or $\cos x$ or any function of x.

Second differentiation, $f''(x) = n(n-1)x^{n-2}$

Third differentiation, $f'''(x) = n(n-1)(n-$

Fourth differentiation, $f''''(x) = n(n-1)(n-$

etc., etc.

But this is not the only way of indicating success
For,

if the original function be $y = f(x)$;

once differentiating gives $\dfrac{dy}{dx} = f'(x)$;

twice differentiating gives $\dfrac{d\left(\dfrac{dy}{dx}\right)}{dx} = f''(x)$;

and this is more conveniently written as $\dfrac{d^2y}{(dx)^2}$, or

Similarly, we may write as the result of thrice diff
$f'''(x)$.

$$\frac{d^5y}{dx^5} = f''''(x) = 0.$$

In a similar manner if $y = \phi(x) = 3x(x^2 - 4)$,

$$\phi'(x) = \frac{dy}{dx} = 3\big[x \times 2x + (x^2 - 4) \times 1\big] =$$
$$\phi''(x) = \frac{d^2y}{dx^2} = 3 \times 6x = 18x,$$
$$\phi'''(x) = \frac{d^3y}{dx^3} = 18,$$
$$\phi''''(x) = \frac{d^4y}{dx^4} = 0.$$

Exercises IV. (See page 253 for Answers.)

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for the following expressions:

(1) $y = 17x + 12x^2.$

(2) $y = \frac{x^2 +$
$x +$

(3) $y = 1 + \frac{x}{1} + \frac{x^2}{1 \times 2} + \frac{x^3}{1 \times 2 \times 3} + \frac{x^4}{1 \times 2 \times 3 \times$

time is the independent variable, and we have to find
of some other quantity that varies when the time v
grow larger as time goes on; some other things grow
tance that a train has got from its starting place goes
as time goes on. Trees grow taller as the years go by
at the greater rate; a plant 12 inches high which in o
14 inches high, or a tree 12 feet high which in a ye
high?

In this chapter we are going to make much use
Nothing to do with poor-rate, or water-rate (except
word suggests a proportion—a ratio—so many per
Nothing to do even with birth-rate or death-rate, t
suggest so many births or deaths per thousand of the
a motor-car whizzes by us, we say: What a terri
spendthrift is flinging about his money, we remar
man is living at a prodigious rate. What do we
both these cases we are making a mental comparison
is happening, and the length of time that it takes

Take another example. A man may have only a possession, and yet be able to spend money at th a year—provided he goes on spending money at t minutes only. Suppose you hand a shilling over t for some goods; and suppose the operation lasts e Then, during that brief operation, you are partin at the rate of 1 shilling per second, which is the sa minute, or £180 per hour, or £4320 per day, or £1 If you have £10 in your pocket, you can go on spen rate of a million a year for just $5\frac{1}{4}$ minutes.

It is said that Sandy had not been in London when "bang went saxpence." If he were to spend m all day long, say for 12 hours, he would be spending or £3. 12*s.* per day, or £21. 12*s.* a week, not counti

Now try to put some of these ideas into differen

Let y in this case stand for money, and let t sta

If you are spending money, and the amount yo

speed of a moving body. From London (Euston stat
200 miles. If a train leaves London at 7 o'clock, and
at 11 o'clock, you know that, since it has travelled 20
its average rate must have been 50 miles per hour;
Here you are really making a mental comparison be
passed over and the time taken to pass over it. Yo
by the other. If y is the whole distance, and t the
the average rate is $\frac{y}{t}$. Now the speed was not actua
way: at starting, and during the slowing up at the
the speed was less. Probably at some part, when
the speed was over 60 miles an hour. If, during any
of time dt, the corresponding element of distance p
then at that part of the journey the speed was $\frac{dy}{dt}$.
one quantity (in the present instance, *distance*) is c
to the other quantity (in this case, *time*) is proper
by stating the differential coefficient of one with re
A *velocity*, scientifically expressed, is the rate at
distance in any given direction is being passed over;

but dv is itself $d\left(\frac{dy}{dt}\right)$. Hence we may put

$$a = \frac{d\left(\frac{dy}{dt}\right)}{dt};$$

and this is usually written $a = \frac{d^2y}{dt^2}$;
or the acceleration is the second differential coefficie
with respect to time. Acceleration is expressed as a
in unit time, for instance, as being so many feet per
the notation used being feet $\div$ second^2.

When a railway train has just begun to move, its
but it is rapidly gaining speed—it is being hurried
by the effort of the engine. So its $\frac{d^2y}{dt^2}$ is large. Whe
top speed it is no longer being accelerated, so that t
to zero. But when it nears its stopping place its sp
down; may, indeed, slow down very quickly if the

or $$f = m\frac{d^2y}{dt^2}.$$

The product of a mass by the speed at which it i
momentum, and is in symbols mv. If we differentiat
respect to time we shall get $\frac{d(mv)}{dt}$ for the rate of cha
But, since m is a constant quantity, this may be w
we see above is the same as f. That is to say, force
either as mass times acceleration, or as rate of chan

Again, if a force is employed to move something
and opposite counter-force), it does *work*; and the ar
is measured by the product of the force into the d
direction) through which its point of application m
a force f moves forward through a length y, the wo
may call w) will be

$$w = f \times y;$$

where we take f as a constant force. If the force
parts of the range y, then we must find an expressio

or $$dw = m\frac{d^2y}{dt^2} \cdot dy;$$

or $$dw = m\frac{dv}{dt} \cdot dy.$$

Further, we may transpose the expression and w

$$\frac{dw}{dy} = f.$$

This gives us yet a third definition of *force*; that to produce a displacement in any direction, the force is equal to the rate at which work is being done pe that direction. In this last sentence the word *rate* is its time-sense, but in its meaning as ratio or propor

Sir Isaac Newton, who was (along with Leibnitz) methods of the calculus, regarded all quantities th *flowing*; and the ratio which we nowadays call the dif he regarded as the rate of flowing, or the *fluxion* of th tion. He did not use the notation of the dy and dx, a to Leibnitz), but had instead a notation of his own.

$dx \qquad dt \qquad dz$

this fluxional notation is less informing than the di
and has in consequence largely dropped out of use.
gives it an advantage if only we will agree to use it
clusively where *time* is the independent variable.
mean $\frac{dy}{dt}$ and $\dot{u}$ will mean $\frac{du}{dt}$; and $\ddot{x}$ will mean $\frac{d^2x}{dt^2}$

Adopting this fluxional notation we may write th
tions considered in the paragraphs above, as follows

distance	x,
velocity	$v = \dot{x}$,
acceleration	$a = \dot{v} = \ddot{x}$,
force	$f = m\dot{v} = m\dot{}$
work	$w = x \times m\ddot{x}$.

Examples.

(1) A body moves so that the distance x (in fee
from a certain point O, is given by the relation $x =$
t is the time in seconds elapsed since a certain instant

the instant the body started.

When $t = 5$, $v = 0.4 \times 5 = 2$ ft./sec.; $a = 0.4$ ft.

When $x = 100$, $100 = 0.2t^2 + 10.4$, or $t^2 = 448$,

$v = 0.4 \times 21.17 = 8.468$ ft./sec.

When $t = 10$,

$$\text{distance travelled} = 0.2 \times 10^2 + 10.4 - 10.$$

$$\text{Average velocity} = \tfrac{20}{10} = 2 \text{ ft./sec}$$

(It is the same velocity as the velocity at the mi
$t = 5$; for, the acceleration being constant, the
uniformly from zero when $t = 0$ to 4 ft./sec. when t

(2) In the above problem let us suppose

$$x = 0.2t^2 + 3t + 10.4.$$

$$v = \dot{x} = \frac{dx}{dt} = 0.4t + 3; \quad a = \ddot{x} = \frac{d^2x}{dt^2} = 0.4$$

When $t = 0$, $x = 10.4$ and $v = 3$ ft./sec, the tim
the instant at which the body passed a point 10.4 ft

motion one must know how far the body was from t
started.

When $t = -7.5$,

$$x = 0.2 \times (-7.5)^2 - 3 \times 7.5 + 10.4 = -0$$

that is 0.85 ft. to the left of the point O.

Now, when $t = 2.5$,

$$x = 0.2 \times 2.5^2 + 3 \times 2.5 + 10.4 = 19$$

So, in 10 seconds, the distance travelled was 19.
and

$$\text{the average velocity } = \tfrac{20}{10} = 2 \text{ ft./s}$$

(3) Consider a similar problem when the distan
$0.2t^2 - 3t + 10.4$. Then $v = 0.4t - 3$, $a = 0.4 =$
$t = 0$, $x = 10.4$ as before, and $v = -3$; so that the
in the direction opposite to its motion in the prev
acceleration is positive, however, we see that this ve

When v is zero, $x = 0.2 \times 7.5^2 - 3 \times 7.5 + 10.4 =$ us that the body moves back to 0.85 ft. beyond th stops. Ten seconds later

$$t = 17.5 \text{ and } x = 0.2 \times 17.5^2 - 3 \times 17.5 + 10$$

The distance travelled $= .85 + 19.15 = 20.0$, and t is again 2 ft./sec.

(4) Consider yet another problem of the same so $3t^2 + 10.4$; $v = 0.6t^2 - 6t$; $a = 1.2t - 6$. The accel constant.

When $t = 0$, $x = 10.4$, $v = 0$, $a = -6$. The body ready to move with a negative acceleration, that is towards the point O.

(5) If we have $x = 0.2t^3 - 3t + 10.4$, then $v = 0.6t$

When $t = 0$, $x = 10.4$; $v = -3$; $a = 0$.

The body is moving towards the point O with a v and just at that instant the velocity is uniform.

1 second; (b) after it has performed one revolution.
at rest, and how many revolutions has it performed

Writing for the acceleration

$$\omega = \dot{\theta} = \frac{d\theta}{dt} = 2 - 0.3t^2, \quad \alpha = \ddot{\theta} = \frac{d^2\theta}{dt^2} =$$

When $t = 0$, $\theta = 3$; $\omega = 2$ rad./sec.; $\alpha = 0$.

When $t = 1$,

$$\omega = 2 - 0.3 = 1.7 \text{ rad./sec.}; \quad \alpha = -0.6 \text{ r}$$

This is a retardation; the wheel is slowing down

After 1 revolution

$$\theta = 2\pi = 6.28; \quad 6.28 = 3 + 2t - 0.1$$

By plotting the graph, $\theta = 3 + 2t - 0.1t^3$, we ca
values of t for which $\theta = 6.28$; these are 2.11 and 3.
negative value).

The velocity is reversed. The wheel is evident
these two instants; it is at rest when $\omega = 0$, that is
or when $t = 2.58$ sec., it has performed

$$\frac{\theta}{2\pi} = \frac{3 + 2 \times 2.58 - 0.1 \times 2.58^3}{6.28} = 1.025 \text{ re}$$

Exercises V. (See page 255 for Answers.)

(1) If $y = a + bt^2 + ct^4$; find $\frac{dy}{dt}$ and $\frac{d^2y}{dt^2}$.

Ans. $\frac{dy}{dt} = 2bt + 4ct^3$; $\frac{d^2y}{dt^2} = 2b + 1$

(2) A body falling freely in space describes in t
in feet, expressed by the equation $s = 16t^2$. Dra
the relation between s and t. Also determine the v
at the following times from its being let drop: $t =$
seconds; $t = 0.01$ second.

(3) If $x = at - \frac{1}{2}gt^2$; find $\dot{x}$ and $\ddot{x}$.

by the law

$$\theta = 2.1 - 3.2t + 4.8t^2.$$

Find the angular velocity (in radians per second) $1\frac{1}{2}$ seconds have elapsed. Find also its angular acce

(7) A slider moves so that, during the first par distance s in inches from its starting point is given

$$s = 6.8t^3 - 10.8t; \quad t \text{ being in secon}$$

Find the expression for the velocity and the a time; and hence find the velocity and the acceleratio

(8) The motion of a rising balloon is such that its is given at any instant by the expression $h = 0.5 +$ in seconds.

Find an expression for the velocity and the accele Draw curves to show the variation of height, velocit during the first ten minutes of the ascent.

(10) A body moves in such a way that the space time t from starting is given by $s = t^n$, where n is a value of n when the velocity is doubled from the 5th find it also when the velocity is numerically equal at the end of the 10th second.

entiated is too complicated to tackle directly.

Thus, the equation

$$y = (x^2 + a^2)^{\frac{3}{2}}$$

is awkward to a beginner.

Now the dodge to turn the difficulty is this: W
such as u, for the expression $x^2 + a^2$; then the equa

$$y = u^{\frac{3}{2}},$$

which you can easily manage; for

$$\frac{dy}{du} = \frac{3}{2}u^{\frac{1}{2}}.$$

Then tackle the expression

$$u = x^2 + a^2,$$

and differentiate it with respect to x,

$$\frac{du}{dx} = 2x.$$

cosines, and exponentials, you will find this dodge o
ness.

Examples.

Let us practise this dodge on a few examples.

(1) Differentiate $y = \sqrt{a+x}$.

Let $a + x = u$.

$$\frac{du}{dx} = 1; \quad y = u^{\frac{1}{2}}; \quad \frac{dy}{du} = \tfrac{1}{2}u^{-\frac{1}{2}} = \tfrac{1}{2}(a -$$
$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \frac{1}{2\sqrt{a+x}}.$$

(2) Differentiate $y = \dfrac{1}{\sqrt{a+x^2}}$.

Let $a + x^2 = u$.

$$\frac{du}{dx} = 2x; \quad y = u^{-\frac{1}{2}}; \quad \frac{dy}{du} = -\tfrac{1}{2}u^{-}$$
$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = -\frac{x}{\sqrt{(a+x^2)^3}}.$$

(4) Differentiate $y = \dfrac{1}{\sqrt{x^3 - a^2}}$.

Let $u = x^3 - a^2$.

$$\frac{du}{dx} = 3x^2; \quad y = u^{-\frac{1}{2}}; \quad \frac{dy}{du} = -\frac{1}{2}(x^3 -$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = -\frac{3x^2}{2\sqrt{(x^3 - a^2)^3}}.$$

(5) Differentiate $y = \sqrt{\dfrac{1-x}{1+x}}$.

Write this as $y = \dfrac{(1-x)^{\frac{1}{2}}}{(1+x)^{\frac{1}{2}}}$.

$$\frac{dy}{dx} = \frac{(1+x)^{\frac{1}{2}} \frac{d(1-x)^{\frac{1}{2}}}{dx} - (1-x)^{\frac{1}{2}} \frac{d(1}{}}{1+x}$$

(We may also write $y = (1-x)^{\frac{1}{2}}(1+x)^{-\frac{1}{2}}$ and
product.)

Proceeding as in example (1) above, we get

$$\frac{d(1-x)^{\frac{1}{2}}}{dx} = -\frac{1}{2\sqrt{1-x}}; \quad \text{and} \quad \frac{d(1+x)^{\frac{1}{2}}}{dx}$$

We may write this

$$y = x^{\frac{3}{2}}(1+x^2)^{-\frac{1}{2}};$$

$$\frac{dy}{dx} = \tfrac{3}{2}x^{\frac{1}{2}}(1+x^2)^{-\frac{1}{2}} + x^{\frac{3}{2}} \times \frac{d\left[(1+x^2\right.}{dx}$$

Differentiating $(1+x^2)^{-\frac{1}{2}}$, as shown in example

$$\frac{d\left[(1+x^2)^{-\frac{1}{2}}\right]}{dx} = -\frac{x}{\sqrt{(1+x^2)^3}};$$

so that

$$\frac{dy}{dx} = \frac{3\sqrt{x}}{2\sqrt{1+x^2}} - \frac{\sqrt{x^5}}{\sqrt{(1+x^2)^3}} = \frac{\sqrt{x}(3}{2\sqrt{(1}$$

(7) Differentiate $y = (x + \sqrt{x^2+x+a})^3$.

Let $x + \sqrt{x^2+x+a} = u$.

$$\frac{du}{dx} = 1 + \frac{d\left[(x^2+x+a)^{\frac{1}{2}}\right]}{dx}.$$

$$y = u^3; \quad \text{and} \quad \frac{dy}{du} = 3u^2 = 3\left(x + \sqrt{x^2+\right.$$

(8) Differentiate $y = \sqrt{\dfrac{a^2+x^2}{a^2-x^2}}\sqrt[3]{\dfrac{a^2-x^2}{a^2+x^2}}$.
We get

$$y = \frac{(a^2+x^2)^{\frac{1}{2}}(a^2-x^2)^{\frac{1}{3}}}{(a^2-x^2)^{\frac{1}{2}}(a^2+x^2)^{\frac{1}{3}}} = (a^2+x^2)^{\frac{1}{6}}(a$$

$$\frac{dy}{dx} = (a^2+x^2)^{\frac{1}{6}}\frac{d\left[(a^2-x^2)^{-\frac{1}{6}}\right]}{dx} + \frac{d\left[(a^2+$$
$$(a^2-x^2$$

Let $u = (a^2-x^2)^{-\frac{1}{6}}$ and $v = (a^2-x^2)$.

$$u = v^{-\frac{1}{6}}; \quad \frac{du}{dv} = -\frac{1}{6}v^{-\frac{7}{6}}; \quad \frac{dv}{dx} = -$$
$$\frac{du}{dx} = \frac{du}{dv}\times\frac{dv}{dx} = \frac{1}{3}x(a^2-x^2)^{-\frac{7}{6}}.$$

Let $w = (a^2+x^2)^{\frac{1}{6}}$ and $z = (a^2+x^2)$.

$$w = z^{\frac{1}{6}}; \quad \frac{dw}{dz} = \frac{1}{6}z^{-\frac{5}{6}}; \quad \frac{dz}{dx} = 2$$
$$\frac{dw}{dx} = \frac{dw}{dz}\times\frac{dz}{dx} = \frac{1}{3}x(a^2+x^2)^{-\frac{5}{6}}.$$

(10) Find the first and second differential coeffic
of $y = \frac{x}{b}\sqrt{(a-x)x}$.

$$\frac{dy}{dx} = \frac{x}{b}\,\frac{d\left\{\left[(a-x)x\right]^{\frac{1}{2}}\right\}}{dx} + \frac{\sqrt{(a-x}}{b}$$

Let $\left[(a-x)x\right]^{\frac{1}{2}} = u$ and let $(a-x)x = w$; then

$$\frac{du}{dw} = \frac{1}{2}w^{-\frac{1}{2}} = \frac{1}{2w^{\frac{1}{2}}} = \frac{1}{2\sqrt{(a-x)}}$$

$$\frac{dw}{dx} = a - 2x.$$

$$\frac{du}{dw} \times \frac{dw}{dx} = \frac{du}{dx} = \frac{a-2x}{2\sqrt{(a-x)x}}.$$

Hence

$$\frac{dy}{dx} = \frac{x(a-2x)}{2b\sqrt{(a-x)x}} + \frac{\sqrt{(a-x)x}}{b} = \frac{x(3}{2b\sqrt{(}$$

Exercises VI. (See page 255 for Answers.)

Differentiate the following:

(1) $y = \sqrt{x^2 + 1}$.

(2) $y = \sqrt{x^2$

(3) $y = \dfrac{1}{\sqrt{a + x}}$.

(4) $y = \dfrac{c}{\sqrt{a}$

(5) $y = \dfrac{\sqrt{x^2 - a^2}}{x^2}$.

(6) $y = \dfrac{\sqrt[3]{x^4}}{\sqrt[2]{x^3}}$

(7) $y = \dfrac{a^2 + x^2}{(a + x)^2}$.

(8) Differentiate y^5 with respect to y^2.

(9) Differentiate $y = \dfrac{\sqrt{1 - \theta^2}}{1 - \theta}$.

The process can be extended to three or more cients, so that $\dfrac{dy}{dx} = \dfrac{dy}{dz} \times \dfrac{dz}{dv} \times \dfrac{dv}{dx}$.

$$\frac{dv}{dx} = \frac{7x(5x-6)}{3\sqrt[3]{(x-1)^4}}; \quad \frac{dx}{dt} = 3t^2 + \tfrac{1}{2}; \quad \frac{dv}{d\theta} =$$

Hence $$\frac{dv}{d\theta} = -\frac{7x(5x-6)(3t^2+\frac{1}{2})}{30\sqrt[3]{(x-1)^4}\sqrt{\theta^3}},$$

an expression in which x must be replaced by its value in terms of θ.

(3) If $\theta = \dfrac{3a^2x}{\sqrt{x^3}}$; $\quad \omega = \dfrac{\sqrt{1-\theta^2}}{1+\theta}$; $\quad$ and $\phi = \sqrt{3} -$

We get

$$\theta = 3a^2x^{-\frac{1}{2}}; \quad \omega = \sqrt{\frac{1-\theta}{1+\theta}}; \quad \text{and} \quad \phi = \sqrt{3}$$

$$\frac{d\theta}{dx} = -\frac{3a^2}{2\sqrt{x^3}}; \quad \frac{d\omega}{d\theta} = -\frac{1}{(1+\theta)\sqrt{1-}$$

(see example 5, p. 68); and

$$\frac{d\phi}{d\omega} = \frac{1}{\sqrt{2}\omega^2}.$$

So that $\dfrac{d\theta}{dx} = \dfrac{1}{\sqrt{2}\times\omega^2} \times \dfrac{1}{(1+\theta)\sqrt{1-\theta^2}} \times \dfrac{3a^2}{2\sqrt{x}}$

Replace now first ω, then θ by its value.

---

differential coefficient.

In the first place, any function of x, such, for exa
or $ax + b$, can be plotted as a curve; and nowadays
familiar with the process of curve-plotting.

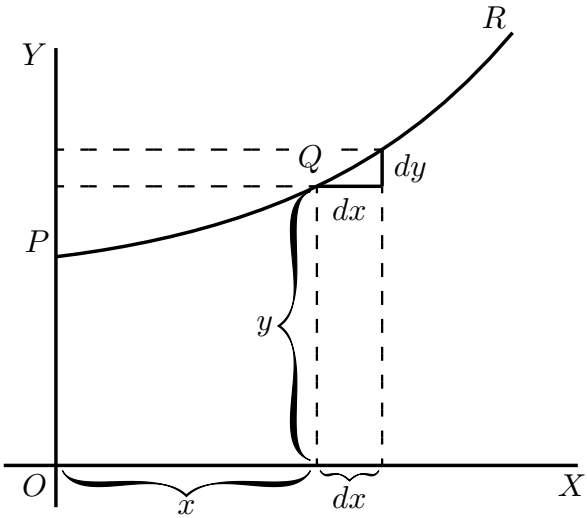


FIG. 7.

Let PQR, in Fig. 7, be a portion of a curve pl
to the axes of coordinates OX and OY. Consider a
curve, where the abscissa of the point is x and its
observe how y changes when x is varied. If x is m

the ratio $\frac{dy}{dx}$ is the slope of the curve along QT. Th
produced on either side touches the curve along th
and if this portion is indefinitely small, the straight
curve at practically one point only, and be therefor
curve.

This tangent to the curve has evidently the sam
that $\frac{dy}{dx}$ is the slope of the tangent to the curve at th
the value of $\frac{dy}{dx}$ is found.

We have seen that the short expression "the slo
no precise meaning, because a curve has so many slo
small portion of a curve has a different slope. "The
a point" is, however, a perfectly defined thing; it is
small portion of the curve situated just at that point
that this is the same as "the slope of the tangent t
point."

Observe that dx is a short step to the right, and
ing short step upwards. These steps must be cons

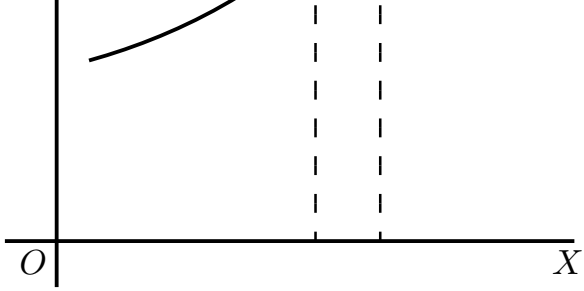


FIG. 8.

If a curve is sloping up at 45° at a particular poi
and dx will be equal, and the value of $\frac{dy}{dx} = 1$.

If the curve slopes up steeper than 45° (Fig. 9),
than 1.

If the curve slopes up very gently, as in Fig. 10, $\frac{d}{d}$
smaller than 1.

For a horizontal line, or a horizontal place in a
therefore $\frac{dy}{dx} = 0$.

If a curve slopes *downward*, as in Fig. 11, dy w
and must therefore be reckoned of negative value;
negative sign also.

If the "curve" happens to be a straight line, like
value of $\frac{dy}{dx}$ will be the same at all points along it.
slope is constant.

If a curve is one that turns more upwards as it
right, the values of $\frac{dy}{dx}$ will become greater and gr
creasing steepness, as in Fig. 13.

If a curve is one that gets flatter and flatter as
values of $\frac{dy}{dx}$ will become smaller and smaller as

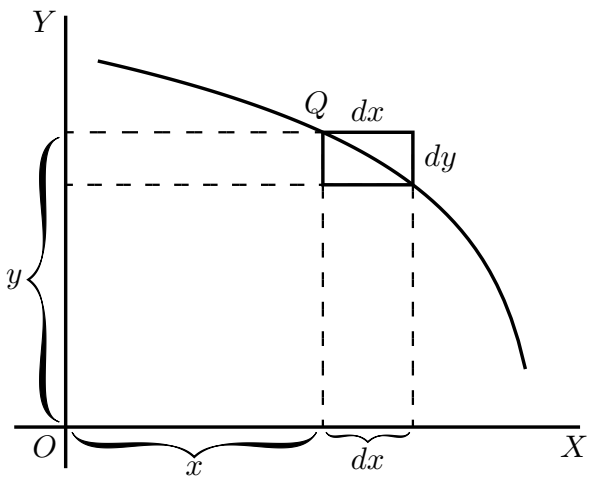


FIG. 11.

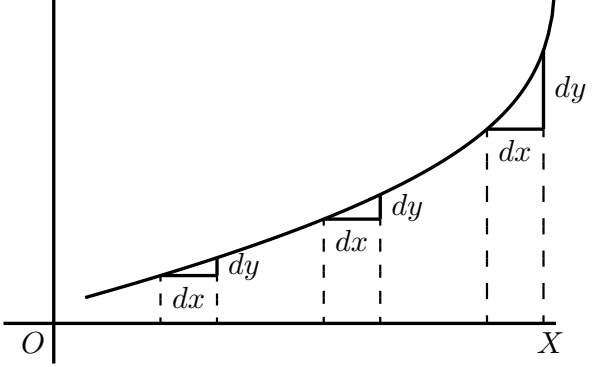


FIG. 13.

reached, as in Fig. 14.

If a curve first descends, and then goes up ag presenting a concavity upwards, then clearly $\frac{dy}{dx}$ wi with diminishing values as the curve flattens, then point where the bottom of the trough of the curve is this point onward $\frac{dy}{dx}$ will have positive values that g such a case y is said to pass by a *minimum*. The mi not necessarily the smallest value of y, it is that value to the bottom of the trough; for instance, in Fig. 28 of y corresponding to the bottom of the trough i

elsewhere values which are smaller than this. The

minimum is that y must increase *on either side* of i

N.B.—For the particular value of x that makes

value of $\frac{dy}{dx} = 0$.

If a curve first ascends and then descends, the v

positive at first; then zero, as the summit is reach

as the curve slopes downwards, as in Fig. 16. In th

pass by a *maximum*, but the maximum value of y is

greatest value of y. In Fig. 28, the maximum of y

by no means the greatest value y can have at some

curve.

N.B.—For the particular value of x that makes

value of $\frac{dy}{dx} = 0$.

If a curve has the peculiar form of Fig. 17, the

always be positive; but there will be one particula

slope is least steep, where the value of $\frac{dy}{dx}$ will be a

less than it is at any other part of the curve.

If a curve has the form of Fig. 18, the value of $\frac{dy}{dx}$ in the upper part, and positive in the lower part; w the curve where it becomes actually perpendicular, t be infinitely great.

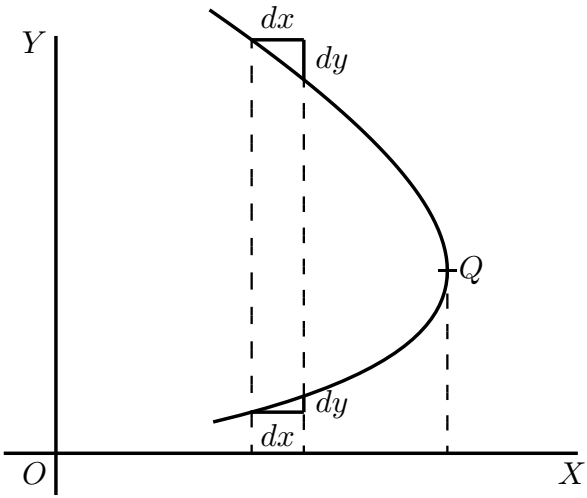
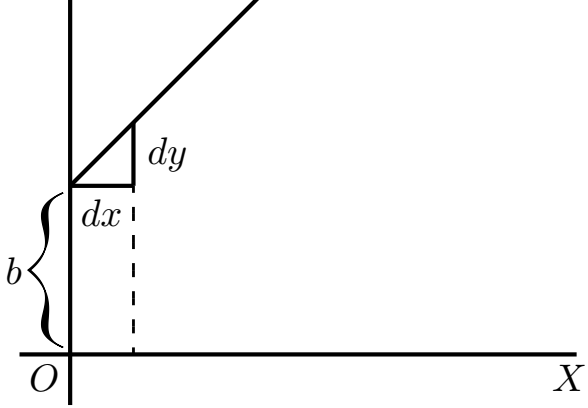


FIG. 18.

Now that we understand that $\frac{dy}{dx}$ measures the s at any point, let us turn to some of the equations whi learned how to differentiate.



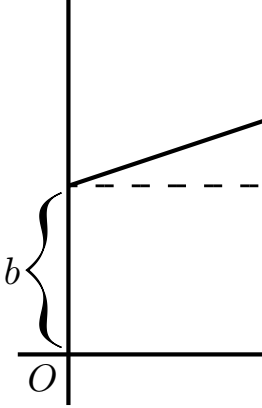


FIG. 19.

for whatever values we give to x to the right, we l
ascend. The line has a gradient of 1 in 1.

Now differentiate $y = x + b$, by the rules we ha
(pp. 21 and 25 *ante*), and we get $\frac{dy}{dx} = 1$.

The slope of the line is such that for every little s
we go an equal little step dy upward. And this slope i
the same slope.

(2) Take another case:

$$y = ax + b.$$



FIG. 21.

(3) Now for a slightly harder case.

Let $$y = ax^2 + b.$$

Again the curve will start on the y-axis at a l
origin.

Now differentiate. [If you have forgotten, turn
rather, *don't* turn back, but think out the differenti

$$\frac{dy}{dx} = 2ax.$$

This shows that the steepness will not be const
x increases. At the starting point P, where $x = 0$,
has no steepness—that is, it is level. On the left of
has negative values, $\frac{dy}{dx}$ will also have negative valu
from left to right, as in the Figure.

Let us illustrate this by working out a particula

the equation

$$y = \tfrac{1}{4}x^2 + 3,$$

and differentiating it, we get

$$\frac{dy}{dx} = \tfrac{1}{2}x.$$

Now assign a few successive values, say from 0 to 5,

the corresponding values of y by the first equation;

second equation. Tabulating results, we have:

x	0	1	2	3	4
y	3	$3\frac{1}{4}$	4	$5\frac{1}{4}$	7
$\frac{dy}{dx}$	0	$\frac{1}{2}$	1	$1\frac{1}{2}$	2

Then plot them out in two curves, Figs. 23 and 24,

the values of y against those of x and in Fig. 24 t

those of x. For any assigned value of x, the *height*

the second curve is proportional to the *slope* of the

If a curve comes to a sudden cusp, as in Fig. 25 point suddenly changes from a slope upward to a s

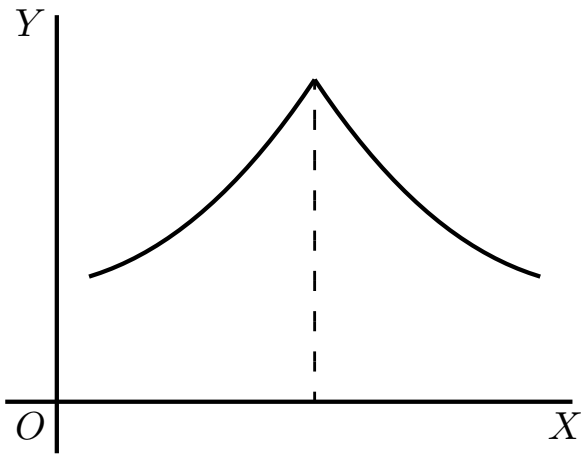


FIG. 25.

that case $\frac{dy}{dx}$ will clearly undergo an abrupt change a negative value.

The following examples show further applicatio just explained.

a straight line, has for equation $y = ax + b$, and it

hence $a = -\frac{1}{2}$. Also if $x = -1$, $y = \frac{1}{2(-1)} + 3$

tangent passes by this point, the coordinates of the

the equation of the tangent, namely

$$y = -\frac{1}{2}x + b,$$

so that $2\frac{1}{2} = -\frac{1}{2} \times (-1) + b$ and $b = 2$; the equatio

therefore $y = -\frac{1}{2}x + 2$.

Now, when two curves meet, the intersection b

mon to both curves, its coordinates must satisfy th

one of the two curves; that is, it must be a solutio

simultaneous equations formed by coupling togethe

the curves. Here the curves meet one another at p

solution of

$$\begin{cases} y = 2x^2 + 2, \\ y = -\frac{1}{2}x + 2 \quad \text{or} \quad 2x^2 + 2 = -\frac{1}{2}x \end{cases}$$

hence the curve at that point slopes downwards to
an angle θ with the horizontal that $\tan\theta = 1$; tha
horizontal.

The slope of the straight line is $-\frac{1}{2}$; that is, it sl
the right and makes with the horizontal an angle ϕ s
that is, an angle of $26^\circ\ 34'$. It follows that at the fi
cuts the straight line at an angle of $26^\circ\ 34'$, while at
it at an angle of $45^\circ - 26^\circ\ 34' = 18^\circ\ 26'$.

(5) A straight line is to be drawn, through a point
are $x = 2$, $y = -1$, as tangent to the curve $y = x^2$ -
coordinates of the point of contact.

The slope of the tangent must be the same as th
that is, $2x - 5$.

The equation of the straight line is $y = ax+b$, and
the values $x = 2$, $y = -1$, then $-1 = a\times 2+b$; also, i

The x and the y of the point of contact must al
equation of the tangent and the equation of the cur

Replacing a and b by their value in this, we get

$$x^2 - 5x + 6 = (2x - 5)x - 1 - 2(2x -$$

which simplifies to $x^2 - 4x + 3 = 0$, the solutions o
and $x = 1$. Replacing in (i), we get $y = 0$ and $y =$
two points of contact are then $x = 1$, $y = 2$, and $x =$

Note.—In all exercises dealing with curves, stud
tremely instructive to verify the deductions obtaine
ting the curves.

Exercises VIII. (See page 256 for Answers.)

(1) Plot the curve $y = \frac{3}{4}x^2 - 5$, using a scale of mi
at points corresponding to different values of x, the

Find, by differentiating the equation, the expres
see, from a Table of Natural Tangents, whether th
measured angle.

$x = 1$, $x = 2$.

(5) In the curve to which the equation is $x^2 + y^2 =$ of x at those points where the slope $= 1$.

(6) Find the slope, at any point, of the curve $\frac{x^2}{3^2} + \frac{y^2}{2^2} = 1$; and give the numerical value of the where $x = 0$, and at that where $x = 1$.

(7) The equation of a tangent to the curve $y = 5$ of the form $y = mx + n$, where m and n are consta of m and n if the point where the tangent touches t for abscissa.

(8) At what angle do the two curves

$$y = 3.5x^2 + 2 \quad \text{and} \quad y = x^2 - 5x +$$

cut one another?

(9) Tangents to the curve $y = \pm\sqrt{25 - x^2}$ are d which $x = 3$ and $x = 4$. Find the coordinates of the p of the tangents and their mutual inclination.

under what conditions the value of the thing differ
maximum, or a minimum. This is often exceedingly
neering questions, where it is most desirable to kno
will make the cost of working a minimum, or will m
maximum.

Now, to begin with a concrete case, let us take t

$$y = x^2 - 4x + 7.$$

By assigning a number of successive values to x, a
responding values of y, we can readily see that the e
a curve with a minimum.

x	0	1	2	3	4
y	7	4	3	4	7

These values are plotted in Fig. 26, which shows
ently a minimum value of 3, when x is made equal
sure that the minimum occurs at 2, and not at $2\frac{1}{4}$ o

FIG. 26.

Of course it would be possible with any algebraic
out a lot of values, and in this way arrive graduall
value that may be a maximum or a minimum.

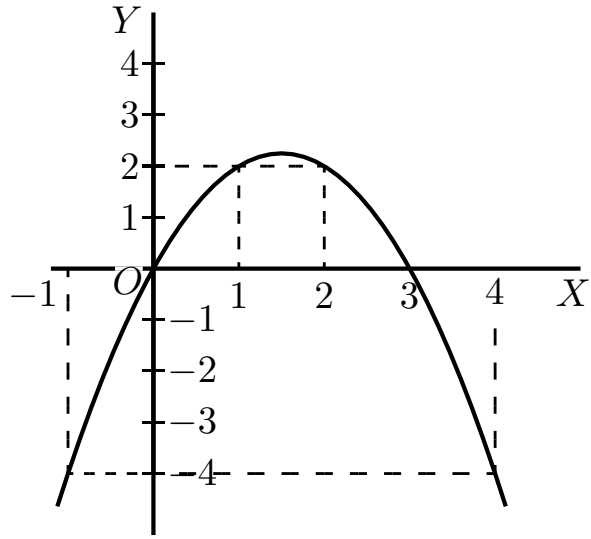


FIG. 27.

Here is another example:

Let $y = 3x - x^2$.

$x = 1\frac{1}{2}$, $y = 2.25$, if $x = 1.6$, $y = 2.24$. How can we

the real maximum, or that it occurs exactly when x

Now it may sound like juggling to be assured t

by which one can arrive straight at a maximum (c

without making a lot of preliminary trials or guess

depends on differentiating. Look back to an earlier

remarks about Figs. 14 and 15, and you will see tha

gets either to its maximum or to its minimum heigh

$\frac{dy}{dx} = 0$. Now this gives us the clue to the dodge tha

there is put before you an equation, and you want

of x that will make its y a minimum (or a maximum

it, and having done so, write its $\frac{dy}{dx}$ as *equal to ze*

for x. Put this particular value of x into the original

will then get the required value of y. This process i

"equating to zero."

To see how simply it works, take the example wi

ter opens, namely

$$y = x^2 - 4x + 7.$$

Now, we know that the maximum (or minimum)
when $x = 2$.

Putting the value $x = 2$ into the original equatio

$$\begin{aligned} y &= 2^2 - (4 \times 2) + 7 \\ &= 4 - 8 + 7 \\ &= 3. \end{aligned}$$

Now look back at Fig. 26, and you will see that th
when $x = 2$, and that this minimum of $y = 3$.

Try the second example (Fig. 24), which is

$$y = 3x - x^2.$$

Differentiating, $$\frac{dy}{dx} = 3 - 2x.$$

Equating to zero,

$$3 - 2x = 0,$$

whence $$x = 1\tfrac{1}{2};$$

is if you have any wits of your own) a kind of resent
know that $\frac{dy}{dx}$ has all sorts of different values at di
curve, according to whether it is sloping up or down
suddenly told to write

$$\frac{dy}{dx} = 0,$$

you resent it, and feel inclined to say that it can't be
have to understand the essential difference between
"an equation of condition." Ordinarily you are deal
that are true in themselves, but, on occasions, of wh
examples, you have to write down equations that
true, but are only true if certain conditions are to b
write them down in order, by solving them, to find th
make them true. Now we want to find the particula
when the curve is neither sloping up nor sloping do
particular place where $\frac{dy}{dx} = 0$. So, writing $\frac{dy}{dx} = 0$ o
it always is $= 0$; but you write it down *as a condit*
how much x will come out if $\frac{dy}{dx}$ is to be zero.

$$y = 4x + \frac{1}{x}.$$

Without stopping to think what curve it corresp
ate it, and equate to zero:

$$\frac{dy}{dx} = 4 - x^{-2} = 4 - \frac{1}{x^2} = 0;$$

whence $x = \frac{1}{2}$;

and, inserting this value,

$$y = 4$$

will be either a maximum or else a minimum. Bu
hereafter be told a way, depending upon a second
Chap. XII., p. 109). But at present it is enough
try any other value of x differing a little from the
whether with this altered value the corresponding
greater than that already found.

the biggest product will be got by dividing into two

Now see what the calculus tells you. Let the num
two parts be called n. Then if x is one part, the o
and the product will be $x(n-x)$ or $nx-x^2$. So we
Now differentiate and equate to zero;

$$\frac{dy}{dx} = n - 2x = 0$$

Solving for x, we get

$$\frac{n}{2} = x.$$

So now we *know* that whatever number n may be,
into two equal parts if the product of the parts is
and the value of that maximum product will always

This is a very useful rule, and applies to any nu
that if $m+n+p =$ a constant number, $m \times n \times p$ is
$m = n = p$.

Equating to zero, we get

$$2x - 1 = 0,$$

whence $$2x = 1,$$

or $$x = \tfrac{1}{2}.$$

That is to say, when x is made $= \frac{1}{2}$, the corres
will be either a maximum or a minimum. Accordin
in the original equation, we get

$$y = (\tfrac{1}{2})^2 - \tfrac{1}{2},$$

or $$y = -\tfrac{1}{4}.$$

Is this a maximum or a minimum? To test it, tr
bigger than $\frac{1}{2}$,—say make $x = 0.6$. Then

$$y = (0.6)^2 - 0.6 = 0.36 - 0.6 = -0.$$

which is higher up than -0.25; showing that $y = -$

Plot the curve for yourself, and verify the calcul

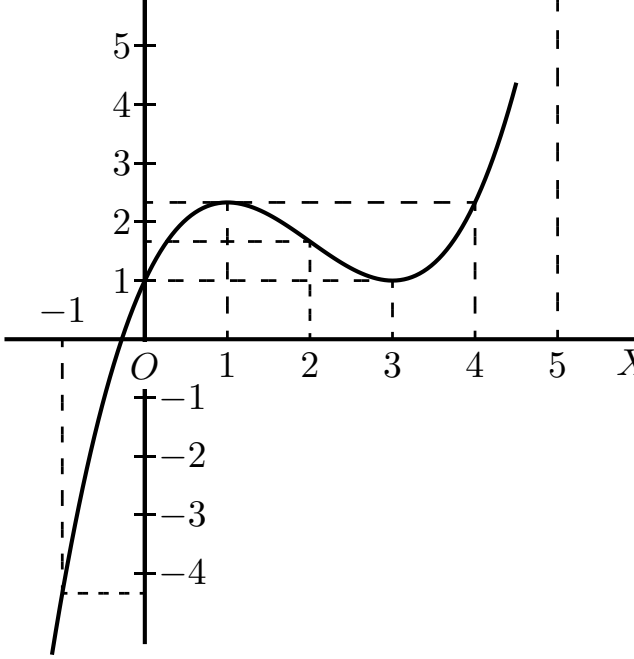


FIG. 28.

Equating to zero, we get the quadratic,

$$x^2 - 4x + 3 = 0;$$

and solving the quadratic gives us *two* roots, viz.

$$\begin{cases} x = 3 \\ x = 1. \end{cases}$$

The equation to a circle of radius r, having its ce
whose coordinates are $x = a$, $y = b$, as depicted in F

$$(y - b)^2 + (x - a)^2 = r^2.$$

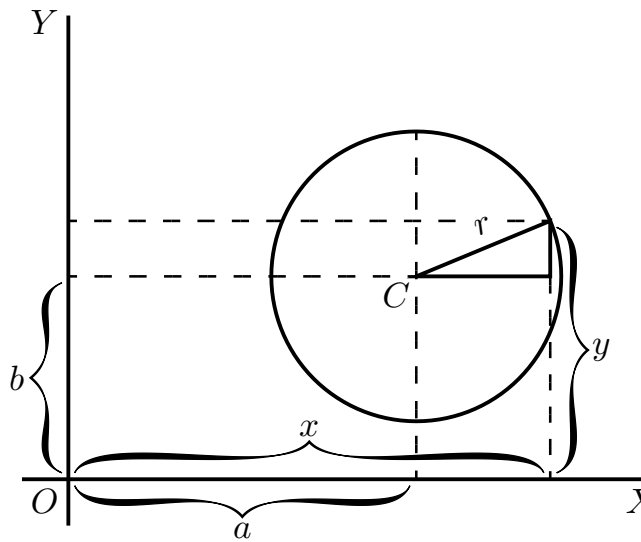


FIG. 29.

This may be transformed into

$$y = \sqrt{r^2 - (x - a)^2} + b.$$

$$\frac{dy}{dx} = \frac{a - x}{\sqrt{r^2 - (x - a)^2}}.$$

Then the condition for y being maximum or mi

$$\frac{a - x}{\sqrt{r^2 - (x - a)^2}} = 0.$$

Since no value whatever of x will make the de
the only condition to give zero is

$$x = a.$$

Inserting this value in the original equation for t

$$y = \sqrt{r^2} + b;$$

and as the root of r^2 is either $+r$ or $-r$, we have t
of y,

$$\begin{cases} y = b + r \\ y = b - r. \end{cases}$$

$$x^2 = \frac{-b}{3a}, \quad \text{and} \quad x = \sqrt{\frac{-b}{3a}}, \text{ which is im}$$

Therefore y has no maximum nor minimum.

A few more worked examples will enable you to this most interesting and useful application of the c

(1) What are the sides of the rectangle of maxi in a circle of radius R?

If one side be called x,

$$\text{the other side} = \sqrt{(\text{diagonal})^2 - x}$$

and as the diagonal of the rectangle is necessarily a side $= \sqrt{4R^2 - x^2}$.

Then, area of rectangle $S = x\sqrt{4R^2 - x^2}$,

$$\frac{dS}{dx} = x \times \frac{d\left(\sqrt{4R^2 - x^2}\right)}{dx} + \sqrt{4R^2 - x^2}$$

If you have forgotten how to differentiate $\sqrt{4R^2}$ write $4R^2 - x^2 = w$ and $y = \sqrt{w}$, and seek $\frac{dy}{dw}$ an and only if you can't get on refer to page 66.

constructed on the radius. In this case it is, of cours
which we are dealing.

(2) What is the radius of the opening of a conica
side of which has a length l when the capacity of the

If R be the radius and H the corresponding heig

$$\text{Volume } V = \pi R^2 \times \frac{H}{3} = \pi R^2 \times \frac{\sqrt{l^2 -}}{3}$$

Proceeding as in the previous problem, we get

$$\frac{dV}{dR} = \pi R^2 \times -\frac{R}{3\sqrt{l^2 - R^2}} + \frac{2\pi R}{3}\sqrt{l^2}$$
$$= \frac{2\pi R(l^2 - R^2) - \pi R^3}{3\sqrt{l^2 - R^2}} = 0$$

for maximum or minimum.

Or, $2\pi R(l^2 - R^2) - \pi R^2 = 0$, and $R = l\sqrt{\frac{2}{3}}$
obviously.

(3) Find the maxima and minima of the functio

$$y = \frac{x}{4 - x} + \frac{4 - x}{x}.$$

$$\text{for} \quad x = 1.5, \quad y = 2.27,$$

$$\text{for} \quad x = 2.5, \quad y = 2.27;$$

it is therefore a minimum. (It is instructive to plo
function.)

(4) Find the maxima and minima of the funct
$\sqrt{1-x}$. (It will be found instructive to plot the gra

Differentiating gives at once (see example No. 1,

$$\frac{dy}{dx} = \frac{1}{2\sqrt{1+x}} - \frac{1}{2\sqrt{1-x}} = 0$$

for maximum or minimum.

Hence $\sqrt{1+x} = \sqrt{1-x}$ and $x = 0$, the only sol

For $x = 0, \quad y = 2.$

For $x = \pm 0.5$, $y = 1.932$, so this is a maximum.

(5) Find the maxima and minima of the functio

$$y = \frac{x^2 - 5}{2x - 4}.$$

These being imaginary, there is no real value of x hence there is neither maximum nor minimum.

(6) Find the maxima and minima of the functio

$$(y - x^2)^2 = x^5.$$

This may be written $y = x^2 \pm x^{\frac{5}{2}}$.

$$\frac{dy}{dx} = 2x \pm \tfrac{5}{2}x^{\frac{3}{2}} = 0 \quad \text{for maximum or m}$$

that is, $x(2 \pm \frac{5}{2}x^{\frac{1}{2}}) = 0$, which is satisfied for $x = 0$, a that is for $x = \frac{16}{25}$. So there are two solutions.

Taking first $x = 0$. If $x = -0.5$, $y = 0.25 \pm$ $x = +0.5$, $y = 0.25 \pm \sqrt[2]{(.5)^5}$. On one side y is i there is no value of y that can be represented by a therefore entirely on the right side of the axis of y (

On plotting the graph it will be found that the origin, as if there were a minimum there; but ins beyond, as it should do for a minimum, it retraces

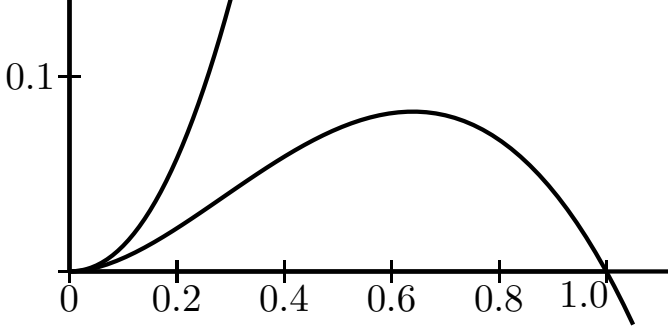


FIG. 30.

Now, if we take $x = \frac{16}{25} = 0.64$. If $x = 0.64$
$y = 0.0819$; if $x = 0.6$, y becomes 0.6389 and 0.08
y becomes 0.8996 and 0.0804.

This shows that there are two branches of the cu
does not pass through a maximum, but the lower o

(7) A cylinder whose height is twice the radius of
ing in volume, so that all its parts keep always in th
to each other; that is, at any instant, the cylinde
original cylinder. When the radius of the base is
area is increasing at the rate of 20 square inches p

The volume changes at the rate of $10r$ cubic inc

Make other examples for yourself. There are f
offer such a wealth for interesting examples.

Exercises IX. (See page 257 for Answers.)

(1) What values of x will make y a maximum a
$y = \dfrac{x^2}{x+1}$?

(2) What value of x will make y a maximum in
$\dfrac{x}{a^2+x^2}$?

(3) A line of length p is to be cut up into 4 part
as a rectangle. Show that the area of the rectangle
if each of its sides is equal to $\frac{1}{4}p$.

(4) A piece of string 30 inches long has its two e
and is stretched by 3 pegs so as to form a triangle.
triangular area that can be enclosed by the string?

square?

(8) Inscribe in a given cone, the height of which is
of the base, a cylinder (a) whose volume is a ma
lateral area is a maximum; (c) whose total area is a

(9) Inscribe in a sphere, a cylinder (a) whose
mum; (b) whose lateral area is a maximum; (c) wh
maximum.

(10) A spherical balloon is increasing in volume.
is r feet, its volume is increasing at the rate of 4 cub
at what rate is its surface then increasing?

(11) Inscribe in a given sphere a cone whose volu

(12) The current C given by a battery of N sim
$C = \dfrac{n \times E}{R + \dfrac{rn^2}{N}}$, where E, R, r, are constants and n
cells coupled in series. Find the proportion of n t
current is greatest.

asked. Why does anybody want to differentiate twi
that when the variable quantities are space and time
twice over we get the acceleration of a moving bod
geometrical interpretation, as applied to curves, $\frac{dy}{dx}$
the curve. But what can $\frac{d^2y}{dx^2}$ mean in this case? C
rate (per unit of length x) at which the slope is cha
is *a measure of the curvature of the slope.*

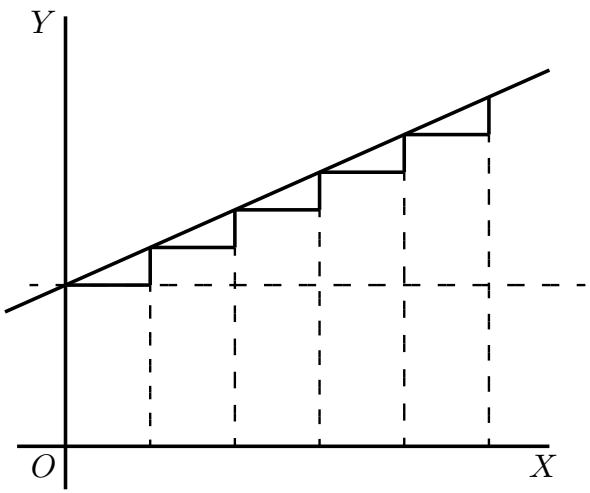
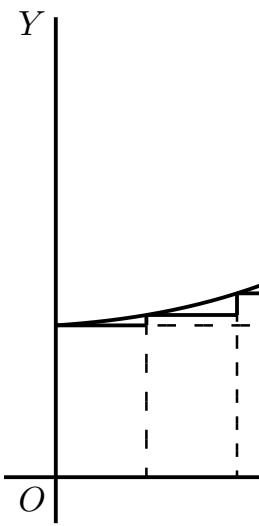


FIG. 31.



Suppose a slope constant, as in Fig. 31.

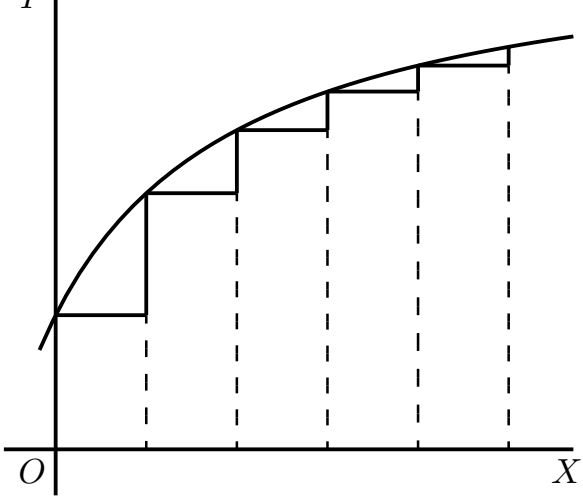


FIG. 33.

It is now time to initiate you into another s
whether the result that you get by "equating to zero
a minimum. The trick is this: After you have diffe
get the expression which you equate to zero), you t
second time, and look whether the result of the sec
is *positive* or *negative*. If $\frac{d^2y}{dx^2}$ comes out *positive*, t
the value of y which you got was a *minimum*; but
negative, then the value of y which you got must be a

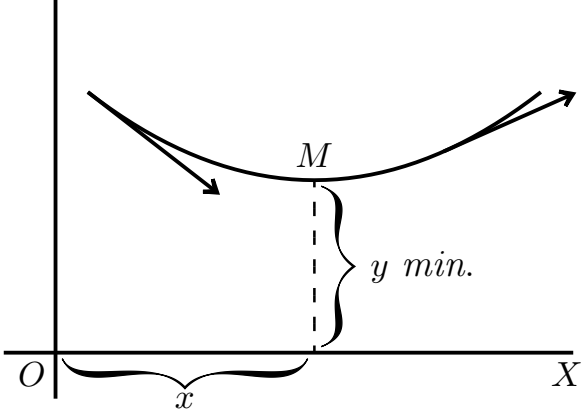


FIG. 34.

curve passes through M is such that $\frac{d^2y}{dx^2}$ is *positive* as x increases toward the right, is to convert a down upward one.

Similarly, consider any curve that has a maximu Fig. 16, p. 81), or like Fig. 35, where the curve is *co* imum point is marked M. In this case, as the curve from left to right, its upward slope is converted in negative slope, so that in this case the "slope of *negative*.

Go back now to the examples of the last chapte

$$(a)\quad \frac{dy}{dx} = 8x - 9 = 0; \quad x = 1\tfrac{1}{8}, \quad \text{and } y =$$

$$\frac{d^2y}{dx^2} = 8; \quad \text{it is } +; \text{ hence it is a minim}$$

$$(b)\quad \frac{dy}{dx} = 9 - 8x = 0; \quad x = 1\tfrac{1}{8}; \quad \text{and } y =$$

$$\frac{d^2y}{dx^2} = -8; \quad \text{it is } -; \text{ hence it is a maxi}$$

(2) Find the maxima and minima of the functio

$$\frac{dy}{dx} = 3x^2 - 3 = 0; \quad x^2 = 1; \quad \text{and } x =$$

$$\frac{d^2y}{dx^2} = 6x; \quad \text{for } x = 1; \text{ it is } +;$$

hence $x = 1$ corresponds to a minimum $y = 14$. F

hence $x = -1$ corresponds to a maximum $y = +18$.

(3) Find the maxima and minima of $y = \dfrac{x-1}{x^2+2}$

$$\frac{dy}{dx} = \frac{(x^2+2)\times 1 - (x-1)\times 2x}{(x^2+2)^2} = \frac{2x - x}{(x^2 +}$$

If we put $x = -0.73$, the numerator is positi

$y = -0.683$.

(4) The expense C of handling the products o

varies with the weekly output P according to the

$\dfrac{b}{c+P} + d$, where a, b, c, d are positive constants. Fo

the expense be least?

$$\frac{dC}{dP} = a - \frac{b}{(c+P)^2} = 0 \quad \text{for maximum or}$$

hence $a = \dfrac{b}{(c+P)^2}$ and $P = \pm\sqrt{\dfrac{b}{a}} - c$.

As the output cannot be negative, $P = +\sqrt{\dfrac{b}{a}} -$

Now $$\frac{d^2C}{dP^2} = +\frac{b(2c+2P)}{(c+P)^4},$$

which is positive for all the values of P; hence $P = +$

to a minimum.

C_e = cost of energy per 1000 watts per

Moreover, the relation connecting the average l
the commercial efficiency at which it is run is appro
where m and n are constants depending on the kin

Find the commercial efficiency for which the to
will be least.

We have $$C = N\left(\frac{C_l}{m}E^{-n} + \frac{PC_e}{1000}E\right),$$

$$\frac{dC}{dE} = \frac{PC_e}{1000} - \frac{nC_l}{m}E^{-(n+1)} = 0$$

for maximum or minimum.

$$E^{n+1} = \frac{1000 \times nC_l}{mPC_e} \quad \text{and} \quad E = \sqrt[n+1]{\frac{100}{n}}$$

This is clearly for minimum, since

$$\frac{d^2C}{dE^2} = (n+1)\frac{nC_l}{m}E^{-(n+2)},$$

(1) Find the maxima and minima of

$$y = x^3 + x^2 - 10x + 8.$$

(2) Given $y = \frac{b}{a}x - cx^2$, find expressions for $\frac{dy}{dx}$ find the value of x which makes y a maximum or a m whether it is maximum or minimum.

(3) Find how many maxima and how many mini curve, the equation to which is

$$y = 1 - \frac{x^2}{2} + \frac{x^4}{24};$$

and how many in that of which the equation is

$$y = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720}.$$

(4) Find the maxima and minima of

$$y = 2x + 1 + \frac{5}{x^2}.$$

$x^2 - 3 \quad 2$

(8) Divide a number N into two parts in such a w
the square of one part plus twice the square of the
a minimum.

(9) The efficiency u of an electric generator at
output x is expressed by the general equation:

$$u = \frac{x}{a + bx + cx^2};$$

where a is a constant depending chiefly on the energ
and c a constant depending chiefly on the resistance
Find an expression for that value of the output at w
will be a maximum.

(10) Suppose it to be known that consumption
steamer may be represented by the formula $y = 0.3+$
the number of tons of coal burned per hour and v is t
in nautical miles per hour. The cost of wages, inter
depreciation of that ship are together equal, per h

We have seen that when we differentiate a fraction
a rather complicated operation; and, if the fraction i
one, the result is bound to be a complicated expre
split the fraction into two or more simpler fractio
sum is equivalent to the original fraction, we coul
differentiating each of these simpler expressions.
differentiating would be the sum of two (or more) dif
of which is relatively simple; while the final expressio
it will be the same as that which could be obtaine
to this dodge, is thus obtained with much less effor
simplified form.

Let us see how to reach this result. Try first
two fractions together to form a resultant fraction.
the two fractions $\frac{1}{x+1}$ and $\frac{2}{x-1}$. Every school
together and find their sum to be $\frac{3x+1}{x^2-1}$. And in th
add together three or more fractions. Now this pr
be reversed: that is to say, that if this last expres

we can simplify it by division, since it is equivalent

$\frac{3}{x^2-1}$ is a proper algebraic fraction to which the op
into partial fractions can be applied, as explained h

Case I. If we perform many additions of two
the denominators of which contain only terms in x
x^2, x^3, or any other powers of x, we *always* find tha
of the final resulting fraction is the product of th
the fractions which were added to form the result.
factorizing the denominator of this final fraction, we
of the denominators of the partial fractions of which

Suppose we wish to go back from $\frac{3x+1}{x^2-1}$ to the
we know are $\frac{1}{x+1}$ and $\frac{2}{x-1}$. If we did not know
nents were we can still prepare the way by writing:

$$\frac{3x+1}{x^2-1} = \frac{3x+1}{(x+1)(x-1)} = \frac{\quad}{x+1} + \frac{\quad}{x}$$

leaving blank the places for the numerators until we

$(x+1)(x-1)$

And, as the denominators in these two expressions numerators must be equal, giving us:

$$3x + 1 = A(x-1) + B(x+1).$$

Now, this is an equation with two unknown quan seem that we need another equation before we can s A and B. But there is another way out of this diffic must be true for all values of x; therefore it must be of x as will cause $x-1$ and $x+1$ to become zero, th for $x = -1$ respectively. If we make $x = 1$, we get $4 =$ so that $B = 2$; and if we make $x = -1$, we get $-2 =$ so that $A = 1$. Replacing the A and B of the partia new values, we find them to become $\frac{1}{x+1}$ and $\frac{2}{x-}$ done.

As a farther example, let us take the fraction $\frac{4x}{x^3-}$ denominator becomes zero when x is given the valu a factor of it, and obviously then the other factor w

Now, if we make $x = 1$, we get:

$$-8 = (A \times 0) + B(2 \times 4) + (C \times 0); \quad \text{that i}$$

If $x = -1$, we get:

$$-12 = A(-2 \times 2) + (B \times 0) + (C \times 0); \quad \text{wh}$$

If $x = -3$, we get:

$$16 = (A \times 0) + (B \times 0) + C(-2 \times -4); \quad \text{wh}$$

So then the partial fractions are:

$$\frac{3}{x+1} - \frac{1}{x-1} + \frac{2}{x+3},$$

which is far easier to differentiate with respect to x th
expression from which it is derived.

$$(x^2+1)(x+1) \qquad x^2+1 \qquad x+$$

$$-x^2-3 = (Ax+B)(x+1) + C(x^2+$$

Putting $x=-1$, we get $-4 = C \times 2$; and $C = -$

hence $$-x^2-3 = (Ax+B)(x+1) - 2x^2 -$$

and $$x^2-1 = Ax(x+1) + B(x+1).$$

Putting $x=0$, we get $-1 = B$;

hence

$$x^2-1 = Ax(x+1) - x - 1; \quad \text{or } x^2+x =$$

and $$x+1 = A(x+1),$$

so that $A=1$, and the partial fractions are:

$$\frac{x-1}{x^2+1} - \frac{2}{x+1}.$$

Take as another example the fraction

$$\frac{x^3-2}{(x^2+1)(x^2+2)}.$$

same. In such a case, and for such algebraical expres
which we are dealing here, *the coefficients of the sa*
equal and of same sign.

Hence, since

$$x^3 - 2 = (Ax + B)(x^2 + 2) + (Cx + D)(x^2 +$$
$$= (A + C)x^3 + (B + D)x^2 + (2A + C)$$

we have $1 = A + C$; $0 = B + D$ (the coefficien
expression being zero); $0 = 2A + C$; and $-2 = 2$
four equations, from which we readily obtain $A = -$
$D = 0$; so that the partial fractions are $\frac{2(x+1)}{x^2+2} - \frac{x}{x^2}$
can always be used; but the method shown first
quickest in the case of factors in x only.

Case III. When, among the factors of the den
some which are raised to some power, one must a
ble existence of partial fractions having for denom
powers of that factor up to the highest. For instan

we get four unknowns, and we have only three re
them, yet

$$\frac{3x^2-2x+1}{(x+1)^2(x-2)} = \frac{x-1}{(x+1)^2} + \frac{1}{x+1} +$$

But if we write

$$\frac{3x^2-2x+1}{(x+1)^2(x-2)} = \frac{A}{(x+1)^2} + \frac{B}{x+1} +$$

we get

$$3x^2-2x+1 = A(x-2) + B(x+1)(x-2) -$$

which gives $C = 1$ for $x = 2$. Replacing C by its
gathering like terms and dividing by $x-2$, we get $-$
which gives $A = -2$ for $x = -1$. Replacing A by its

$$2x = -2 + B(x+1).$$

Hence $B = 2$; so that the partial fractions are:

$$\frac{2}{x+1} - \frac{2}{(x+1)^2} + \frac{1}{x-2},$$

We see that it is sufficient to allow for one num
numerator, and that we always get the ultimate pa

When there is a power of a factor of x^2 in the den
the corresponding numerators must be of the form A

$$\frac{3x-1}{(2x^2-1)^2(x+1)} = \frac{Ax+B}{(2x^2-1)^2} + \frac{Cx+D}{2x^2-1}$$

which gives

$$3x-1 = (Ax+B)(x+1) + (Cx+D)(x+1)(2x^2 -$$

For $x=-1$, this gives $E=-4$. Replacing, tra
like terms, and dividing by $x+1$, we get

$$16x^3 - 16x^2 + 3 = 2Cx^3 + 2Dx^2 + x(A-C)$$

Hence $2C=16$ and $C=8$; $2D=-16$ and $D=$
$A-8=0$ and $A=8$, and finally, $B-D=3$ or B
obtain as the partial fractions:

$$\frac{(8x-5)}{(2x^2-1)^2} + \frac{8(x-1)}{2x^2-1} - \frac{4}{x+1}.$$

The partial fractions are, therefore,

$$\frac{4}{(x+1)^2} - \frac{3}{(x+1)^3}.$$

Application to differentiation. Let it be requi
$y = \dfrac{5-4x}{6x^2+7x-3}$; we have

$$\frac{dy}{dx} = -\frac{(6x^2+7x-3)\times 4 + (5-4x)(1}{(6x^2+7x-3)^2}$$
$$= \frac{24x^2-60x-23}{(6x^2+7x-3)^2}.$$

If we split the given expression into

$$\frac{1}{3x-1} - \frac{2}{2x+3},$$

we get, however,

$$\frac{dy}{dx} = -\frac{3}{(3x-1)^2} + \frac{4}{(2x+3)^2},$$

(1) $\dfrac{}{(x-3)(x+4)}$.

(2) $\dfrac{}{(x-1)($

(3) $\dfrac{3x+5}{x^2+x-12}$.

(4) $\dfrac{x+}{x^2-7x}$

(5) $\dfrac{x-8}{(2x+3)(3x-2)}$.

(6) $\dfrac{x^2-}{(x-2)($

(7) $\dfrac{x^2-3x+1}{(x-1)(x+2)(x-3)}$.

(8) $\dfrac{5x^2+7x+1}{(2x+1)(3x-2)(3x+1)}$.

(9) $\dfrac{x^2}{x^3-1}$.

(10) $\dfrac{x^4+1}{x^3+1}$.

(11) $\dfrac{5x^2+6x+4}{(x+1)(x^2+x+1)}$.

(12) $\dfrac{x}{(x-1)($

(13) $\dfrac{x}{(x^2-1)(x+1)}$.

(14) $\dfrac{x+}{(x+2)^2}$

If $y = 3x$, $\dfrac{dy}{dx} = 3$; if $x = \dfrac{y}{3}$, $\dfrac{dx}{dy} = \dfrac{1}{3}$, and we

$$\frac{dy}{dx} = \frac{1}{\dfrac{dx}{dy}} \quad \text{or} \quad \frac{dy}{dx} \times \frac{dx}{dy} = 1.$$

Consider $y = 4x^2$, $\dfrac{dy}{dx} = 8x$; the inverse function

$$x = \frac{y^{\frac{1}{2}}}{2}, \quad \text{and} \quad \frac{dx}{dy} = \frac{1}{4\sqrt{y}} = \frac{1}{4 \times 2x}$$

Here again $$\frac{dy}{dx} \times \frac{dx}{dy} = 1.$$

It can be shown that for all functions which ca
inverse form, one can always write

$$\frac{dy}{dx} \times \frac{dx}{dy} = 1 \quad \text{or} \quad \frac{dy}{dx} = \frac{1}{\dfrac{dx}{dy}}.$$

$$\frac{dy}{dx} = -\frac{1}{2x^2\sqrt{\frac{3}{x} - 1}}.$$

If we had forgotten how to proceed by this me
check our result by some other way of obtaining th
ficient, or for any other reason we could not use the
we can proceed as follows: The inverse function is x

$$\frac{dx}{dy} = -\frac{3 \times 2y}{(1+y^2)^2} = -\frac{6y}{(1+y^2)^2};$$

hence

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = -\frac{(1+y^2)^2}{6y} = -\frac{\left(1 + \frac{3}{x} - 1\right)^2}{6 \times \sqrt[2]{\frac{3}{x} - 1}} =$$

Let us take as an other example $y = \frac{1}{\sqrt[3]{\theta + 5}}$.

The inverse function is $\theta = \frac{1}{y^3} - 5$ or $\theta = y^{-3} -$

$$\frac{d\theta}{dy} = -3y^{-4} = -3\sqrt[3]{(\theta+5)^4}.$$

many respects the calculus is an art rather than a s
to be acquired, as all other arts are, by practice.
work many examples, and set yourself other example
work them out, until the various artifices become fa

LET there be a quantity growing in such a way tha
its growth, during a given time, shall always be prop
magnitude. This resembles the process of reckoning
at some fixed rate; for the bigger the capital, the bi
interest on it in a given time.

Now we must distinguish clearly between two ca
tion, according as the calculation is made by what th
call "simple interest," or by what they call "compo
in the former case the capital remains fixed, whil
interest is added to the capital, which therefore incr
additions.

(1) *At simple interest.* Consider a concrete case.
start be £100, and let the rate of interest be 10 pe
Then the increment to the owner of the capital will
Let him go on drawing his interest every year, and l
it by in a stocking, or locking it up in his safe. The
10 years, by the end of that time he will have rece

$$y + n\frac{y}{n} = 2y.$$

(2) *At compound interest.* As before, let the o
capital of £100, earning interest at the rate of 10 pe
but, instead of hoarding the interest, let it be ad
each year, so that the capital grows year by year.
of one year, the capital will have grown to £110;
year (still at 10%) this will earn £11 interest. He
year with £121, and the interest on that will be £
starts the fourth year with £133. 2*s.*, and so on.
it out, and find that at the end of the ten years th
have grown to £259. 7*s.* 6*d.* In fact, we see that
year, each pound will have earned $\frac{1}{10}$ of a pound
this is always added on, each year multiplies the ca
continued for ten years (which will multiply by th
over) will multiply the original capital by 2.59374. I
symbols. Put y_0 for the original capital; $\frac{1}{n}$ for the fr

calling it 5% per half-year; with 20 operations, th
which the capital is multiplied by $\frac{21}{20}$. If reckoned th
of ten years the capital would have grown to £265.

$$(1 + \tfrac{1}{20})^{20} = 2.653.$$

But, even so, the process is still not quite fair;
the first month, there will be some interest earned
reckoning assumes that the capital remains stationar
a time. Suppose we divided the year into 10 parts, an
cent. interest for each tenth of the year. We now h
lasting over the ten years; or

$$y_n = £100 \left(1 + \tfrac{1}{100}\right)^{100};$$

which works out to £270. 9*s*. $7\frac{1}{2}d$.

Even this is not final. Let the ten years be divide
each of $\frac{1}{100}$ of a year; the interest being $\frac{1}{10}$ per cent. fo
then

$$y_n = £100 \left(1 + \tfrac{1}{1000}\right)^{1000};$$

and closer to a particular limiting value. However b
value of this expression grows nearer and nearer to

$$2.71828\ldots$$

a number *never to be forgotten.*

Let us take geometrical illustrations of these t
OP stands for the original value. OT is the whole
the value is growing. It is divided into 10 periods
there is an equal step up. Here $\frac{dy}{dx}$ is a constant; a
is $\frac{1}{10}$ of the original OP, then, by 10 such steps, the
If we had taken 20 steps, each of half the height sho
height would still be just doubled. Or n such steps
original height OP, would suffice to double the heig
of simple interest. Here is 1 growing till it becomes

In Fig. 37, we have the corresponding illustration
progression. Each of the successive ordinates is to
$\frac{n+1}{n}$ times as high as its predecessor. The steps

because each step up is now $\frac{}{n}$ of the ordinate *at the*
If we had literally 10 steps, with $\left(1+\frac{1}{10}\right)$ for the
the final total would be $(1+\frac{1}{10})^{10}$ or 2.594 times th
only we take n sufficiently large (and the correspon
small), then the final value $\left(1+\frac{1}{n}\right)^n$ to which u
be 2.71828.

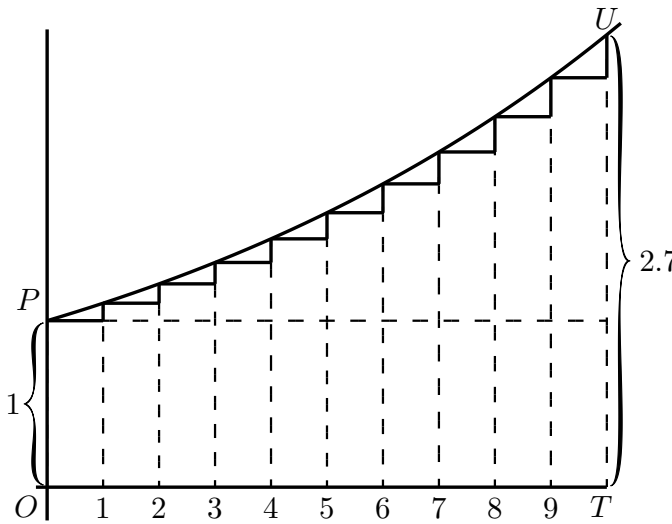


FIG. 37.

Epsilon. To this mysterious number 2.7182818 et

This process of growing proportionately, at eve
magnitude at that instant, some people call *a logari*
ing. Unit logarithmic rate of growth is that rate whi
cause 1 to grow to 2.718281. It might also be calle
of growing: because it is characteristic of organic
circumstances) that the increment of the organism
proportional to the magnitude of the organism itsel

If we take 100 per cent. as the unit of rate, an
as the unit of time, then the result of letting 1 gro
unit rate, for unit time, will be 2, while the resul
logarithmically at unit rate, for the same time, will

A little more about Epsilon. We have seen that
what value is reached by the expression $\left(1+\frac{1}{n}\right)^n$
indefinitely great. Arithmetically, here are tabula
(which anybody can calculate out by the help of a
logarithms) got by assuming $n = 2$; $n = 5$; $n = 10$

$$\left(1+\frac{1}{10{,}000}\right) \qquad = 2.7181.$$

It is, however, worth while to find another way
immensely important figure.

Accordingly, we will avail ourselves of the bino
expand the expression $\left(1+\frac{1}{n}\right)^n$ in that well-know

The binomial theorem gives the rule that

$$(a+b)^n = a^n + n\frac{a^{n-1}b}{1!} + n(n-1)\frac{a^{n-2}b^2}{2!}$$
$$+ n(n-1)(n-2)\frac{a^{n-3}b^3}{3!} +$$

Putting $a=1$ and $b=\frac{1}{n}$, we get

$$\left(1+\frac{1}{n}\right)^n = 1+1+\frac{1}{2!}\left(\frac{n-1}{n}\right) + \frac{1}{3!}\frac{(n-}{}$$
$$+ \frac{1}{4!}\frac{(n-1)(n-2)(n-}{n^3}$$

	1.000000
dividing by 1	1.000000
dividing by 2	0.500000
dividing by 3	0.166667
dividing by 4	0.041667
dividing by 5	0.008333
dividing by 6	0.001389
dividing by 7	0.000198
dividing by 8	0.000025
dividing by 9	0.000002
Total	2.718281

ϵ is incommensurable with 1, and resembles π minable non-recurrent decimal.

The Exponential Series. We shall have need of y

Let us, again making use of the binomial theore pression $\left(1+\frac{1}{n}\right)^{nx}$, which is the same as ϵ^x when

$$= 1 + x + \frac{n}{2!} + \frac{n}{3!} + \text{etc.}$$

But, when n is made indefinitely great, this sim
following:

$$\epsilon^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \text{etc.}.$$

This series is called *the exponential series.*

The great reason why ϵ is regarded of importance
a property, not possessed by any other function of
differentiate it its value remains unchanged; or, i
differential coefficient is the same as itself. This ca
by differentiating it with respect to x, thus:

$$\frac{d(\epsilon^x)}{dx} = 0 + 1 + \frac{2x}{1 \cdot 2} + \frac{3x^2}{1 \cdot 2 \cdot 3} + \frac{4x^3}{1 \cdot 2 \cdot 3 \cdot 4}$$

$$+ \frac{5x^4}{1 \cdot 2 \cdot 3 \cdot 4}$$

or $$= 1 + x + \frac{x^2}{1 \cdot 2} + \frac{x^3}{1 \cdot 2 \cdot 3} + \frac{x^4}{1 \cdot 2 \cdot 3 \cdot 4}$$

which is exactly the same as the original series.

$$\frac{dy}{dx} = B + 2Cx + 3Dx^2 + 4Ex^3 + \text{etc}$$

Now, if this new expression is really to be the
which it was derived, it is clear that A *must* $= B$; th
that $D = \frac{C}{3} = \frac{A}{1 \cdot 2 \cdot 3}$; that $E = \frac{D}{4} = \frac{A}{1 \cdot 2 \cdot 3 \cdot 4}$, e

The law of change is therefore that

$$y = A\left(1 + \frac{x}{1} + \frac{x^2}{1 \cdot 2} + \frac{x^3}{1 \cdot 2 \cdot 3} + \frac{x^4}{1 \cdot 2 \cdot 3 \cdot}\right.$$

If, now, we take $A = 1$ for the sake of further sir

$$y = 1 + \frac{x}{1} + \frac{x^2}{1 \cdot 2} + \frac{x^3}{1 \cdot 2 \cdot 3} + \frac{x^4}{1 \cdot 2 \cdot 3 \cdot 4}$$

Differentiating it any number of times will giv
series over again.

If, now, we take the particular case of $A = 1$

thus finally demonstrating that

$$\epsilon^x = 1 + \frac{x}{1} + \frac{x^2}{1 \cdot 2} + \frac{x^3}{1 \cdot 2 \cdot 3} + \frac{x^4}{1 \cdot 2 \cdot 3 \cdot 4}$$

[NOTE.—*How to read exponentials.* For the be
have no tutor at hand it may be of use to state
"*epsilon to the eksth power*;" or some people read it
So ϵ^{pt} is read "*epsilon to the pee-teeth-power*" or "*e*
Take some similar expressions:—Thus, ϵ^{-2} is read "*e*
two power" or "*exponential minus two.*" ϵ^{-ax} is re
minus ay-eksth" or "*exponential minus ay-eks.*"]

Of course it follows that ϵ^y remains unchanged if
respect to y. Also ϵ^{ax}, which is equal to $(\epsilon^a)^x$, will,
with respect to x, be $a\epsilon^{ax}$, because a is a constant.

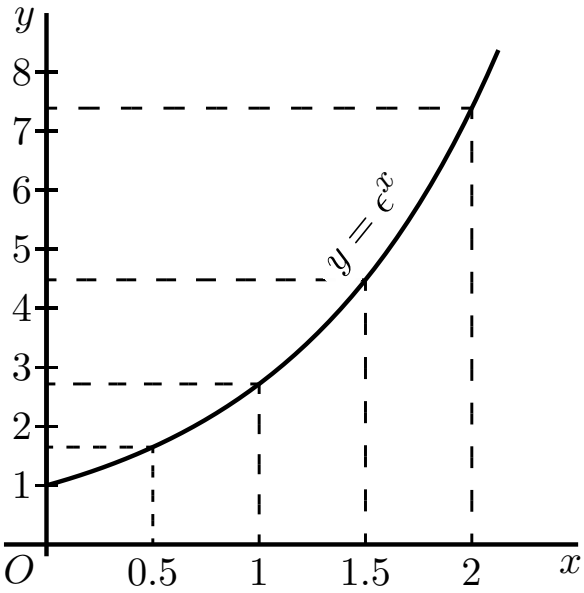
Natural or Naperian Logarithms.

Another reason why ϵ is important is because it w
the inventor of logarithms, the basis of his system.

y	1	1.65	2.71	4

For FIG. 39

y	1	2	3	4
x	0	0.69	1.10	1



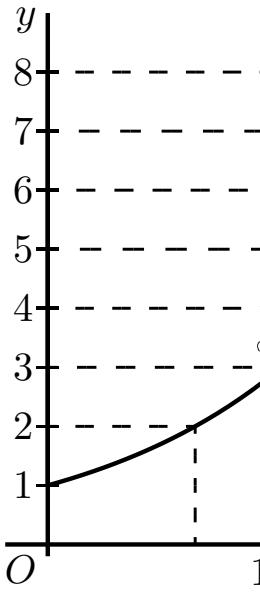


FIG. 38.

F

It will be seen that, though the calculations yie
for plotting, yet the result is identical. The two equ
the same thing.

But as 10 is no longer the basis, one cannot multi
by merely adding 2 or 3 to the index. One can
logarithm to the ordinary logarithm simply by multi
or

$$\log_{10} x = 0.4343 \times \log_{\epsilon} x,$$

and conversely, $$\log_{\epsilon} x = 2.3026 \times \log_{10} x.$$

Exponential and Logarithmic Equations.

Now let us try our hands at differentiating certa
contain logarithms or exponentials.

Take the equation:

$$y = \log_{\epsilon} x.$$

First transform this into

$$\epsilon^y = x,$$

whence, since the differential of ϵ^y with regard t
function unchanged (see p. 139),

$$\frac{dx}{dy} = \epsilon^y,$$

1.2	0.1823	8	2.
1.5	0.4055	9	2.
1.7	0.5306	10	2.
2.0	0.6931	20	2.
2.2	0.7885	50	3.
2.5	0.9163	100	4.
2.7	0.9933	200	5.
2.8	1.0296	500	6.
3.0	1.0986	1,000	6.
3.5	1.2528	2,000	7.
4.0	1.3863	5,000	8.
4.5	1.5041	10,000	9.
5.0	1.6094	20,000	9.

and reduce the power by 1. Thus, differentiating x
differentiating x^2 gave $2x^1$. But differentiating x^0
x^{-1} or $0 \times x^{-1}$, because x^0 is itself $= 1$, and is a c
have to come back to this curious fact that differen
us $\frac{1}{x}$ when we reach the chapter on integrating.

Now, try to differentiate

$$y = \log_\epsilon(x + a),$$

that is $$\epsilon^y = x + a;$$

we have $\frac{d(x+a)}{dy} = \epsilon^y$, since the differential of ϵ^y re

This gives $$\frac{dx}{dy} = \epsilon^y = x + a;$$

hence, reverting to the original function (see p. 128)

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{x+a}.$$

The next thing is not quite so simple. Try this:

$$y = a^x.$$

Taking the logarithm of both sides, we get

$$\log_\epsilon y = x \log_\epsilon a,$$

or
$$x = \frac{\log_\epsilon y}{\log_\epsilon a} = \frac{1}{\log_\epsilon a} \times \log_\epsilon y.$$

Since $\dfrac{1}{\log_\epsilon a}$ is a constant, we get

$$\frac{dx}{dy} = \frac{1}{\log_\epsilon a} \times \frac{1}{y} = \frac{1}{a^x \times \log_\epsilon a};$$

hence, reverting to the original function.

$$\frac{dy}{dx} = \frac{1}{\dfrac{dx}{dy}} = a^x \times \log_\epsilon a.$$

Let us now attempt further examples.

Examples.

(1) $y = \epsilon^{-ax}$. Let $-ax = z$; then $y = \epsilon^z$.

$$\frac{dy}{dx} = \epsilon^z; \quad \frac{dz}{dx} = -a; \quad \text{hence} \quad \frac{dy}{dx} = -$$

Or thus:

$$\log_\epsilon y = -ax; \quad \frac{1}{y}\frac{dy}{dx} = -a; \quad \frac{dy}{dx} = -ay =$$

(2) $y = \epsilon^{\frac{x^2}{3}}$. Let $\dfrac{x^2}{3} = z$; then $y = \epsilon^z$.

$$\frac{dy}{dz} = \epsilon^z; \quad \frac{dz}{dx} = \frac{2x}{3}; \quad \frac{dy}{dx} = \frac{2x}{3}\,\epsilon^{\frac{x}{3}}$$

Or thus:

$$\log_\epsilon y = \frac{x^2}{3}; \quad \frac{1}{y}\frac{dy}{dx} = \frac{2x}{3}; \quad \frac{dy}{dx} = \frac{2x}{3}$$

$$\frac{1}{y}\frac{dy}{dx} = \frac{}{(x^2+a)^{\frac{1}{2}}} \quad \text{and} \quad \frac{dy}{dx} = \frac{}{(x^2+}$$

For if $(x^2+a)^{\frac{1}{2}} = u$ and $x^2+a = v$, $u = v^{\frac{1}{2}}$,

$$\frac{du}{dv} = \frac{1}{2v^{\frac{1}{2}}}; \quad \frac{dv}{dx} = 2x; \quad \frac{du}{dx} = \frac{x}{(x^2+}$$

Check by writing $\sqrt{x^2+a} = z$.

(5) $y = \log(a+x^3)$. Let $(a+x^3) = z$; then $y =$

$$\frac{dy}{dz} = \frac{1}{z}; \quad \frac{dz}{dx} = 3x^2; \quad \text{hence} \quad \frac{dy}{dx} = \frac{}{a}$$

(6) $y = \log_\epsilon\{3x^2+\sqrt{a+x^2}\}$. Let $3x^2+\sqrt{a+}$ $\log_\epsilon z$.

$$\frac{dy}{dz} = \frac{1}{z}; \quad \frac{dz}{dx} = 6x + \frac{x}{\sqrt{x^2+a}};$$

$$\frac{dy}{dx} = \frac{6x+\dfrac{x}{\sqrt{x^2+a}}}{3x^2+\sqrt{a+x^2}} = \frac{x(1+6\sqrt{x^2+}}{(3x^2+\sqrt{x^2+a})\sqrt{}}$$

$$\frac{1}{y}\frac{dy}{dx} = 3\frac{2x}{(x^2+3)} + \frac{2}{3}\frac{3x^2}{x^3-2} = \frac{6x}{x^2+3} +$$

For if $y = \log_\epsilon(x^2+3)$, let $x^2+3 = z$ and $u = \log_\epsilon z$

$$\frac{du}{dz} = \frac{1}{z}; \quad \frac{dz}{dx} = 2x; \quad \frac{du}{dx} = \frac{2x}{x^2+3}$$

Similarly, if $v = \log_\epsilon(x^3-2)$, $\frac{dv}{dx} = \frac{3x^2}{x^3-2}$ and

$$\frac{dy}{dx} = (x^2+3)^3(x^3-2)^{\frac{2}{3}}\left\{\frac{6x}{x^2+3} + \frac{2}{x^3}\right.$$

(9) $y = \dfrac{\sqrt[2]{x^2+a}}{\sqrt[3]{x^3-a}}$.

$$\log_\epsilon y = \frac{1}{2}\log_\epsilon(x^2+a) - \frac{1}{3}\log_\epsilon(x^3-a).$$

$$\frac{1}{y}\frac{dy}{dx} = \frac{1}{2}\frac{2x}{x^2+a} - \frac{1}{3}\frac{3x^2}{x^3-a} = \frac{x}{x^2+a} -$$

and

$$\frac{dy}{dx} = \frac{\sqrt[2]{x^2+a}}{\sqrt[3]{x^3-a}}\left\{\frac{x}{x^2+a} - \frac{x^2}{x^3-a}\right\}.$$

$$\left(\frac{1}{a^x}\right)$$

$$\log_\epsilon y = ax(\log_\epsilon 1 - \log_\epsilon a^x) = -ax \log_\epsilon a$$

$$\frac{1}{y}\frac{dy}{dx} = -ax \times a^x \log_\epsilon a - a \log_\epsilon a^x.$$

and
$$\frac{dy}{dx} = -\left(\frac{1}{a^x}\right)^{ax} (x \times a^{x+1} \log_\epsilon a + a \log$$

Try now the following exercises.

Exercises XII. (See page 260 for Answers.)

(1) Differentiate $y = b(\epsilon^{ax} - \epsilon^{-ax})$.

(2) Find the differential coefficient with respect to $u = at^2 + 2\log_\epsilon t$.

(3) If $y = n^t$, find $\dfrac{d(\log_\epsilon y)}{dt}$.

(4) Show that if $y = \dfrac{1}{b} \cdot \dfrac{a^{bx}}{\log_\epsilon a}$, $\dfrac{dy}{dx} = a^{bx}$.

(5) If $w = pv^n$, find $\dfrac{dw}{dv}$.

Differentiate

through a submarine cable depends on the value
external diameter of the core to the diameter of th
wire. If this ratio is called y, then the number of sig
sent per minute can be expressed by the formula

$$s = ay^2 \log_\epsilon \frac{1}{y};$$

where a is a constant depending on the length and
materials. Show that if these are given, s will be a
$1 \div \sqrt{\epsilon}$.

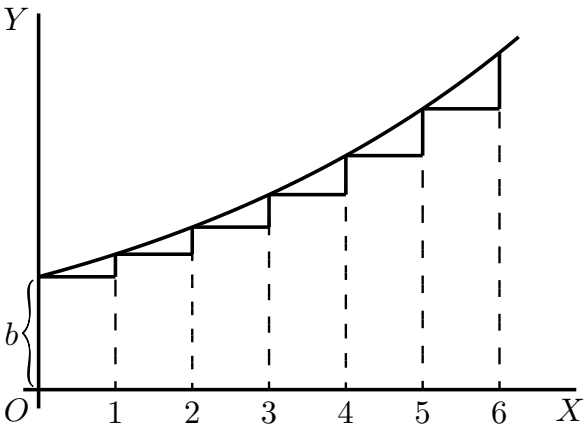
(14) Find the maximum or minimum of

$$y = x^3 - \log_\epsilon x.$$

(15) Differentiate $y = \log_\epsilon(ax\epsilon^x)$.

(16) Differentiate $y = (\log_\epsilon ax)^3$.

Also, we see that p is the numerical value of the height of any ordinate and that of the next preceding have taken p as $\frac{6}{5}$; each ordinate being $\frac{6}{5}$ as high as



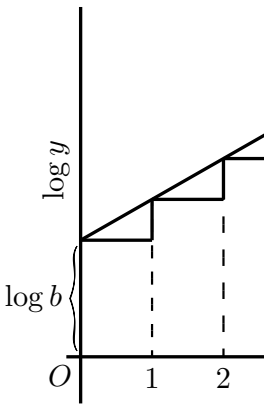


FIG. 40.

If two successive ordinates are related together ratio, their logarithms will have a constant differe should plot out a new curve, Fig. 41, with values of it would be a straight line sloping up by equal steps

$$y = b\epsilon^{ax}.$$

The Die-away Curve.

If we were to take p as a proper fraction (less tha
would obviously tend to sink downwards, as in F
successive ordinate is $\frac{3}{4}$ of the height of the precedi

The equation is still

$$y = bp^x;$$

but since p is less than one, $\log_\epsilon p$ will be a negative
be written $-a$; so that $p = \epsilon^{-a}$, and now our equa
takes the form

$$y = b\epsilon^{-ax}.$$

The importance of this expression is that, in t
independent variable is *time*, the equation represe

away. Thus, the cooling of a hot body is represe
celebrated "law of cooling") by the equation

$$\theta_t = \theta_0 \epsilon^{-at};$$

where θ_0 is the original excess of temperature of a
of its surroundings, θ_t the excess of temperature at
and a is a constant—namely, the constant of decren
the amount of surface exposed by the body, and o
conductivity and emissivity, etc.

A similar formula,

$$Q_t = Q_0 \epsilon^{-at},$$

is used to express the charge of an electrified body,
charge Q_0, which is leaking away with a constant of
constant depends in this case on the capacity of th
resistance of the leakage-path.

Oscillations given to a flexible spring die out aft
dying-out of the amplitude of the motion may be ex

$$y = b\epsilon$$

or $$\log_\epsilon y = \log_\epsilon b - ax \log_\epsilon \epsilon = \log_\epsilon b -$$

and, differentiating, $$\frac{1}{y}\frac{dy}{dx} = -a;$$

hence $$\frac{dy}{dx} = b\epsilon^{-ax} \times (-a) = -ay;$$

or, in words, the slope of the curve is downward, an
and to the constant a.

We should have got the same result if we had tal
the form

$$y = bp^x;$$

for then $$\frac{dy}{dx} = bp^x \times \log_\epsilon p.$$

But $$\log_\epsilon p = -a;$$

giving us $$\frac{dy}{dx} = y \times (-a) = -ay,$$

as before.

ical tables, some of the values are tabulated on p. 1

As an example of the use of this table, suppose t
cooling, and that at the beginning of the experimen
it is 72° hotter than the surrounding objects, and if
of its cooling is 20 minutes (that is, if it takes 20 mi
of temperature to fall to $\frac{1}{\epsilon}$ part of 72°), then we can
it will have fallen in any given time t. For instance, l
Then $\frac{t}{T} = 60 \div 20 = 3$, and we shall have to find
and then multiply the original 72° by this. The ta
is 0.0498. So that at the end of 60 minutes the exc
will have fallen to $72° \times 0.0498 = 3.586°$.

1.10	3.0042	0.3329
1.20	3.3201	0.3012
1.25	3.4903	0.2865
1.50	4.4817	0.2231
1.75	5.755	0.1738
2.00	7.389	0.1353
2.50	12.182	0.0821
3.00	20.086	0.0498
3.50	33.115	0.0302
4.00	54.598	0.0183
4.50	90.017	0.0111
5.00	148.41	0.0067
5.50	244.69	0.0041
6.00	403.43	0.00248
7.50	1808.04	0.00055
10.00	22026.5	0.000045

$$R$$

Its value at any time may be written:

$$C = 10 - 10\epsilon^{-\frac{t}{0.01}},$$

the time-constant being 0.01. This means that it tak
variable term to fall by $\frac{1}{\epsilon} = 0.3678$ of its initial valu

To find the value of the current when $t = 0.001$
$\epsilon^{-0.1} = 0.9048$ (from table).

It follows that, after 0.001 sec., the variable ter
9.048, and the actual current is $10 - 9.048 = 0.952$.

Similarly, at the end of 0.1 sec.,

$$\frac{t}{T} = 10; \quad \epsilon^{-10} = 0.000045;$$

the variable term is $10 \times 0.000045 = 0.00045$, the cu

(2) The intensity I of a beam of light which ha
thickness l cm. of some transparent medium is $I =$
the initial intensity of the beam and K is a "consta

$$\log 0.5 = -0.02 \times l \times \log \epsilon,$$

which gives

$$l = \frac{-0.3010}{-0.02 \times 0.4343} = 34.7 \text{ centimetres}$$

(3) The quantity Q of a radio-active substance wh
dergone transformation is known to be related to the
of the substance by the relation $Q = Q_0 \epsilon^{-\lambda t}$, where
t the time in seconds elapsed since the transformati

For "Radium A," if time is expressed in seconds
that $\lambda = 3.85 \times 10^{-3}$. Find the time required for tra
substance. (This time is called the "mean life" of th

We have $0.5 = \epsilon^{-0.00385t}$.

$$\log 0.5 = -0.00385t \times \log \epsilon;$$

and $t = 3$ minutes very nearly.

---

(4) The following equations give very similar cur

$$\text{(i) } y = \frac{ax}{x+b};$$

$$\text{(ii) } y = a(1 - \epsilon^{-\frac{x}{b}});$$

$$\text{(iii) } y = \frac{a}{90^\circ} \operatorname{arc\,tan}\left(\frac{x}{b}\right).$$

Draw all three curves, taking $a = 100$ millimetr

tres.

(5) Find the differential coefficient of y with resp

$$(a)\ y = x^x; \quad (b)\ y = (\epsilon^x)^x; \quad (c)\ y =$$

(6) For "Thorium A," the value of λ is 5; find th

is, the time taken by the transformation of a quantity

equal to half the initial quantity Q_0 in the expressio

$$Q = Q_0 \epsilon^{-\lambda t};$$

t being in seconds.

(9) The damping on a telephone line can be ascer
lation $i = i_0\epsilon^{-\beta l}$, where i is the strength, after t seco
current of initial strength i_0; l is the length of the line
β is a constant. For the Franco-English submarine
$\beta = 0.0114$. Find the damping at the end of the cab
and the length along which i is still 8% of the origin
value of very good audition).

(10) The pressure p of the atmosphere at an altit
given by $p = p_0\epsilon^{-kh}$; p_0 being the pressure at sea-leve

The pressures at 10, 20 and 50 kilometres bein
respectively, find k in each case. Using the mean v
percentage error in each case.

(11) Find the minimum or maximum of $y = x^x$.

(12) Find the minimum or maximum of $y = x^{\frac{1}{x}}$.

(13) Find the minimum or maximum of $y = xa^{\frac{1}{x}}$

letter for any variable angle the letter θ ("theta").

Let us consider the function

$$y = \sin\theta.$$

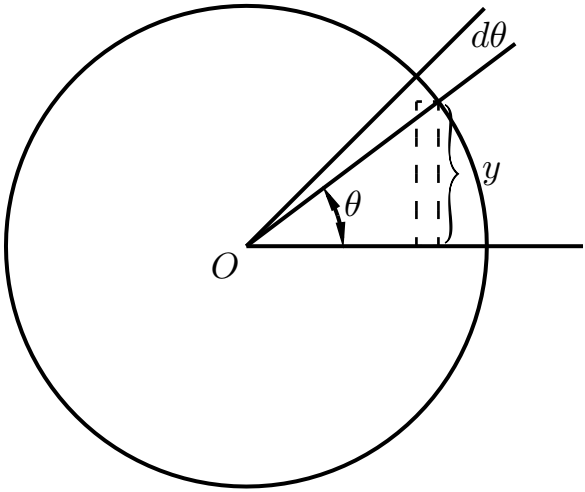


FIG. 43.

What we have to investigate is the value of $\frac{d(}{}$ words, if the angle θ varies, we have to find the re increment of the sine and the increment of the angl being indefinitely small in themselves. Examine Fig.

The quantity on the right-hand side is the diffe
sines, and books on trigonometry tell us how to work
tell us that if M and N are two different angles,

$$\sin M - \sin N = 2\cos\frac{M+N}{2} \cdot \sin\frac{M}{}$$

If, then, we put $M = \theta + d\theta$ for one angle, and N
we may write

$$dy = 2\cos\frac{\theta + d\theta + \theta}{2} \cdot \sin\frac{\theta + d\theta -}{2}$$

or, $$dy = 2\cos(\theta + \tfrac{1}{2}d\theta) \cdot \sin\tfrac{1}{2}d\theta.$$

But if we regard $d\theta$ as indefinitely small, then i
neglect $\frac{1}{2}d\theta$ by comparison with θ, and may also ta
the same as $\frac{1}{2}d\theta$. The equation then becomes:

$$dy = 2\cos\theta \times \tfrac{1}{2}d\theta;$$

$$dy = \cos\theta \cdot d\theta,$$

and, finally, $$\frac{dy}{d\theta} = \cos\theta.$$



FIG. 44.

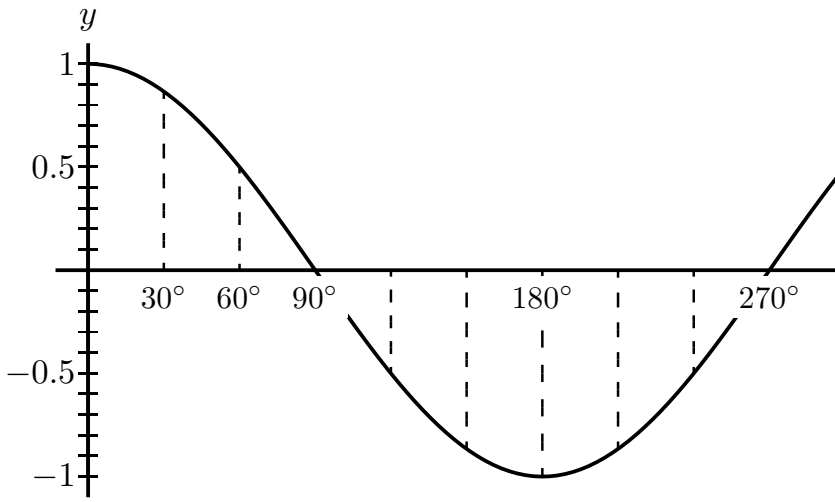


FIG. 45.

Take next the cosine.

Let $y = \cos\theta$.

$$\frac{dy}{d\theta} = -\sin\theta.$$

Lastly, take the tangent.

Let
$$y = \tan\theta,$$
$$dy = \tan(\theta + d\theta) - \tan\theta.$$

Expanding, as shown in books on trigonometry,

$$\tan(\theta + d\theta) = \frac{\tan\theta + \tan d\theta}{1 - \tan\theta \cdot \tan d\theta};$$

whence
$$dy = \frac{\tan\theta + \tan d\theta}{1 - \tan\theta \cdot \tan d\theta} - \text{ta}$$
$$= \frac{(1 + \tan^2\theta)\tan d\theta}{1 - \tan\theta \cdot \tan d\theta}.$$

Now remember that if $d\theta$ is indefinitely dim
of $\tan d\theta$ becomes identical with $d\theta$, and $\tan\theta \cdot d\theta$

$\sin\theta$	$\cos\theta$
$\cos\theta$	$-\sin\theta$
$\tan\theta$	$\sec^2\theta$

Sometimes, in mechanical and physical question in simple harmonic motion and in wave-motions, we angles that increase in proportion to the time. Thu of one complete *period*, or movement round the circ angle all round the circle is 2π radians, or $360°$, th moved through in time t, will be

$$\theta = 2\pi\frac{t}{T}, \quad \text{in radians,}$$

or

$$\theta = 360\frac{t}{T}, \quad \text{in degrees.}$$

If the *frequency*, or number of periods per second then $n = \dfrac{1}{T}$, and we may then write:

$$\theta = 2\pi nt.$$

$$\frac{dy}{dt} = \cos\theta \times 2\pi n$$
$$= 2\pi n \cdot \cos 2\pi nt.$$

Similarly, it follows that

$$\frac{d(\cos 2\pi nt)}{dt} = -2\pi n \cdot \sin 2\pi nt.$$

Second Differential Coefficient of Sine

We have seen that when $\sin\theta$ is differentiated w
becomes $\cos\theta$; and that when $\cos\theta$ is differentiated
it becomes $-\sin\theta$; or, in symbols,

$$\frac{d^2(\sin\theta)}{d\theta^2} = -\sin\theta.$$

So we have this curious result that we have fou
that if we differentiate it twice over, we get the sam
we started, but with the sign changed from + to −.

With what we have so far learned we can now d
sions of a more complex nature.

(1) $y = \arcsin x$.

If y is the arc whose sine is x, then $x = \sin y$.

$$\frac{dx}{dy} = \cos y.$$

Passing now from the inverse function to the origi

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{\cos y}.$$

Now $$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2};$$

hence $$\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}},$$

a rather unexpected result.

(2) $y = \cos^3 \theta$.

This is the same thing as $y = (\cos \theta)^3$.

(4) $y = \log_\epsilon \sin\theta$.

Let $\sin\theta = v$; $\quad y = \log_\epsilon v$.

$$\frac{dy}{dv} = \frac{1}{v}; \quad \frac{dv}{d\theta} = \cos\theta;$$
$$\frac{dy}{d\theta} = \frac{1}{\sin\theta} \times \cos\theta = \cot\theta.$$

(5) $y = \cot\theta = \dfrac{\cos\theta}{\sin\theta}$.

$$\frac{dy}{d\theta} = \frac{-\sin^2\theta - \cos^2\theta}{\sin^2\theta}$$
$$= -(1 + \cot^2\theta) = -\operatorname{cosec}^2\theta.$$

(6) $y = \tan 3\theta$.

Let $3\theta = v$; $\quad y = \tan v$; $\quad \dfrac{dy}{dv} = \sec^2 v$.

$$\frac{dv}{d\theta} = 3; \quad \frac{dy}{d\theta} = 3\sec^2 3\theta.$$

(7) $y = \sqrt{1 + 3\tan^2\theta}$; $\quad y = (1 + 3\tan^2\theta)^{\frac{1}{2}}$.

$d\theta$

hence $$\frac{dy}{d\theta} = \frac{6\tan\theta\sec^2\theta}{2\sqrt{1+3\tan^2\theta}}.$$

(8) $y = \sin x \cos x.$

$$\frac{dy}{dx} = \sin x(-\sin x) + \cos x \times \cos x$$
$$= \cos^2 x - \sin^2 x.$$

Exercises XIV. (See page 261 for Answers.)

(1) Differentiate the following:

(i) $y = A\sin\left(\theta - \frac{\pi}{2}\right).$

(ii) $y = \sin^2\theta;$ and $y = \sin 2\theta$

(iii) $y = \sin^3\theta;$ and $y = \sin 3\theta$

(2) Find the value of θ for which $\sin\theta \times \cos\theta$ is

(3) Differentiate $y = \frac{1}{2\pi}\cos 2\pi nt.$

respect to θ.

(10) Differentiate $y = \epsilon^x \sin^2 x$.

(11) Differentiate the three equations of Exerci
No. 4, and compare their differential coefficients, a
are equal, or nearly equal, for very small values of
values of x, or for values of x in the neighbourhood

(12) Differentiate the following:

(i) $y = \sec x$. (ii) $y =$

(iii) $y = \text{arc}\tan x$. (iv) $y =$

(v) $y = \tan x \times \sqrt{3 \sec x}$.

(13) Differentiate $y = \sin(2\theta + 3)^{2.3}$.

(14) Differentiate $y = \theta^3 + 3\sin(\theta + 3) - 3^{\sin\theta} -$

(15) Find the maximum or minimum of $y = \theta \cos$

one independent variable. Thus, we may find a cas
on two other variable quantities, one of which we
other v. In symbols

$$y = f(u, v).$$

Take the simplest concrete case.

Let $$y = u \times v.$$

What are we to do? If we were to treat v as a constan
with respect to u, we should get

$$dy_v = v\,du;$$

or if we treat u as a constant, and differentiate wit
should have:

$$dy_u = u\,dv.$$

The little letters here put as subscripts are to sh
has been taken as constant in the operation.

$$\left.\begin{aligned} dy_v &= \frac{\partial y}{\partial u}\,du, \\ dy_u &= \frac{\partial y}{\partial v}\,dv, \end{aligned}\right\} \quad \text{which are } \textit{partial diffe}$$

But, if you think of it, you will observe that the
depends on *both* these things at the same time. Tha
are varying, the real dy ought to be written

$$dy = \frac{\partial y}{\partial u}\,du + \frac{\partial y}{\partial v}\,dv;$$

and this is called a *total differential*. In some books
$\left(\frac{dy}{du}\right)\,du + \left(\frac{dy}{dv}\right)\,dv$.

Example (1). Find the partial differential coeffic
sion $w = 2ax^2 + 3bxy + 4cy^3$. The answers are:

$$\left.\begin{aligned} \frac{\partial w}{\partial x} &= 4ax + 3by. \\ \frac{\partial w}{\partial y} &= 3bx + 12cy^2. \end{aligned}\right\}$$

$$\left.\begin{aligned} \frac{\partial z}{\partial x} &= \\ \frac{\partial z}{\partial y} &= x^y \times \log_\epsilon x, \end{aligned}\right\}$$

so that $dz = yx^{y-1}\,dx + x^y \log_\epsilon x\,dy$.

Example (3). A cone having height h and ra volume $V = \frac{1}{3}\pi r^2 h$. If its height remains constant the ratio of change of volume, with respect to radiu ratio of change of volume with respect to height wh the height were varied and the radius kept constant

$$\left.\begin{aligned} \frac{\partial V}{\partial r} &= \frac{2\pi}{3} rh, \\ \frac{\partial V}{\partial h} &= \frac{\pi}{3} r^2. \end{aligned}\right\}$$

The variation when both the radius and the heig by $dV = \frac{2\pi}{3} rh\,dV + \frac{\pi}{3} r^2\,dh$.

Example (4). In the following example F and bitrary functions of any form whatsoever. For exa sine-functions, or exponentials, or mere algebraic fu

(where the figure 1 is simply the coefficient of x in

and $$\frac{\partial^2 y}{\partial x^2} = F''(w) + f''(v).$$

Also $$\frac{\partial y}{\partial t} = \frac{\partial F(w)}{\partial w} \cdot \frac{\partial w}{\partial t} + \frac{\partial f(v)}{\partial v} \cdot \frac{\partial v}{\partial t}$$
$$= F'(w) \cdot a - f'(v)a;$$

and $$\frac{\partial^2 y}{\partial t^2} = F''(w)a^2 + f''(v)a^2;$$

whence $$\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}.$$

This differential equation is of immense importan
physics.

Maxima and Minima of Functions of two Variables.

Example (5). Let us take up again Exercise IX.,

Let x and y be the length of two of the port
The third is $30 - (x + y)$, and the area of the

For a maximum (clearly it will not be a minimum
must have simultaneously

$$\frac{\partial P}{\partial x} = 0 \quad \text{and} \quad \frac{\partial P}{\partial y} = 0;$$

that is,

$$2xy - 30x + y^2 - 45y + 450 = 0,$$
$$2xy - 30y + x^2 - 45x + 450 = 0.$$

An immediate solution is $x = y$.

If we now introduce this condition in the value o

$$P = (15 - x)^2(2x - 15) = 2x^3 - 75x^2 + 900$$

For maximum or minimum, $\frac{dP}{dx} = 6x^2 - 150x + 90$
$x = 15$ or $x = 10$.

Clearly $x = 15$ gives minimum area; $x = 10$ gives
$\frac{d^2P}{dx^2} = 12x - 150$, which is $+30$ for $x = 15$ and -30

For minimum (clearly it won't be a maximum here)

$$y - \frac{2V}{x^2} = 0, \quad x - \frac{2V}{y^2} = 0.$$

Here also, an immediate solution is $x = y$, so t

$\frac{dS}{dx} = 2x - \frac{4V}{x^2} = 0$ for minimum, and

$$x = \sqrt[3]{2V}.$$

Exercises XV. (See page 263 for Answers.)

(1) Differentiate the expression $\frac{x^3}{3} - 2x^3y - 2y^2$ to x alone, and with respect to y alone.

(2) Find the partial differential coefficients wit and z, of the expression

$$x^2yz + xy^2z + xyz^2 + x^2y^2z^2.$$

a constant k, is maximum when these three quantit

(7) Find the maximum or minimum of the funct

$$u = x + 2xy + y.$$

(8) The post-office regulations state that no par
a size that its length plus its girth exceeds 6 feet.
est volume that can be sent by post (a) in the ca
rectangular cross section; (b) in the case of a packa
section.

(9) Divide π into 3 parts such that the continue
sines may be a maximum or minimum.

(10) Find the maximum or minimum of $u = \dfrac{\epsilon^{x+y}}{xy}$

(11) Find maximum and minimum of

$$u = y + 2x - 2\log_\epsilon y - \log_\epsilon x.$$

bol $\int$, which is after all only a long S, merely mean
"the sum of all such quantities as." It therefore re
symbol $\sum$ (the Greek *Sigma*), which is also a sign of
is this difference, however, in the practice of mathe
the use of these signs, that while $\sum$ is generally u
sum of a number of finite quantities, the integral s
used to indicate the summing up of a vast number of
indefinitely minute magnitude, mere elements in fac
up the total required. Thus $\int dy = y$, and $\int dx =$

Any one can understand how the whole of anythi
of as made up of a lot of little bits; and the smaller t
them there will be. Thus, a line one inch long may be
up of 10 pieces, each $\frac{1}{10}$ of an inch long; or of 100 par
$\frac{1}{100}$ of an inch long; or of $1,000,000$ parts, each of
an inch long; or, pushing the thought to the limits
may be regarded as made up of an infinite number
which is infinitesimally small.

Yes, you will say, but what is the use of thinkin

$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} + \text{etc}$

Here each member of the series is formed by taki
of the preceding. What is the value of the total if
an infinite number of terms? Every schoolboy knov
is 2. Think of it, if you like, as a line. Begin with o



FIG. 46.

inch, add a quarter; add an eighth; and so on. If a
operation we stop, there will still be a piece wanti
whole 2 inches; and the piece wanting will always b
the last piece added. Thus, if after having put toget
stop, there will be $\frac{1}{4}$ wanting. If we go on till we ha
will still be $\frac{1}{64}$ wanting. The remainder needed will
the last term added. By an infinite number of oper
we reach the actual 2 inches. Practically we should

rapid and easy way of getting at a result that would an interminable lot of elaborate working out. So w time in learning *how to integrate*.

Slopes of Curves, and the Curves the

Let us make a little preliminary enquiry about t For we have seen that differentiating a curve means sion for its slope (or for its slopes at different points the reverse process of reconstructing the whole cur slopes) are prescribed for us?

Go back to case (2) on p. 82. Here we have the a sloping line with the equation

$$y = ax + b.$$

We know that here b represents the initial heigh and that a, which is the same as $\frac{dy}{dx}$, is the "slope" o

FIG. 47.

has a constant slope. All along it the elementary t
have the same proportion between height and base
to take the dx's, and dy's of finite magnitude, so tha
one inch, then there would be ten little triangles lik



Now, suppose that we were ordered to recons
starting merely from the information that $\frac{dy}{dx} = a$
do? Still taking the little d's as of finite size, we coul
all with the same slope, and then put them togethe
this: And, as the slope is the same for all, they wou
in Fig. 48, a sloping line sloping with the correct s
whether we take the dy's and dx's as finite or infini
are all alike, clearly $\frac{y}{x} = a$, if we reckon y as the to
and x as the total of all the dx's. But whereabouts

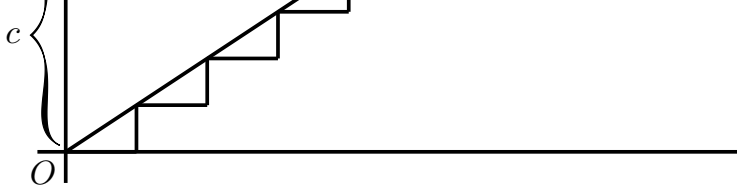


FIG. 48.

sloping line? Are we to start at the origin O, or high
information we have is as to the slope, we are witho
as to the particular height above O; in fact the init
termined. The slope will be the same, whatever the
us therefore make a shot at what may be wanted, ar
line at a height C above O. That is, we have the eq

$$y = ax + C.$$

It becomes evident now that in this case the add
the particular value that y has when $x = 0$.

Now let us take a harder case, that of a line, th
not constant, but turns up more and more. Let u
upward slope gets greater and greater in proporti

When $x = 0, \quad \frac{dy}{dx} = 0,$

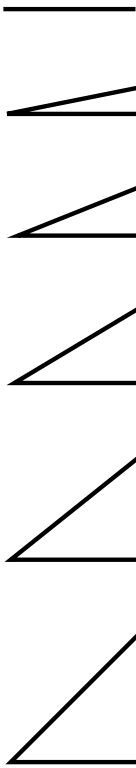
$x = 1, \quad \frac{dy}{dx} = 0.2,$

$x = 2, \quad \frac{dy}{dx} = 0.4,$

$x = 3, \quad \frac{dy}{dx} = 0.6,$

$x = 4, \quad \frac{dy}{dx} = 0.8,$

$x = 5, \quad \frac{dy}{dx} = 1.0.$



Now try to put the pieces together, setting each
of its base is the proper distance to the right, an
together at the corners; thus (Fig. 49). The result
a smooth curve: but it is an approximation to one
bits half as long, and twice as numerous, like Fig. 50

better approximation. But for a perfect curve we ou
and its corresponding dy infinitesimally small, and i

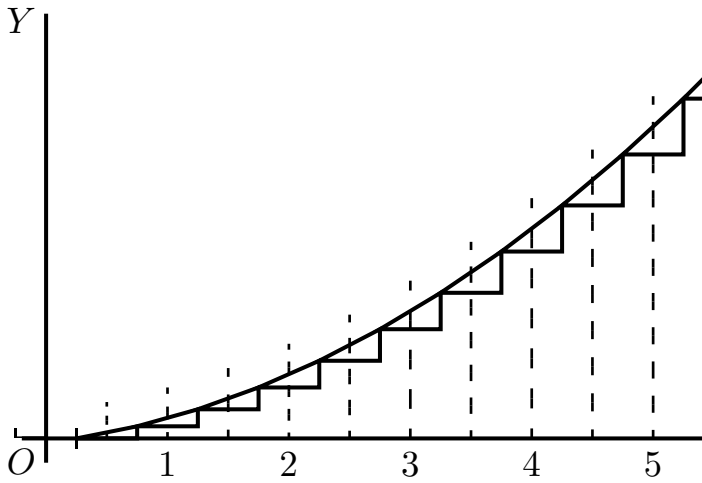


FIG. 50.

Then, how much ought the value of any y to b
point P of the curve, the value of y will be the sum
from 0 up to that level, that is to say, $\int dy = y$.
equal to $\frac{1}{5}x \cdot dx$, it follows that the whole y will be e
all such bits as $\frac{1}{5}x \cdot dx$, or, as we should write it, $\int$

Now if x had been constant, $\int \frac{1}{5}x \cdot dx$ would h

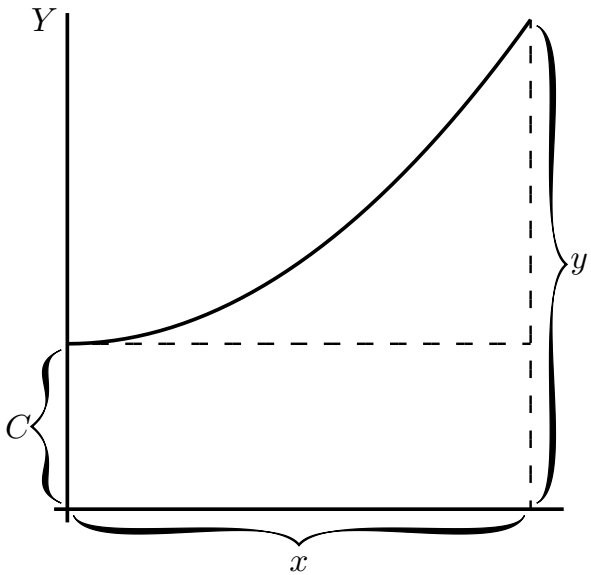


FIG. 51.

Exercises XVI. (See page 264 for Answers.)

(1) Find the ultimate sum of $\frac{2}{3} + \frac{1}{3} + \frac{1}{6} + \frac{1}{12} + \frac{1}{24}$

(2) Show that the series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7}$ and find its sum to 8 terms.

DIFFERENTIATING is the process by which when y function of x), we can find $\frac{dy}{dx}$.

Like every other mathematical operation, the pr tion may be reversed; thus, if differentiating $y = x^4$ if one begins with $\frac{dy}{dx} = 4x^3$ one would say that re would yield $y = x^4$. But here comes in a curious po $\frac{dy}{dx} = 4x^3$ if we had begun with *any* of the follow or $x^4 + c$, or x^4 with *any* added constant. So it is cl backwards from $\frac{dy}{dx}$ to y, one must make provision of there being an added constant, the value of w termined until ascertained in some other way. So, i yields nx^{n-1}, going backwards from $\frac{dy}{dx} = nx^{n-1}$ will where C stands for the yet undetermined possible c

Clearly, in dealing with powers of x, the rule for will be: Increase the power by 1, then divide by tha and add the undetermined constant.

$$\frac{dy}{dx} = anx^{n-1},$$

and reversing the process, will give us

$$y = ax^n.$$

So, when we are dealing with a multiplying constar
put the constant as a multiplier of the result of the

Thus, if $\frac{dy}{dx} = 4x^2$, the reverse process gives us y

But this is incomplete. For we must remember th
with

$$y = ax^n + C,$$

where C is any constant quantity whatever, we sh
found

$$\frac{dy}{dx} = anx^{n-1}.$$

So, therefore, when we reverse the process we mus
to add on this undetermined constant, even if we do
its value will be.

$$dy = x^2\,dx.$$

Now this is a "differential equation" which infor
ment of y is equal to the corresponding element of
Now, what we want is the integral; therefore, wr
proper symbol the instructions to integrate both sic

$$\int dy = \int x^2\,dx.$$

[Note as to reading integrals: the above would b

"*Integral dee-wy* equals *integral eks-squared d*

We haven't yet integrated: we have only written
to integrate—if we can. Let us try. Plenty of other fo
not we also? The left-hand side is simplicity itself.
bits of y is the same thing as y itself. So we may at

$$y = \int x^2\,dx.$$

by the same number as this increased power." That
be changed* to $\frac{1}{3}x^3$. Put this into the equation; but
the "constant of integration" C at the end. So we g

$$y = \tfrac{1}{3}x^3 + C.$$

You have actually performed the integration. H
Let us try another simple case.

Let $$\frac{dy}{dx} = ax^{12},$$

where a is any constant multiplier. Well, we found
ing (see p. 27) that any constant factor in the valu

*You may ask, what has become of the little dx at the
that it was really part of the differential coefficient, and whe
right-hand side, as in the $x^2\,dx$, serves as a reminder that
variable with respect to which the operation is to be effected
the product being totalled up, the power of x has increased
become familiar with all this.

So that is done. How easy!

We begin to realize now that integrating is a pro
way back, as compared with differentiating. If ever,
ing, we have found any particular expression—in this
can find our way back to the y from which it was der
between the two processes may be illustrated by th
due to a well-known teacher. If a stranger were set
Square, and told to find his way to Euston Station
task hopeless. But if he had previously been persona
Euston Station to Trafalgar Square, it would be con
him to find his way back to Euston Station.

Integration of the Sum or Difference of tw

Let $$\frac{dy}{dx} = x^2 + x^3,$$

then $$dy = x^2\,dx + x^3\,dx.$$

$$= \int x^2\,dx + \int x^3\,dx$$

$$y = \tfrac{1}{3}x^3 + \tfrac{1}{4}x^4 + C.$$

If either of the terms had been a negative quanti
ing term in the integral would have also been negat
ences are as readily dealt with as sums.

How to deal with Constant Ter

Suppose there is in the expression to be integrate
such as this:

$$\frac{dy}{dx} = x^n + b.$$

This is laughably easy. For you have only to re
you differentiated the expression $y = ax$, the result w
when you work the other way and integrate, the

Examples.

(1) Given $\dfrac{dy}{dx} = 24x^{11}$. Find y. *Ans.* $y = 2x^{1}$

(2) Find $\displaystyle\int (a+b)(x+1)\,dx$. It is $(a+b)\int ($

or $(a+b)\left[\displaystyle\int x\,dx + \int dx\right]$ or $(a+b)\left(\dfrac{x^2}{2} + x\right.$

(3) Given $\dfrac{du}{dt} = gt^{\frac{1}{2}}$. Find u. *Ans.* $u = \frac{2}{3}gt^{\frac{3}{2}}$

(4) $\dfrac{dy}{dx} = x^3 - x^2 + x$. Find y.

$dy = (x^3 - x^2 + x)\,dx$ or

$dy = x^3\,dx - x^2\,dx + x\,dx$; $y = \displaystyle\int x^3\,dx - \int x$

and $y = \frac{1}{4}x^4 - \frac{1}{3}x^3 + \frac{1}{2}x^2 + C.$

(5) Integrate $9.75x^{2.25}\,dx$. *Ans.* $y = 3x^{3.25} +$

If you look back amongst the results of differe
and x^n, etc., you will find we never got x^{-1} from a
the value of $\frac{dy}{dx}$. We got $3x^2$ from x^3; we got $2x$
from x^1 (that is, from x itself); but we did not ge
two very good reasons. *First*, x^0 is simply $= 1$, and
could not have a differential coefficient. *Secondly*,
differentiated, its differential coefficient (got by slav
usual rule) would be $0 \times x^{-1}$, and that multiplicati
zero value! Therefore when we now come to try to i
see that it does not come in anywhere in the powers
by the rule:

$$\int x^n \, dx = \frac{1}{n+1} x^{n+1}.$$

It is an exceptional case.

Well; but try again. Look through all the vario
tained from various functions of x, and try to find a
A sufficient search will show that we actually did g
result of differentiating the function $y = \log_\epsilon x$ (see

responding differentiation. If no one had found out
$\log_\epsilon x$ gave x^{-1}, we should have been utterly stuck b
to integrate $x^{-1}\,dx$. Indeed it should be frankly ad
one of the curious features of the integral calculus:–
tegrate anything before the reverse process of differe
else has yielded that expression which you want to
even to-day, is able to find the general integral of th

$$\frac{dy}{dx} = a^{-x^2},$$

because a^{-x^2} has never yet been found to result f
anything else.

Another simple case.

Find $\int (x+1)(x+2)\,dx$.

On looking at the function to be integrated, y
is the product of two different functions of x. You
integrate $(x+1)\,dx$ by itself, or $(x+2)\,dx$ by itself. C
But what to do with a product? None of the differe

And performing the integrations, we get

$$\tfrac{1}{3}x^3 + \tfrac{3}{2}x^2 + 2x + C.$$

Some other Integrals.

Now that we know that integration is the reverse
we may at once look up the differential coefficients we
see from what functions they were derived. This giv
integrals ready made:

x^{-1}	(p. 145);	$\int x^{-1}\,dx$	$= \log_\epsilon x + C.$
$\dfrac{1}{x+a}$	(p. 145);	$\int \dfrac{1}{x+a}\,dx$	$= \log_\epsilon(x+a)$
ϵ^x	(p. 139);	$\int \epsilon^x\,dx$	$= \epsilon^x + C.$
ϵ^{-x}		$\int \epsilon^{-x}\,dx$	$= -\epsilon^{-x} + C$

(for if $y = x \log_\epsilon x - x$, $\dfrac{dy}{dx} = \dfrac{x}{x} + \log_\epsilon x - 1 = \log_\epsilon$

$\log_{10} x$; $\displaystyle\int \log_{10} x\, dx = 0.4343x(\log$

a^x (p. 146); $\displaystyle\int a^x\, dx = \frac{a^x}{\log_\epsilon a} + C$

$\cos ax$; $\displaystyle\int \cos ax\, dx = \frac{1}{a} \sin ax +$

(for if $y = \sin ax$, $\dfrac{dy}{dx} = a \cos ax$; hence to get $\cos ax$
tiate $y = \dfrac{1}{a} \sin ax$).

$\sin ax$; $\displaystyle\int \sin ax\, dx = -\frac{1}{a} \cos ax$

Try also $\cos^2 \theta$; a little dodge will simplify matte

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta -$$

hence $\cos^2 \theta = \frac{1}{2}(\cos 2\theta + 1)$,

which you have successfully differentiated and integr
it grows steadily!

On Double and Triple Integral

In many cases it is necessary to integrate some ex
more variables contained in it; and in that case the
appears more than once. Thus,

$$\iint f(x, y,)\, dx\, dy$$

means that some function of the variables x and y h
for each. It does not matter in which order they ar
the function $x^2 + y^2$. Integrating it with respect to

$$\int (x^2 + y^2)\, dx = \tfrac{1}{3}x^3 + xy^2.$$

Now, integrate this with respect to y:

$$\int (\tfrac{1}{3}x^3 + xy^2)\, dy = \tfrac{1}{3}x^3y + \tfrac{1}{3}xy^3,$$

a little rectangle dx long and dy broad) has to be su
whole length and whole breadth.

Similarly in the case of solids, where we deal
sions. Consider any element of volume, the small c
sions are $dx\ dy\ dz$. If the figure of the solid be expres
$f(x, y, z)$, then the whole solid will have the *volume*

$$\text{volume} = \iiint f(x, y, z) \cdot dx \cdot dy \cdot d$$

Naturally, such integrations have to be taken betwee
its* in each dimension; and the integration cannot b
one knows in what way the boundaries of the surfa
and z. If the limits for x are from x_1 to x_2, those f
and those for z from z_1 to z_2, then clearly we have

$$\text{volume} = \int_{z1}^{z2} \int_{y1}^{y2} \int_{x1}^{x2} f(x, y, z) \cdot dx \cdot d$$

*See p. 206 for integration between limits.

(4) Find $\int (x^2 + a)\, dx$.

(5) Integrate

(6) Find $\int (4x^3 + 3x^2 + 2x + 1)\, dx$.

(7) If $\dfrac{dy}{dx} = \dfrac{ax}{2} + \dfrac{bx^2}{3} + \dfrac{cx^3}{4}$; find y.

(8) Find $\int \left(\dfrac{x^2 + a}{x + a}\right) dx$.

(9) Find $\int$

(10) Find $\int (x + 2)(x - a)\, dx$.

(11) Find $\int (\sqrt{x} + \sqrt[3]{x}) 3a^2\, dx$.

(12) Find $\int (\sin\theta - \frac{1}{2})\, \dfrac{d\theta}{3}$.

(13) Find $\int \cos^2 a\theta\, d\theta$.

(14) Find $\int$

of areas bounded by curves.

Let us try to get at the subject bit by bit.

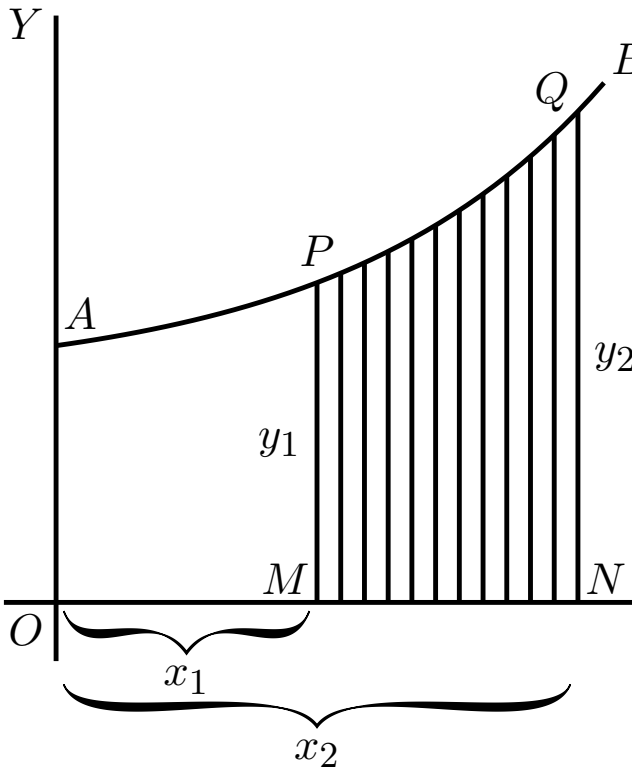


FIG. 52.

Let AB (Fig. 52) be a curve, the equation to wh
is, y in this curve is some known function of x. Thi
curve from the point P to the point Q.

the areas of all such strips. Our business will then
expression for the area of any one narrow strip, and
as to add together all the strips. Now think of any o
will be like this: being bounded between two vert
with a flat bottom dx, and with a slightly curved slo
Suppose we take its *average* height as being y; th
width is dx, its area will be $y\,dx$. And seeing that we
the width as narrow as we please, if we only take
enough its average height will be the same as the hei
middle of it. Now let us call the unknown value of
area S, meaning surface. The area of one strip will
the whole area, and may therefore be called dS. So

$$\text{area of 1 strip} = dS = y \cdot dx.$$

If then we add up all the strips, we get

$$\text{total area } S = \int dS = \int y\,dx.$$

on the left by PM, and on the right by QN, it fol
do something to define our area between those '*lim*

This introduces us to a new notion, namely tha
tween limits. We suppose x to vary, and for the pres
not require any value of x below x_1 (that is OM),
above x_2 (that is ON). When an integral is to be th
two limits, we call the lower of the two values *the inf*
upper value *the superior limit*. Any integral so limit
a *definite integral*, by way of distinguishing it from
to which no limits are assigned.

In the symbols which give instructions to integ
marked by putting them at the top and bottom resp
of integration. Thus the instruction

$$\int_{x=x_1}^{x=x_2} y \cdot dx$$

will be read: find the integral of $y \cdot dx$ between the i
the superior limit x_2.

the area $APMO$. If then we were to subtract the
the larger, we should have left as a remainder the a
is what we want. Here we have the clue as to what
integral between the two limits is *the difference* be
worked out for the superior limit and the integral
lower limit.

Let us then go ahead. First, find the general int

$$\int y\,dx,$$

and, as $y = b + ax^2$ is the equation to the curve (Fi

$$\int (b + ax^2)\,dx$$

is the general integral which we must find.

Doing the integration in question by the rule (p

$$bx + \frac{a}{3}x^3 + C;$$

and this will be the whole area from 0 up to any
may assign.

area $S = b(x_2 - x_1) + \frac{a}{3}(x_2^3 - x_1^3)$.

This is the answer we wanted. Let us give some numerical example. Suppose $b = 10$, $a = 0.06$, and $x_2 = 8$ and $x_1 = 6$. Then the area is equal to

$$\begin{aligned} & 10(8-6) + \frac{0.06}{3}(8^3 - 6^3) \\ &= 20 + 0.02(512 - 216) \\ &= 20 + 0.02 \times 296 \\ &= 20 + 5.92 \\ &= 25.92. \end{aligned}$$

Let us here put down a symbolic way of stating the result we have ascertained about limits:

$$\int_{x=x_1}^{x=x_2} y\,dx = y_2 - y_1,$$

where y_2 is the integrated value of $y\,dx$ corresponding to x_2, and y_1 is the integrated value of $y\,dx$ corresponding to x_1.

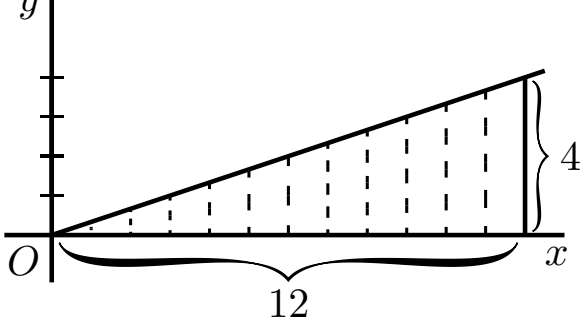


FIG. 53.

Now, here we have as the "curve" a sloping line f
tion is

$$y = \frac{x}{3}.$$

The area in question will be

$$\int_{x=0}^{x=12} y \cdot dx = \int_{x=0}^{x=12} \frac{x}{3} \cdot dx.$$

Integrating $\frac{x}{3}\,dx$ (p. 192), and putting down the
integral in square brackets with the limits marked al

Let us satisfy ourselves about this rather surpri
culation, by testing it on a simple example. Get so
preferably some that is ruled in little squares of one-

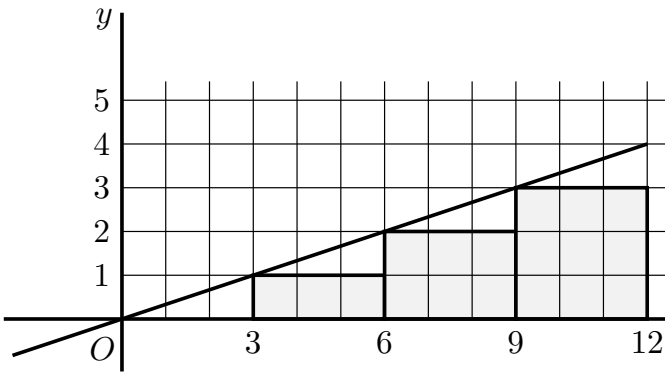


FIG. 54.

tenth inch each way. On this squared paper plot ou
equation,

$$y = \frac{x}{3}.$$

The values to be plotted will be:

x	0	3	6	9	
y	0	1	2	3	

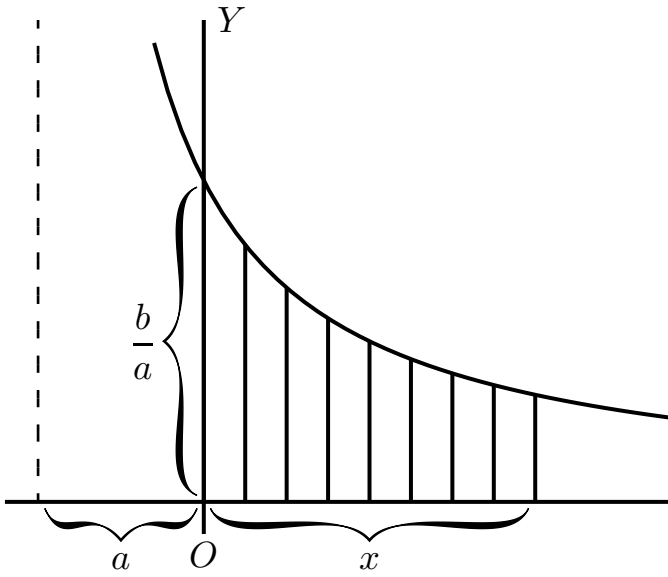


FIG. 55.

(2) Find the area, between limits $x = x_1$ and x
$y = \dfrac{b}{x+a}$.

$$\text{Area} = \int_{x=0}^{x=x_1} y \cdot dx = \int_{x=0}^{x=x_1} \frac{b}{x+a}$$
$$= b\left[\log_\epsilon(x+a)\right]_0^{x_1} + C$$
$$= b\left[\log_\epsilon(x_1+a) - \log_\epsilon(0+a)\right]$$

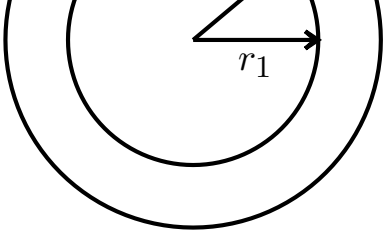


FIG. 56.

and the inner radius is r_1? You know from mensura
of the outer circle is πr_2^2; then you find the area of th
then you subtract the latter from the former, and
$= \pi(r_2^2 - r_1^2)$; which may be written

$$\pi(r_2 + r_1)(r_2 - r_1)$$

= mean circumference of ring × width of ring.

(3) Here's another case—that of the *die-away cu*
the area between $x = 0$ and $x = a$, of the curve (Fig.
is

$$y = b\epsilon^{-x}.$$

$$\text{Area} = b \int_{x=0}^{x=a} \epsilon^{-x} \cdot dx.$$

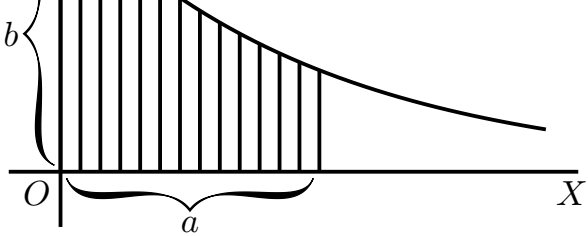
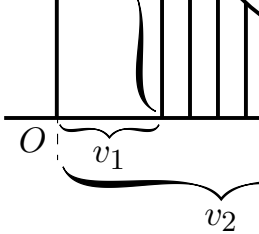


FIG. 57.



(4) Another example is afforded by the adiabatic gas, the equation to which is $pv^n = c$, where p stands volume, and n is of the value 1.42 (Fig. 58).

Find the area under the curve (which is propor done in suddenly compressing the gas) from volume

Here we have

$$\begin{aligned}\text{area} &= \int_{v=v_1}^{v=v_2} cv^{-n} \cdot dv \\ &= c\left[\frac{1}{1-n}v^{1-n}\right]_{v_1}^{v_2} \\ &= c\frac{1}{1-n}(v_2^{1-n} - v_1^{1-n}) \\ &= \frac{-c}{0.42}\left(\frac{1}{v_2^{0.42}} - \frac{1}{v_1^{0.42}}\right).\end{aligned}$$

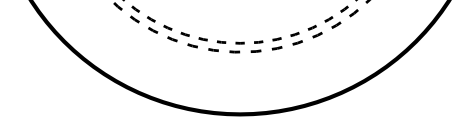


FIG. 59.

Consider an elementary zone or annulus of the s
breadth dr, situated at a distance r from the centre
the entire surface as consisting of such narrow zor
area A will simply be the integral of all such elem
centre to margin, that is, integrated from $r = 0$ to r

We have therefore to find an expression for the e
of the narrow zone. Think of it as a strip of breadth
that is the periphery of the circle of radius r, that
Then we have, as the area of the narrow zone,

$$dA = 2\pi r\, dr.$$

Hence the area of the whole circle will be:

$$A = \int dA = \int_{r=0}^{r=R} 2\pi r \cdot dr = 2\pi \int_{r=0}^{r=R}$$

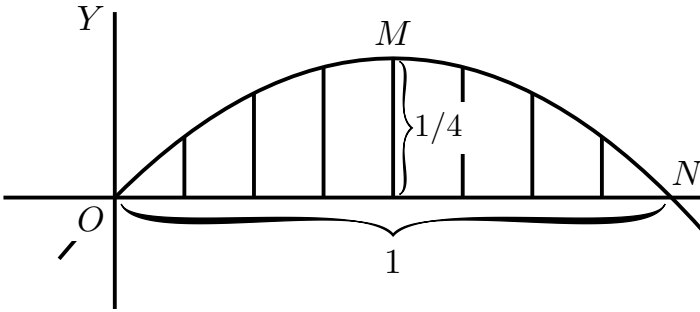


FIG. 60.

shall have to find the area of the piece OMN, and the length of the base ON. But before we can find ascertain the length of the base, so as to know up to to integrate. At N the ordinate y has zero value; look at the equation and see what value of x will clearly, if x is 0, y will also be 0, the curve passing th but also, if $x = 1$, $y = 0$; so that $x = 1$ gives us point N.

[N.B.—It will be a pretty and simple exercise in i
to find by differentiation what is the height of the r
It *must* be greater than the average.]

The mean ordinate of any curve, over a range fro
is given by the expression,

$$\text{mean } y = \frac{1}{x_1} \int_{x=0}^{x=x_1} y \cdot dx.$$

One can also find in the same way the surface
revolution.

Example.

The curve $y = x^2 - 5$ is revolving about the axis
of the surface generated by the curve between $x = 0$

A point on the curve, the ordinate of which is y,
ference of length $2\pi y$, and a narrow belt of the su
corresponding to this point, has for area $2\pi y\, dx$. Th

$$2\pi \int_{x=0}^{x=6} y\, dx = 2\pi \int_{x=0}^{x=6} (x^2 - 5)\, dx = 2\pi \Big[$$

$$= 6.28 \times 42 = 263.76.$$

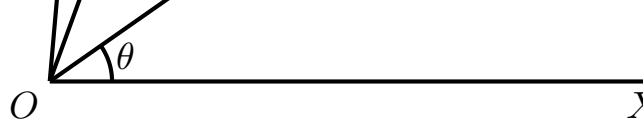


FIG. 61.

direction OX, the process just explained can be ap
with a small modification. Instead of a strip of a
small triangle OAB, the angle at O being $d\theta$, and
all the little triangles making up the required area.

The area of such a small triangle is approxi
$\frac{r\,d\theta}{2} \times r$; hence the portion of the area included bet
two positions of r corresponding to the angles θ_1 a

$$\tfrac{1}{2}\int_{\theta=\theta_1}^{\theta=\theta_2} r^2\,d\theta.$$

Examples.

(1) Find the area of the sector of 1 radian in
radius a inches.

$$= \frac{a^2}{2}\left[\theta + 2\sin\theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]$$
$$= \frac{a^2(3\pi + 8)}{8}.$$

Volumes by Integration.

What we have done with the area of a little str
can, of course, just as easily do with the volume o
solid. We can add up all the little strips that make
and find its volume, just as we have added up all t
that made up an area to find the final area of the fig

Examples.

(1) Find the volume of a sphere of radius r.

A thin spherical shell has for volume $4\pi x^2\,dx$ (s
summing up all the concentric shells which make

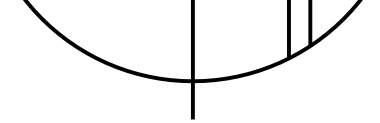


FIG. 62.

We can also proceed as follows: a slice of the sphe has for volume $\pi y^2\,dx$ (see Fig. 62). Also x and y expression

$$y^2 = r^2 - x^2.$$

$$\text{Hence} \quad \text{volume sphere} = 2\int_{x=0}^{x=r} \pi(r^2 - x^2)\,dx$$

$$= 2\pi\left[\int_{x=0}^{x=r} r^2\,dx - \int_{x=0}^{x=}\right.$$

$$= 2\pi\left[r^2x - \frac{x^3}{3}\right]_0^r = \frac{4\pi}{3}r^3$$

(2) Find the volume of the solid generated by th curve $y^2 = 6x$ about the axis of x, between $x = 0$ a

mean of a variable quantity. By "quadratic mean" is
root of the mean of the squares of all the values betw
sidered. Other names for the quadratic mean of a
"virtual" value, or its "R.M.S." (meaning root-mean-
French term is *valeur efficace*. If y is the function u
and the quadratic mean is to be taken between the
$x = l$; then the quadratic mean is expressed as

$$\sqrt[2]{\frac{1}{l}\int_0^l y^2\,dx}.$$

Examples.

(1) To find the quadratic mean of the function y
Here the integral is $\displaystyle\int_0^l a^2x^2\,dx$, which is $\frac{1}{3}a^2l^3$.
Dividing by l and taking the square root, we ha

$$\text{quadratic mean} = \frac{1}{\sqrt{3}}\,al.$$

(2) To find the quadratic mean of the function y

The integral is $\int_{x=0}^{x=l} x^{2a}\,dx$, that is $\dfrac{l^{2a+1}}{2a+1}$.

Hence quadratic mean $= \sqrt[2]{\dfrac{l^{2a}}{2a+1}}$.

(3) To find the quadratic mean of the function y

The integral is $\int_{x=0}^{x=l} (a^{\frac{x}{2}})^2\,dx$, that is $\int_{x=0}^{x=l} a^x\,dx$

or $\left[\dfrac{a^x}{\log_\epsilon a}\right]_{x=0}^{x=l}$,

which is $\dfrac{a^l - 1}{\log_\epsilon a}$.

Hence the quadratic mean is $\sqrt[2]{\dfrac{a^l - 1}{l \log_\epsilon a}}$.

Exercises XVIII. (See p. 265 for Answers.)

(1) Find the area of the curve $y = x^2 + x - 5$ b

$x = 6$, and the mean ordinates between these limits

$y = x^2 \pm x^{\frac{3}{2}}$ from $x = 0$ to $x = 1$, also the area of t

of the lower branch of the curve (see Fig. 30, p. 106

(6) Find the volume of a cone of radius of base r

(7) Find the area of the curve $y = x^3 - \log_\epsilon x$ b

$x = 1$.

(8) Find the volume generated by the curve y

revolves about the axis of x, between $x = 0$ and $x =$

(9) Find the volume generated by a sine curve r

axis of x. Find also the area of its surface.

(10) Find the area of the portion of the curve xy

tween $x = 1$ and $x = a$. Find the mean ordinate be

(11) Show that the quadratic mean of the function

the limits of 0 and π radians, is $\frac{\sqrt{2}}{2}$. Find also the

of the same function between the same limits; and s

factor is $= 1.11$.

(15) Show that the radius of a circle, the area of
area of a polar diagram, is equal to the quadratic me
of r for that polar diagram.

(16) Find the volume generated by the curve y
rotating about the axis of x.

licking them into some shape that can be integrated
by this is meant the serious books—on the Integral
plans and methods and dodges and artifices for this
following are a few of them.

Integration by Parts. This name is given to a dod
which is

$$\int u\,dx = ux - \int x\,du + C.$$

It is useful in some cases that you can't tackle dir
that if in any case $\int x\,du$ can be found, then $\int u\,dx$
The formula can be deduced as follows. From p. 37

$$d(ux) = u\,dx + x\,du,$$

which may be written

$$u(dx) = d(ux) - x\,du,$$

which by direct integration gives the above expressi

(2) Find $\int$

Write $u = x, \qquad \epsilon^x\,dx = dv;$

then $du = dx, \qquad v = \epsilon^x,$

and $$\int x\epsilon^x\,dx = x\epsilon^x - \int \epsilon^x\,dx \quad \text{(by the for}$$
$$= x\epsilon^x - \epsilon^x = \epsilon^x(x-1) + C.$$

(3) Try $\int \cos^2\theta\,d\theta.$

$u = \cos\theta, \qquad \cos\theta\,d\theta = dv$

Hence $du = -\sin\theta\,d\theta, \qquad v = \sin$

$$\int \cos^2\theta\,d\theta = \cos\theta\sin\theta + \int \sin^2\theta\,d\theta$$
$$= \frac{2\cos\theta\sin\theta}{2} + \int (1 - \cos$$
$$= \frac{\sin 2\theta}{2} + \int d\theta - \int \cos^2\theta$$

then $du = 2x\,dx,$ $v = -\cos$

$$\int x^2 \sin x\,dx = -x^2 \cos x + 2\int x \cos$$

Now find $\int x \cos x\,dx$, integrating by parts (as in

$$\int x \cos x\,dx = x \sin x + \cos x + C$$

Hence

$$\int x^2 \sin x\,dx = -x^2 \cos x + 2x \sin x + 2 \cos$$

$$= 2\left[x \sin x + \cos x\left(1 - \frac{x^2}{2}\right.\right.$$

(5) Find $\int \sqrt{1 - x^2}\,dx.$

Write $u = \sqrt{1 - x^2}, \quad dx = dv;$

then $du = -\dfrac{x\,dx}{\sqrt{1 - x^2}}$ (see Chap. IX., p.

$$2\int \sqrt{1-x^2}\,dx = x\sqrt{1-x^2} + \int \frac{d}{\sqrt{1-}}$$

Do you remember meeting $\dfrac{dx}{\sqrt{1-x^2}}$? it is got $y = \arcsin x$ (see p. 168); hence its integral is arc sin

$$\int \sqrt{1-x^2}\,dx = \frac{x\sqrt{1-x^2}}{2} + \tfrac{1}{2}\arcsin x$$

You can try now some exercises by yourself; yo the end of this chapter.

Substitution. This is the same dodge as expla p. 66. Let us illustrate its application to integration

(1) $\displaystyle\int \sqrt{3+x}\,dx.$

Let $\qquad 3+x = u, \quad dx = du;$

replace $\qquad \displaystyle\int u^{\frac{1}{2}}\,du = \tfrac{2}{3}u^{\frac{3}{2}} = \tfrac{2}{3}(3+x)^{\frac{3}{2}}.$

$(x)\ \int \frac{dx}{x^2+2x+3} \qquad \int \frac{dx}{x^2+2x+1+2} \qquad \int \frac{dx}{(x}$

Let $x+1=u, \quad dx=du;$

then the integral becomes $\int \frac{du}{u^2+(\sqrt{2})^2}$; but $\frac{du}{u^2+}$

differentiating $u=\frac{1}{a}\operatorname{arc\,tan}\frac{u}{a}$.

Hence one has finally $\frac{1}{\sqrt{2}}\operatorname{arc\,tan}\frac{x+1}{\sqrt{2}}$ for the integral.

Formulæ of Reduction are special forms applica mial and trigonometrical expressions that have to have to be reduced into some form of which the int

Rationalization, and *Factorization of Denomina* plicable in special cases, but they do not admit of a explanation. Much practice is needed to become f preparatory processes.

The following example shows how the process of s fractions, which we learned in Chap. XIII., p. 118, in integration.

tised hand would avoid; such as the use of factors tha
either zero or infinity, and the occurrence of indete
such as $\frac{0}{0}$. There is no golden rule that will meet e
Nothing but practice and intelligent care will avail
pitfall which had to be circumvented arose in Cha
when we came to the problem of integrating $x^{-1}\,dx$

Triumphs. By triumphs must be understood
which the calculus has been applied to the solution
wise intractable. Often in the consideration of phy
is able to build up an expression for the law gove
tion of the parts or of the forces that govern ther
being naturally in the form of a *differential equatio*
tion containing differential coefficients with or with
quantities. And when such a differential equation h
can get no further until it has been integrated. G
easier to state the appropriate differential equation
the real trouble begins then only when one wants t

$$\frac{dx}{} \quad a^2 - x^2$$

could blossom out into

$$y = \frac{1}{2a} \log_\epsilon \frac{a+x}{a-x} + C?$$

yet the latter is the *solution* of the former.

As a last example, let us work out the above tog

By partial fractions,

$$\frac{1}{a^2 - x^2} = \frac{1}{2a(a+x)} + \frac{1}{2a(a-x)},$$

$$dy = \frac{dx}{2a(a+x)} + \frac{dx}{2a(a-x)},$$

$$y = \frac{1}{2a}\left(\int \frac{dx}{a+x} + \int \frac{dx}{a-x}\right)$$

*This means that the actual result of solving it is called its mathematicians would say, with Professor Forsyth, "every c *considered as solved* when the value of the dependent varia function of the independent variable by means either of k integrals, whether the integrations in the latter can or cannot of functions already known."

(1) Find $\int \sqrt{a^2 - x^2}\, dx$.

(2) Find $\int$

(3) Find $\int x^a \log_\epsilon x\, dx$.

(4) Find $\int$

(5) Find $\int \frac{1}{x} \cos(\log_\epsilon x)\, dx$.

(6) Find $\int$

(7) Find $\int \frac{(\log_\epsilon x)^a}{x}\, dx$.

(8) Find $\int$

(9) Find $\int \frac{5x+1}{x^2+x-2}\, dx$.

(10) Find $\int$

(11) Find $\int \frac{b\, dx}{x^2 - a^2}$.

(12) Find $\int$

(13) Find $\int \frac{dx}{1 - x^4}$.

(14) Find $\int$

differential equations, using for this purpose the pro
preceding chapters.

The beginner, who now knows how easy most
are in themselves, will here begin to realize that in
As in all arts, so in this, facility can be acquired o
regular practice. He who would attain that facil
examples, and more examples, and yet more exa
found abundantly in all the regular treatises on t
purpose here must be to afford the briefest introducti

Example 1. Find the solution of the differential

$$ay + b\frac{dy}{dx} = 0.$$

Transposing we have

$$b\frac{dy}{dx} = -ay.$$

can do nothing until we get both y and dy to one s
other. To do this, we must split our usually insep
dy and dx from one another.

$$\frac{dy}{y} = -\frac{a}{b}\,dx.$$

Having done the deed, we now can see that both
a shape that is integrable, because we recognize $\frac{dy}{y}$,
ential that we have met with (p. 143) when differen
So we may at once write down the instructions to i

$$\int \frac{dy}{y} = \int -\frac{a}{b}\,dx;$$

and doing the two integrations, we have:

$$\log_\epsilon y = -\frac{a}{b}x + \log_\epsilon C,$$

where $\log_\epsilon C$ is the yet undetermined constant* of

*We may write down any form of constant as the "consta
the form $\log_\epsilon C$ is adopted here by preference, because the o
of equation are, or are treated as logarithms; and it saves co
if the added constant be *of the same kind*.

particular value* of y at starting. This we may call y solution as

$$y = y_0 \epsilon^{-\frac{a}{b}x}.$$

Example 2.

Let us take as an example to solve

$$ay + b\frac{dy}{dx} = g,$$

where g is a constant. Again, inspecting the equ (1) that somehow or other ϵ^x will come into the sol if at any part of the curve y becomes either a maxim so that $\frac{dy}{dx} = 0$, then y will have the value $= \frac{g}{a}$. Bu as before, separating the differentials and trying to t

*Compare what was said about the "constant of integrati Fig. 48 on p. 184, and Fig. 51 on p. 187.

integrable?

It is of the same form as the result on p. 145; structions to integrate, we have:

$$\int \frac{dy}{y - \frac{g}{a}} = -\int \frac{a}{b}\, dx;$$

and, doing the integration, and adding the appropri

$$\log_\epsilon \left(y - \frac{g}{a}\right) = -\frac{a}{b}x + \log_\epsilon C;$$

whence

$$y - \frac{g}{a} = C\epsilon^{-\frac{a}{b}x};$$

and finally,

$$y = \frac{g}{a} + C\epsilon^{-\frac{a}{b}x},$$

which is *the solution.*

If the condition is laid down that $y = 0$ when x for then the exponential becomes $= 1$; and we have

$$0 = \frac{g}{a} + C,$$

or

$$C = -\frac{g}{a}.$$

Example 3.

Let $$ay + b\frac{dy}{dt} = g \cdot \sin 2\pi nt.$$

We shall find this much less tractable than th
divide through by b.

$$\frac{dy}{dt} + \frac{a}{b}y = \frac{g}{b}\sin 2\pi nt.$$

Now, as it stands, the left side is not integrab
made so by the artifice—and this is where skill and
plan—of multiplying all the terms by $\epsilon^{\frac{a}{b}t}$, giving us

$$\frac{dy}{dt}\epsilon^{\frac{a}{b}t} + \frac{a}{b}y\epsilon^{\frac{a}{b}t} = \frac{g}{b}\epsilon^{\frac{a}{b}t} \cdot \sin 2\pi nt,$$

which is the same as

$$\frac{dy}{dt}\epsilon^{\frac{a}{b}t} + y\frac{d(\epsilon^{\frac{a}{b}t})}{dt} = \frac{g}{b}\epsilon^{\frac{a}{b}t} \cdot \sin 2\pi nt$$

p. 224) of integration by parts, the general formula fo

$uv - \int vdu$. For this purpose write

$$\begin{cases} u = \epsilon^{\frac{a}{b}t}; \\ dv = \sin 2\pi nt \cdot dt. \end{cases}$$

We shall then have

$$\begin{cases} du = \epsilon^{\frac{a}{b}t} \times \frac{a}{b}\, dt; \\ v = -\frac{1}{2\pi n} \cos 2\pi nt. \end{cases}$$

Inserting these, the integral in question becomes

$$\int \epsilon^{\frac{a}{b}t} \cdot \sin 2\pi nt \cdot dt$$

$$= -\frac{1}{2\pi n} \cdot \epsilon^{\frac{a}{b}t} \cdot \cos 2\pi nt - \int -\frac{1}{2\pi n} \cos 2$$

$$= -\frac{1}{2\pi n} \epsilon^{\frac{a}{b}t} \cos 2\pi nt + \frac{a}{2\pi nb} \int \epsilon^{\frac{a}{b}t} \cdot \cos 2$$

Inserting these, we get

$$\int \epsilon^{\frac{a}{b}t} \cdot \sin 2\pi nt \cdot dt$$

$$= \frac{b}{a} \cdot \epsilon^{\frac{a}{b}t} \cdot \sin 2\pi nt - \frac{2\pi nb}{a} \int \epsilon^{\frac{a}{b}t} \cdot \cos$$

Noting that the final intractable integral in [C] i
in [B], we may eliminate it, by multiplying [B] by $\frac{2}{}$
ing [C] by $\frac{a}{2\pi nb}$, and adding them.

The result, when cleared down, is:

$$\int \epsilon^{\frac{a}{b}t} \cdot \sin 2\pi nt \cdot dt = \epsilon^{\frac{a}{b}t} \left\{ \frac{ab \cdot \sin 2\pi nt - 2\pi nb^2 \cdot}{a^2 + 4\pi^2 n^2 b^2} \right.$$

Inserting this value in [A], we get

$$y = g \left\{ \frac{a \cdot \sin 2\pi nt - 2\pi nb \cdot \cos}{a^2 + 4\pi^2 n^2 b^2} \right.$$

To simplify still further, let us imagine an angle

which may be written

$$y = g\frac{\sin(2\pi nt - \phi)}{\sqrt{a^2 + 4\pi^2 n^2 b^2}},$$

which is *the solution* desired.

This is indeed none other than the equation of an
current, where g represents the amplitude of the e
n the frequency, a the resistance, b the coefficient
the circuit, and ϕ is an angle of lag.

Example 4.

Suppose that $M\,dx + N\,dy = 0.$

We could integrate this expression directly, if M
only, and N a function of y only; but, if both M a
that depend on both x and y, how are we to integ
an exact differential? That is: have M and N ea

Now the test of the matter is this. If the expr
differential, it must be true that

$$\frac{dM}{dy} = \frac{dN}{dx};$$

for then

$$\frac{d(dU)}{dx\,dy} = \frac{d(dU)}{dy\,dx},$$

which is necessarily true.

Take as an illustration the equation

$$(1 + 3xy)\,dx + x^2\,dy = 0.$$

Is this an exact differential or not? Apply the te

$$\begin{cases} \dfrac{d(1+3xy)}{dy} = 3x, \\ \dfrac{d(x^2)}{dx} = 2x, \end{cases}$$

which do not agree. Therefore, it is not an exact d
two functions $1 + 3xy$ and x^2 have not come from
function.

$$\begin{cases} \dfrac{d(2x^3 + 6x^2y)}{dy} = 6x^2, \\ \dfrac{d(2x^3)}{dx} = 6x^2, \end{cases}$$

which agrees. Hence this is an exact differential, and
Now, if $w = 2x^3y$,

$$dw = 6x^2y\,dx + 2x^3\,dy.$$

Hence $$\int 6x^2y\,dx + \int 2x^3\,dy = w = 2x^3y$$

so that we get $$U = x^2 + 2x^3y + C.$$

Example 5. Let $\dfrac{d^2y}{dt^2} + n^2y = 0.$

In this case we have a differential equation of th
which y appears in the form of a second differential
as in person.

Transposing, we have $\dfrac{d^2y}{dt^2} = -n^2y.$

$$2\frac{d^2y}{dt^2}\,\frac{dy}{dt} = \frac{d\left(\frac{dy}{dt}\right)^2}{dt}, \qquad \left(\frac{dy}{dt}\right)^2 + n^2(y^2 -$$

C being a constant. Then, taking the square roots,

$$\frac{dy}{dt} = -n\sqrt{y^2 - C^2} \quad \text{and} \quad \frac{dy}{\sqrt{C^2 - y^2}} =$$

But it can be shown that (see p. 168)

$$\frac{1}{\sqrt{C^2 - y^2}} = \frac{d(\arc\sin\frac{y}{C})}{dy};$$

whence, passing from angles to sines,

$$\arc\sin\frac{y}{C} = nt + C_1 \quad \text{and} \quad y = C\sin(nt$$

where C_1 is a constant angle that comes in by integ

Or, preferably, this may be written

$$y = A\sin nt + B\cos nt, \text{ which is the so}$$

---

and, as $2\dfrac{d^2y}{dx^2}\,\dfrac{dy}{dx} = \dfrac{d\left(\dfrac{dy}{dx}\right)^2}{dx}$, $\left(\dfrac{dy}{dx}\right)^2 - n^2(y^2 +$

$$\frac{dy}{dx} - n\sqrt{y^2+c^2} = 0,$$

where c is a constant, and $\dfrac{dy}{\sqrt{y^2+c^2}} = n\,dx$.

Now, if $w = \log_\epsilon(y + \sqrt{y^2+c^2}) = \log_\epsilon u$,

$$\frac{dw}{du} = \frac{1}{u}, \quad \frac{du}{dy} = 1 + \frac{y}{\sqrt{y^2+c^2}} = \frac{y+\sqrt{}}{\sqrt{y^2}}$$

and $$\frac{dw}{dy} = \frac{1}{\sqrt{y^2+c^2}}.$$

Hence, integrating, this gives us

$$\log_\epsilon(y + \sqrt{y^2+c^2}) = nx + \log_\epsilon C$$

$$y + \sqrt{y^2+c^2} = C\epsilon^{nx}.$$

Now $(y + \sqrt{y^2+c^2}) \times (-y + \sqrt{y^2+c^2})$

whence $$-y + \sqrt{y^2+c^2} = \frac{c^2}{C}\epsilon^{-nx}.$$

which dies away as x increases.

Example 7.

Let $$b\frac{d^2y}{dt^2} + a\frac{dy}{dt} + gy = 0.$$

Examination of this expression will show that, form of Example 1, the solution of which was a ne On the other hand, if $a = 0$, its form becomes th Example 6, the solution of which is the sum of a posi exponential. It is therefore not very surprising to fin of the present example is

$$y = (\epsilon^{-mt})(A\epsilon^{nt} + B\epsilon^{-nt}),$$

where $$m = \frac{a}{2b} \quad \text{and} \quad n = \sqrt{\frac{a^2}{4b^2}} - \frac{g}{b}.$$

The steps by which this solution is reached are n may be found in advanced treatises.

$$\frac{d^2F}{du \cdot dv} = 0,$$

where $u = x + at$, and $v = x - at$, leading to the sam
If we consider a case in which F vanishes, then we h

$$y = f(x - at);$$

and this merely states that, at the time $t = 0$, y is
tion of x, and may be looked upon as denoting tha
relation of y to x has a particular shape. Then any
of t is equivalent simply to an alteration in the orig
reckoned. That is to say, it indicates that, the fo
being conserved, it is propagated along the x direct
velocity a; so that whatever the value of the ordi
ticular time t_0 at any particular point x_0, the sa
appear at the subsequent time t_1 at a point further
of which is $x_0 + a(t_1 - t_0)$. In this case the simplified
the propagation of a wave (of any form) at a unifo
x direction.

enchanted land. And in order that you may have
to the principal results, the author, in bidding yo
present you with a passport in the shape of a conv
standard forms (see pp. 249–251). In the middle co
a number of the functions which most commonly oc
differentiating them are set down on the left; the re
them are set down on the right. May you find them

bad book. Of that there can be, from their point
ble manner of doubt whatever. It commits several
deplorable errors.

First, it shows how ridiculously easy most of th
calculus really are.

Secondly, it gives away so many trade secrets. B
what one fool can do, other fools can do also, it lets
mathematical swells, who pride themselves on havin
awfully difficult subject as the calculus, have no su
be puffed up. They like you to think how terribly
don't want that superstition to be rudely dissipated

Thirdly, among the dreadful things they will sa
is this: that there is an utter failure on the part
demonstrate with rigid and satisfactory completen
sundry methods which he has presented in simple
even *dared to use* in solving problems! But why sh
don't forbid the use of a watch to every person who

is, indeed, why the book has been written—writte
innocents who have hitherto been deterred from acq
of the calculus by the stupid way in which its tea
ways presented. Any subject can be made repulsi
bristling with difficulties. The aim of this book is t
to learn its language, to acquire familiarity with
plicities, and to grasp its powerful methods of solvi
out being compelled to toil through the intricate o
mostly irrelevant) mathematical gymnastics so dear
mathematician.

There are amongst young engineers a number
adage that *what one fool can do, another can*, may
sound. They are earnestly requested not to give th
to tell the mathematicians what a fool he really is.

1	$x \pm a$	$\frac{1}{2}x^2 \pm$
a	ax	$\frac{1}{2}ax$
$2x$	x^2	$\frac{1}{3}x^3$
nx^{n-1}	x^n	$\frac{1}{n+1}x$
$-x^{-2}$	x^{-1}	$\log_\epsilon$
$\frac{du}{dx} \pm \frac{dv}{dx} \pm \frac{dw}{dx}$	$u \pm v \pm w$	$\int u\,dx \pm \int$
$u\frac{dv}{dx} + v\frac{du}{dx}$	uv	No general
$\frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$	$\frac{u}{v}$	No general
$\frac{du}{dx}$	u	$ux - \int$

		$\log_\epsilon a$
	Trigonometrical.	
$\cos x$	$\sin x$	$-\cos x$
$-\sin x$	$\cos x$	$\sin x$
$\sec^2 x$	$\tan x$	$-\log_\epsilon \cos$
	Circular (Inverse).	
$\frac{1}{\sqrt{(1-x^2)}}$	$\arcsin x$	$x \cdot \arcsin x +$
$-\frac{1}{\sqrt{(1-x^2)}}$	$\arccos x$	$x \cdot \arccos x -$
$\frac{1}{1+x^2}$	$\arctan x$	$x \cdot \arctan x - \frac{1}{2}$ l
	Hyperbolic.	
$\cosh x$	$\sinh x$	$\cosh x$
$\sinh x$	$\cosh x$	$\sinh x$
$\operatorname{sech}^2 x$	$\tanh x$	$\log_\epsilon \cosh$

$a \cdot \cos ax$	$\sin ax$	$-\frac{1}{a} \cos$
$-a \cdot \sin ax$	$\cos ax$	$\frac{1}{a} \sin$
$a \cdot \sec^2 ax$	$\tan ax$	$-\frac{1}{a} \log_\epsilon$
$\sin 2x$	$\sin^2 x$	$\frac{x}{2} - \frac{\sin}{}$
$-\sin 2x$	$\cos^2 x$	$\frac{x}{2} + \frac{\sin}{}$
$n \cdot \sin^{n-1} x \cdot \cos x$	$\sin^n x$	$-\frac{\cos x}{n} \sin^{n-1} x + \frac{n}{}$
$-\frac{\cos x}{\sin^2 x}$	$\frac{1}{\sin x}$	$\log_\epsilon \tan$
$-\frac{\sin 2x}{\sin^4 x}$	$\frac{1}{\sin^2 x}$	$-\cot a$
$\frac{\sin^2 x - \cos^2 x}{\sin^2 x \cdot \cos^2 x}$	$\frac{1}{\sin x \cdot \cos x}$	$\log_\epsilon \tan$
$n \cdot \sin mx \cdot \cos nx + m \cdot \sin nx \cdot \cos mx$	$\sin mx \cdot \sin nx$	$\frac{1}{2} \cos(m-n)x -$
$2a \cdot \sin 2ax$	$\sin^2 ax$	$\frac{x}{2} - \frac{\sin}{4}$
$-2a \cdot \sin 2ax$	$\cos^2 ax$	$\frac{x}{2} + \frac{\sin}{4}$

(9) $\frac{dy}{dx} = \frac{3}{q} x^{\frac{3-q}{q}}$.

(10) $\frac{dy}{dx} = -\frac{m}{n} x^{-}$

Exercises II. (p. 31.)

(1) $\frac{dy}{dx} = 3ax^2$.

(2) $\frac{dy}{dx} = 13 \times \frac{3}{2} x^{\frac{1}{2}}$.

(3) $\frac{dy}{dx}$

(4) $\frac{dy}{dx} = \frac{1}{2} c^{\frac{1}{2}} x^{-\frac{1}{2}}$.

(5) $\frac{du}{dz} = \frac{an}{c} z^{n-1}$.

(6) $\frac{dy}{dt}$

(7) $\frac{dl_t}{dt} = 0.000012 \times l_0$.

(8) $\frac{dC}{dV} = abV^{b-1}$, 0.98, 3.00 and 7.47 candle power p

(9) $\frac{dn}{dD} = -\frac{1}{LD^2}\sqrt{\frac{gT}{\pi\sigma}}$, $\frac{dn}{dL} = -\frac{1}{DL^2}\sqrt{\frac{gT}{\pi\sigma}}$,

$\frac{dn}{d\sigma} = -\frac{1}{2DL}\sqrt{\frac{gT}{\pi\sigma^3}}$, $\frac{dn}{dT} = \frac{1}{2DL}\sqrt{\frac{g}{\pi\sigma T}}$.

(2) $\dfrac{dw}{dt} = a - bt.$

(3) $\dfrac{dy}{dx} = 2x.$

(4) $14110x^4 - 65404x^3 - 2244x^2 + 8192x + 1379.$

(5) $\dfrac{dx}{dy} = 2y + 8.$

(6) $185.9022654x^$

(7) $\dfrac{-5}{(3x+2)^2}.$

(8) $\dfrac{6x^4 + 6x^3 + 9}{(1 + x + 2x^2$

(9) $\dfrac{ad - bc}{(cx+d)^2}.$

(10) $\dfrac{anx^{-n-1} + b}{(x^{-n}$

(11) $b + 2ct.$

(12) $R_0(a+2bt), \quad R_0\left(a + \dfrac{b}{2\sqrt{t}}\right), \quad -\dfrac{R_0(a+2bt)}{(1+at+bt^2)^2}$

(13) $1.4340(0.000014t - 0.001024), \quad -0.00117, \quad -0.$

(14) $\dfrac{dE}{dl} = b + \dfrac{k}{i}, \quad \dfrac{dE}{di} = -\dfrac{c+kl}{i^2}.$

(2) $-b$, 0. (3) 2, 0.

(4) $56440x^3 - 196212x^2 - 4488x + 8192.$

$$169320x^2 - 392424x - 4488.$$

(5) 2, 0. (6) $371.80453x$, 371.804

(7) $\dfrac{30}{(3x+2)^3}$, $-\dfrac{270}{(3x+2)^4}$.

(*Examples*, p. 40):

(1) $\dfrac{6a}{b^2}x$, $\dfrac{6a}{b^2}$. (2) $\dfrac{3a\sqrt{b}}{2\sqrt{x}} - \dfrac{6b\sqrt[3]{a}}{x^3}$, $\dfrac{18b\sqrt[3]{}}{x^4}$

(3) $\dfrac{2}{\sqrt[3]{\theta^8}} - \dfrac{1.056}{\sqrt[5]{\theta^{11}}}$, $\dfrac{2.3232}{\sqrt[5]{\theta^{16}}} - \dfrac{16}{3\sqrt[3]{\theta^{11}}}$.

(4) $810t^4 - 648t^3 + 479.52t^2 - 139.968t + 26.64$

$$3240t^3 - 1944t^2 + 959.04t - 139.968.$$

(5) $12x + 2$, 12. (6) $6x^2 - 9x$, 12

(7) $\dfrac{3}{4}\left(\dfrac{1}{\sqrt{\theta}} + \dfrac{1}{\sqrt{\theta^5}}\right) + \dfrac{1}{4}\left(\dfrac{15}{\sqrt{\theta^7}} - \dfrac{1}{\sqrt{\theta^3}}\right)$.

$\dfrac{3}{8}\left(\dfrac{1}{\sqrt{\theta^5}} - \dfrac{1}{\sqrt{\theta^3}}\right) - \dfrac{15}{8}\left(\dfrac{7}{\sqrt{\theta^9}} + \dfrac{1}{\sqrt{\theta^7}}\right)$.

$= 9.6$ radians per second per second.

(7) $v = 20.4t^2 - 10.8$. $a = 40.8t$. 172.8 in./sec., 1

(8) $v = \dfrac{1}{30\sqrt[3]{(t-125)^2}}$, $a = -\dfrac{1}{45\sqrt[3]{(t-125)^5}}$.

(9) $v = 0.8 - \dfrac{8t}{(4+t^2)^2}$, $a = \dfrac{24t^2-32}{(4+t^2)^3}$, 0.7926 an

(10) $n = 2$, $n = 11$.

Exercises VI. (p. 72.)

(1) $\dfrac{x}{\sqrt{x^2+1}}$.

(2) $\dfrac{x}{\sqrt{x^2+a^2}}$.

(3) $-$

(4) $\dfrac{ax}{\sqrt{(a-x^2)^3}}$.

(5) $\dfrac{2a^2-x^2}{x^3\sqrt{x^2-a^2}}$.

(1) $\dfrac{dw}{dx} = \dfrac{3x^2\left(3+3x^3\right)}{27\left(\frac{1}{2}x^3+\frac{1}{4}x^6\right)^3}.$

(2) $\dfrac{dv}{dx} = -\dfrac{12x}{\sqrt{1+\sqrt{2}+3x^2}\left(\sqrt{3}+4\sqrt{1+\sqrt{2}+3x^2}\right.}$

(3) $\dfrac{du}{dx} = -\dfrac{x^2\left(\sqrt{3}+x^3\right)}{\sqrt{\left[1+\left(1+\dfrac{x^3}{\sqrt{3}}\right)^2\right]^3}}$

Exercises VIII. (p. 88.)

(2) 1.44.

(4) $\dfrac{dy}{dx} = 3x^2+3$; and the numerical values are: 3, 3

(5) $\pm\sqrt{2}$.

(6) $\dfrac{dy}{dx} = -\dfrac{4}{9}\dfrac{x}{y}$. Slope is zero where $x = 0$; and is $\mp$

(1) Min.: $x = 0$, $y = 0$; max.: $x = -2$, $y = -4$.

(2) $x = a$.

(4) $25\sqrt{3}$ square i

(5) $\dfrac{dy}{dx} = -\dfrac{10}{x^2} + \dfrac{10}{(8-x)^2}$; $x = 4$; $y = 5$.

(6) Max. for $x = -1$; min. for $x = 1$.

(7) Join the middle points of the four sides.

(8) $r = \frac{2}{3}R$, $r = \dfrac{R}{2}$, no max.

(9) $r = R\sqrt{\dfrac{2}{3}}$, $r = \dfrac{R}{\sqrt{2}}$, $r = 0.8506R$.

(10) At the rate of $\dfrac{8}{r}$ square feet per second.

(11) $r = \dfrac{R\sqrt{8}}{3}$.

(12) $n = \sqrt{\dfrac{NR}{r}}$.

(6) Max.: $x = 1.414$, $y = 1.7675$.

Min.: $x = -1.414$, $y = 1.7675$.

(7) Max.: $x = -3.565$, $y = 2.12$.

Min.: $x = +3.565$, $y = 7.88$.

(8) $0.4N$, $0.6N$.

(9) $x = \sqrt{\frac{a}{c}}$.

(10) Speed 8.66 nautical miles per hour. Time taken

Minimum cost £112. 12s.

(11) Max. and min. for $x = 7.5$, $y = \pm 5.414$. (Se

p. 71.)

(12) Min.: $x = \frac{1}{2}$, $y = 0.25$; max.: $x = -\frac{1}{3}$, $y = 1.408$

(8) $\dfrac{7}{9(3x+1)} + \dfrac{71}{63(3x-2)} - \dfrac{5}{7(2x+1)}.$

(9) $\dfrac{1}{3(x-1)} + \dfrac{2x+1}{3(x^2+x+1)}.$

(10) $x + \dfrac{2}{3(x+1)}$

(11) $\dfrac{3}{(x+1)} + \dfrac{2x+1}{x^2+x+1}.$

(12) $\dfrac{1}{x-1} - \dfrac{1}{x-}$

(13) $\dfrac{1}{4(x-1)} - \dfrac{1}{4(x+1)} + \dfrac{1}{2(x+1)^2}.$

(14) $\dfrac{4}{9(x-1)} - \dfrac{4}{9(x+2)} - \dfrac{1}{3(x+2)^2}.$

(15) $\dfrac{1}{x+2} - \dfrac{x-1}{x^2+x+1} - \dfrac{1}{(x^2+x+1)^2}.$

(16) $\dfrac{5}{x+4} - \dfrac{32}{(x+4)^2} + \dfrac{36}{(x+4)^3}.$

(17) $\dfrac{7}{9(3x-2)^2} + \dfrac{55}{9(3x-2)^3} + \dfrac{73}{9(3x-2)^4}.$

(18) $\dfrac{1}{6(x-2)} + \dfrac{1}{3(x-2)^2} - \dfrac{x}{6(x^2+2x+4)}.$

$\left(3x^2 - 1 \quad 2\left(\sqrt{x} + x\right)\right)$

(11) $\dfrac{1 - \log_\epsilon (x+3)}{(x+3)^2}$.

(12) $a^x \left(ax^{a-1} + x^a \log_\epsilon a\right)$.

(14) Min.: $y = 0.7$

(15) $\dfrac{1+x}{x}$.

(16) $\dfrac{3}{x}(\log_\epsilon ax)^2$.

Exercises XIII. (p. 160.)

(1) Let $\dfrac{t}{T} = x$ $(\therefore t = 8x)$, and use the Table on pag

(2) $T = 34.627$; 159.46 minutes.

(3) Take $2t = x$; and use the Table on page 157.

(5) (a) $x^x \left(1 + \log_\epsilon x\right)$; (b) $2x(\epsilon^x)^x$; (c) $\epsilon^{x^x} \times x^x$ (

(13) Min. for $x = \log_\epsilon a$.

Exercises XIV. (p. 170.)

(1) (i) $\dfrac{dy}{d\theta} = A\cos\left(\theta - \dfrac{\pi}{2}\right)$;

(ii) $\dfrac{dy}{d\theta} = 2\sin\theta\cos\theta = \sin 2\theta$ and $\dfrac{dy}{d\theta} = 2\cos$

(iii) $\dfrac{dy}{d\theta} = 3\sin^2\theta\cos\theta$ and $\dfrac{dy}{d\theta} = 3\cos 3\theta$.

(2) $\theta = 45°$ or $\dfrac{\pi}{4}$ radians.

(3) $\dfrac{dy}{dt} = -n\sin 2$

(4) $a^x \log_\epsilon a \cos a^x$.

(5) $\dfrac{\cos x}{\sin x} = \text{cotan}$

(6) $18.2\cos(x + 26°)$.

(11) $(i)\ \dfrac{dy}{dx} = \dfrac{ab}{(x+b)^2}$; (ii) $\dfrac{a}{b}\epsilon^{-\frac{x}{b}}$; (iii) $\dfrac{1}{90}^{\circ} \times \dfrac{a}{(b^2 -$

(12) (i) $\dfrac{dy}{dx} = \sec x \tan x$;

(ii) $\dfrac{dy}{dx} = -\dfrac{1}{\sqrt{1-x^2}}$;

(iii) $\dfrac{dy}{dx} = \dfrac{1}{1+x^2}$;

(iv) $\dfrac{dy}{dx} = \dfrac{1}{x\sqrt{x^2-1}}$;

(v) $\dfrac{dy}{dx} = \dfrac{\sqrt{3\sec x}\,(3\sec^2 x - 1)}{2}$.

(13) $\dfrac{dy}{d\theta} = 4.6\,(2\theta+3)^{1.3}\cos(2\theta+3)^{2.3}$.

(14) $\dfrac{dy}{d\theta} = 3\theta^2 + 3\cos(\theta+3) - \log_{\epsilon} 3\left(\cos\theta \times 3^{\sin\theta} +$

(15) $\theta = \cot\theta; \theta = \pm 0.86$; is max. for $+\theta$, min. for $-$

(5) $dy = 3 \sin v u^2 \, du + u^3 \cos v \, dv,$

$dy = u \sin x^{u-1} \cos x \, dx + (\sin x)^u \log_\epsilon \sin x du,$

$dy = \frac{1}{v} \frac{1}{u} du - \log_\epsilon u \frac{1}{v^2} dv.$

(7) Minimum for $x = y = -\frac{1}{2}$.

(8) (a) Length 2 feet, width = depth = 1 foot, vol.

(b) Radius = $\frac{2}{\pi}$ feet = 7.46 in., length = 2 f

(9) All three parts equal; the product is maximum.

(10) Minimum for $x = y = 1$.

(11) Min.: $x = \frac{1}{2}$ and $y = 2$.

(12) Angle at apex = 90°; equal sides = length = $\sqrt[3]{}$

Exercises XVII. (p. 202.)

(1) $\dfrac{4\sqrt{a}x^{\frac{3}{2}}}{3} + C.$ (2) $-\dfrac{1}{x^3} + C.$ (3) $\dfrac{x^4}{4a}$

(4) $\frac{1}{3}x^3 + ax + C.$ (5) $-2x^{-\frac{5}{2}} + C.$

(6) $x^4 + x^3 + x^2 + x + C.$ (7) $\dfrac{ax^2}{4} + \dfrac{bx^3}{9} +$

(8) $\dfrac{x^2 + a}{x + a} = x - a + \dfrac{a^2 + a}{x + a}$ by division. Therefo

$\dfrac{x^2}{2} - ax + (a^2 + a)\log_\epsilon(x + a) + C.$ (See pages 1

(9) $\dfrac{x^4}{4} + 3x^3 + \dfrac{27}{2}x^2 + 27x + C.$ (10) $\dfrac{x^3}{3} + \dfrac{2 - a}{2}x^2$

(11) $a^2(2x^{\frac{3}{2}} + \frac{9}{4}x^{\frac{4}{3}}) + C.$ (12) $-\frac{1}{3}\cos\theta - \frac{1}{6}\theta$

(13) $\dfrac{\theta}{2} + \dfrac{\sin 2a\theta}{4a} + C.$ (14) $\dfrac{\theta}{2} - \dfrac{\sin 2\theta}{4} +$

(2) Area $= \frac{2}{3}$ of $a \times 2a\sqrt{a}$.

(3) Area $= 2$; mean ordinate $= \dfrac{2}{\pi} = 0.637$.

(4) Area $= 1.57$; mean ordinate $= 0.5$.

(5) 0.572, 0.0476.

(6) Volume $= \pi r^2$

(7) 1.25.

(8) 79.4.

(9) Volume $= 4.9348$; area of surface $= 12.57$ (from

(10) $a \log_\epsilon a, \quad \dfrac{a}{a-1} \log_\epsilon a.$

(12) Arithmetical mean $= 9.5$; quadratic mean $= 10.$

(13) Quadratic mean $= \dfrac{1}{\sqrt{2}} \sqrt{A_1^2 + A_3^2}$; arithmetical

The first involves a somewhat difficult inte
stated thus: By definition the quadratic mean w

$$\sqrt{\frac{1}{2\pi} \int_0^{2\pi} (A_1 \sin x + A_3 \sin 3x)^2}$$

2

Making these substitutions, and integrating,

$$\frac{A_1^2}{2}\left(x - \frac{\sin 2x}{2}\right) + A_1 A_3 \left(\frac{\sin 2x}{2} - \frac{\sin 4x}{4}\right) +$$

At the lower limit the substitution of 0 for x vanish, whilst at the upper limit the substitutio $A_1^2\pi + A_3^2\pi$. And hence the answer follows.

(14) Area is 62.6 square units. Mean ordinate is 10.4

(16) 436.3. (This solid is pear shaped.)

Exercises XIX. (p. 231.)

(1) $\frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C.$ (2) $\frac{x^2}{2}(\log_\epsilon x - \frac{1}{2})$

(11) $\frac{1}{2a}\log_\epsilon \frac{x-a}{x+a} + C.$ (12) $\log_\epsilon \frac{x-1}{x^2+1} +$

(13) $\frac{1}{4}\log_\epsilon \frac{1+x}{1-x} + \frac{1}{2}\arctan x + C.$

(14) $\frac{1}{\sqrt{a}}\log_\epsilon \frac{\sqrt{a}-\sqrt{a-bx^2}}{x\sqrt{a}}.$ (Let $\frac{1}{x} = v$; then, $\sqrt{v^2 - \frac{b}{a}} = v - u.$)

You had better differentiate now the answer the given expression as a check.

Every earnest student is exhorted to manufactu for himself at every stage, so as to test his powers. he can always test his answer by differentiating it, gets back the expression from which he started.

There are lots of books which give examples for p fice here to name two: R. G. Blaine's *The Calculus a* and F. M. Saxelby's *A Course in Practical Mathem*

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of Development. Vol. I. Part I. General
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TRANSCRIBER'S NOTE

The diagrams have been re-created, using accompan
formulas or descriptions from the text where possib

In Chapter XIV, pages 132–159, numerical values o
ϵ^x, and related quantities of British currency have b
and rounded to the nearest digit.

On page 142 (page 146 in the original), the graphs
natural logarithm and exponential functions, Figure
have been interchanged to match the surrounding t

The vertical dashed lines in the natural logarithm g
Figure 39 (Figure 38 in the original), have been mo
match the data in the corresponding table.

On page 164 (page 167 in the original), the graphs
and cosine functions, Figures 44 and 45, have been
to match the surrounding text.

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